MANUAL BOOK

PENERAPAN *FEW-SHOT ASPECT-BASED SENTIMENT ANALYSIS* UNTUK KOMENTAR AKREDITASI BERBAHASA INDONESIA: PERBANDINGAN MODEL *BASELINE* DAN MODEL TERAUGMENTASI TESAURUS

PEMEGANG HAK CIPTA

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A. Gambaran Umum

Manual book ini disusun sebagai panduan teknis untuk penggunaan dan implementasi model *few-shot Aspect Based Sentiment Analysis* (ABSA) berbasis IndoBERT, yang diperkaya dengan tesaurus domain hasil ekstraksi *large language model* (LLM). Panduan dirancang sebagai langkah untuk menganalisis inkonsistensi penilaian akreditasi pada Kriteria 9.2.A melalui deteksi relevansi aspek (*Aspect Category Detection* / ACD) dan klasifikasi sentimen (*Aspect Sentiment Classification* / ASC). Panduan ini dibuat untuk memandu pengguna dalam melakukan instalasi lingkungan kerja yang dibutuhkan, menjelaskan alur kerja teknis dari pra-pemrosesan data, konstruksi tesaurus, hingga pelatihan dan evaluasi model, serta memberikan instruksi untuk menjalankan setiap *script* kode secara berurutan.

B. Tujuan Sistem

Sistem ini bertujuan untuk mengolah data komentar mentah dari asesor akreditasi, membangun tesaurus dan leksikon sentimen domain menggunakan LLM, menerapkan augmentasi data berbasis tesaurus tersebut pada model IndoBERT yang dilatih dalam kondisi *few-shot*, hingga menghasilkan data terstruktur dan metrik evaluasi yang digunakan untuk menganalisis potensi inkonsistensi penilaian antara tahap *Asesmen Kecukupan* (AK) dan *Asesmen Lapangan* (AL).

C. Spesifikasi Perangkat

Perangkat Keras:

- Prosesor : Intel Core i7 atau setara (multi-*core*, untuk menjalankan Ollama secara lokal)
- RAM : minimal 16 GB — dibutuhkan untuk memuat model LLM (`gemma4:e4b` untuk ekstraksi, serta `qwen3.5:8b` dan `llama3.1:8b` untuk validasi) melalui Ollama secara bersamaan dengan proses Python/pandas
- GPU (disarankan) : NVIDIA GPU dengan dukungan CUDA dan VRAM minimal 8–12 GB (mis. Google Colab GPU runtime) untuk mempercepat inferensi LLM maupun pelatihan IndoBERT
- Penyimpanan : ruang kosong beberapa GB per model untuk mengunduh *weight* model melalui `ollama pull`

Perangkat Lunak:

- Aplikasi : Google Colaboratory atau Jupyter Notebook
- *Runtime* LLM : Ollama (untuk menjalankan model `gemma4:e4b`, `qwen3.5:8b`, dan `llama3.1:8b`)
- Pustaka Utama : pandas, numpy, nltk, transformers, torch, scikit-learn, iterative-stratification, ollama, openpyxl, matplotlib, seaborn, tqdm

D. Menu dan Cara Penggunaan

1. **Persiapan Lingkungan**
   - Siapkan aplikasi yang dapat menjalankan file dengan ekstensi `.ipynb`, misalnya Google Colaboratory.

- Untuk tahap konstruksi tesaurus (ACD/ASC), pastikan Ollama telah terpasang dan model yang dibutuhkan telah diunduh (`ollama pull`).

2. **Persiapan Data**
   - Pastikan file dataset penelitian sudah diunggah ke Google Drive Anda.
   - Jika menggunakan Google Colaboratory, unggah file dataset tersebut ke sesi Colab atau ke Google Drive Anda.
   - Sesuaikan baris kode `BASE_PATH` pada setiap *cell* konfigurasi agar sesuai dengan lokasi file Anda.

3. **Eksekusi Program**
   - Sistem ini terdiri dari beberapa file `.ipynb` yang harus dijalankan secara berurutan sesuai tahapan di bawah, dan untuk menjalankan keseluruhan program pada setiap file, klik menu navigasi *Runtime* > *Run all*.
   - Beberapa tahap (konstruksi tesaurus ACD/ASC dan kombinasi AK+AL) memproses data AK dan AL secara terpisah; sesuaikan konfigurasi sesuai catatan pada tiap file.

Catatan: setiap tahap pada bagian D di bawah disarankan dijalankan sebagai satu *file* `.ipynb` yang terpisah, agar alur kerja tetap modular dan mudah ditelusuri apabila terjadi kesalahan pada salah satu tahap.

Tahapan yang dicakup dalam manual book ini:

1. Tahap Preprocessing
2. Tahap Deteksi dan Penyamaran Identitas Institusi
3. Tahap Konstruksi Tesaurus - ACD (AK/AL)
4. Tahap Konstruksi Leksikon Sentimen - ASC (AK/AL)
5. Tahap Analisis & Kombinasi Tesaurus - ACD (AK + AL)
6. Tahap Analisis & Kombinasi Sentimen - ASC (AK + AL)
7. Tahap *Few-Shot Fine-Tuning* - Baseline (IndoBERT)
8. Tahap *Few-Shot Fine-Tuning* - Tesaurus (IndoBERT)
9. Tahap Analisis Metrik Terpadu - Baseline vs Tesaurus

1 Tahap Preprocessing

Langkah ini dilakukan sebagai tahapan persiapan data mentah sebelum digunakan pada proses ekstraksi maupun pemodelan. Data dipisahkan menjadi dua subset berdasarkan tahap asesmen, yaitu Asesmen Kecukupan (AK) dan Asesmen Lapangan (AL), agar karakteristik duplikasi dan nilai kosong pada masing-masing tahap dapat ditangani dengan aturan yang sesuai.

[ ]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import re
import string
```

[ ]:
```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

[ ]:
```
df = pd.read_excel('/content/drive/MyDrive/file_name.xlsx') # Modifikasi␣
 ↪file_name
```

[ ]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20850 entries, 0 to 20849
Data columns (total 12 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Unnamed: 0        20850 non-null  int64
 1   nomor_usulan      20850 non-null  object
 2   nomor_aspek       20850 non-null  object
 3   keterangan_aspek  20850 non-null  object
 4   komen_AK          20846 non-null  object
 5   nilai_AK          20850 non-null  float64
 6   no_asesor_AK      20850 non-null  int64
 7   tipe_asesor       20850 non-null  int64
 8   komen_AL          20850 non-null  object
 9   nilai_AL          20850 non-null  object
 10  kode_asesor_AL    20850 non-null  object
 11  tipe_asesor_AL    20850 non-null  object
dtypes: float64(1), int64(3), object(8)
memory usage: 1.9+ MB
```

1.1 Filter hanya yang mengandung 'C9.2.A'

[ ]:
```
df = df[df['nomor_aspek'].str.startswith('C9.2.A')]
output_path = '/content/drive/MyDrive/c92a.xlsx'
df.to_excel(output_path, index=False)
```

[ ]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 578 entries, 66 to 20799
Data columns (total 12 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Unnamed: 0        578 non-null    int64
 1   nomor_usulan      578 non-null    object
 2   nomor_aspek       578 non-null    object
 3   keterangan_aspek  578 non-null    object
 4   komen_AK          578 non-null    object
 5   nilai_AK          578 non-null    float64
 6   no_asesor_AK      578 non-null    int64
 7   tipe_asesor       578 non-null    int64
 8   komen_AL          578 non-null    object
 9   nilai_AL          578 non-null    object
 10  kode_asesor_AL    578 non-null    object
 11  tipe_asesor_AL    578 non-null    object
dtypes: float64(1), int64(3), object(8)
memory usage: 58.7+ KB
```

[ ]:
```
selected_columns = ['nomor_usulan',
                    'komen_AK', 'no_asesor_AK',
                    'komen_AL', 'kode_asesor_AL', 'tipe_asesor_AL']
```

[ ]:
```
df[selected_columns] = df[selected_columns].replace('(NULL)', np.nan).
 ↪infer_objects(copy=False)
print(df[selected_columns].isna().sum())
```

```
nomor_usulan        0
komen_AK            0
no_asesor_AK        0
komen_AL           16
kode_asesor_AL    296
tipe_asesor_AL    296
dtype: int64
```

```
/tmp/ipykernel_4883/1467794762.py:1: FutureWarning: Downcasting behavior in
`replace` is deprecated and will be removed in a future version. To retain the
old behavior, explicitly call `result.infer_objects(copy=False)`. To opt-in to
the future behavior, set `pd.set_option('future.no_silent_downcasting', True)`
  df[selected_columns] = df[selected_columns].replace('(NULL)',
```

```
np.nan).infer_objects(copy=False)
```

[ ]:
```
null_komen_al_rows = df[df['komen_AL'].isna()]
display(null_komen_al_rows)
```

1.2 Pisah subset (AK dan AL)

[ ]:
```
ak_columns = [
    'nomor_usulan',
    'komen_AK',
    'no_asesor_AK'
]

al_columns = [
    'nomor_usulan',
    'komen_AL',
    'kode_asesor_AL',
    'tipe_asesor_AL'
]
```

[ ]:
```
df_ak = df[ak_columns].copy()
df_al = df[al_columns].copy()

df_ak.to_excel('/content/drive/MyDrive/komentar_AK.xlsx', index=False)
df_al.to_excel('/content/drive/MyDrive/komentar_AL.xlsx', index=False)
```

[ ]:
```
print(f"Komentar AK rows: {len(df_ak)}")
print(f"Komentar AL rows: {len(df_al)}")
```

```
Komentar AK rows: 578
Komentar AL rows: 578
```

1.3 File AK

[ ]:
```
df_ak.replace('(NULL)', np.nan).infer_objects(copy=False)
print(df_ak.isna().sum())
```

```
nomor_usulan    0
komen_AK        0
no_asesor_AK    0
dtype: int64
```

[ ]:
```
duplicate_count = df_ak.duplicated(
    subset=['nomor_usulan', 'komen_AK']
).sum()

print(f"Number of duplicates: {duplicate_count}")
```

```
Number of duplicates: 2
```

[ ]:
```python
# Remove duplicates based on nomor_usulan and komen_AK
before_rows = len(df_ak)

df_ak = df_ak.drop_duplicates(
    subset=['nomor_usulan', 'komen_AK'],
    keep='first'
).reset_index(drop=True)

after_rows = len(df_ak)

print(f"Removed {before_rows - after_rows} duplicate rows")
print(f"Remaining rows: {after_rows}")

# Save cleaned file
df_ak.to_excel('/content/drive/MyDrive/komentar_AK.xlsx', index=False)
```

```
Removed 2 duplicate rows
Remaining rows: 576
```

[ ]:
```python
print(df_ak.shape)
```

```
(576, 3)
```

1.4 File AL

1.5 Missing Values

[ ]:
```python
df_al.replace('(NULL)', np.nan).infer_objects(copy=False)
print(df_al.isna().sum())
```

```
nomor_usulan        0
komen_AL           16
kode_asesor_AL    296
tipe_asesor_AL    296
dtype: int64
```

[ ]:
```python
komen_al_rows = df_al[df_al['komen_AL'].isna()]
display(komen_al_rows)
```

[ ]:
```python
df_al = df_al.dropna(subset=['komen_AL']).reset_index(drop=True)

df_al.to_excel('/content/drive/MyDrive/komentar_AL.xlsx', index=False)
```

[ ]:
```python
print("\nNull values after deletion:")
print(df_al.isna().sum())

print(f"\nRemaining rows: {len(df_al)}")
```

```
Null values after deletion:
```

```
nomor_usulan        0
komen_AL            0
kode_asesor_AL    281
tipe_asesor_AL    281
dtype: int64

Remaining rows: 562
```

[ ]:
```python
# Check whether all missing kode_asesor_AL rows are exactly
# the same rows as the missing tipe_asesor_AL rows
kode_null = df_al['kode_asesor_AL'].isna()
tipe_null = df_al['tipe_asesor_AL'].isna()

print("\nSame rows?")
print((kode_null == tipe_null).all())

mismatch_rows = df_al[kode_null != tipe_null]
print("\nRows with mismatched missing values:", len(mismatch_rows))
```

```
Same rows?
True

Rows with mismatched missing values: 0
```

1.6 Check Duplicates

[ ]:
```python
print(
    df_al.duplicated(
        subset=['nomor_usulan', 'komen_AL']
    ).sum()
)
```

```
281
```

[ ]:
```python
# Select rows where BOTH assessor-related columns are missing
null_rows = df_al[
    df_al['kode_asesor_AL'].isna() &
    df_al['tipe_asesor_AL'].isna()
]

# Select all rows that belong to a duplicate group based on proposal number,␣
↪aspect, and comment
duplicate_rows = df_al[
    df_al.duplicated(
        subset=['nomor_usulan', 'komen_AL'],
        keep=False
    )
```

```
]

# Print the number of rows with missing assessor information
print(len(null_rows))

# Print the total number of rows involved in duplicate groups
print(len(duplicate_rows))

# Keep only the columns used to define duplicates, then remove repeated␣
↪occurrences of those combinations
duplicate_keys = duplicate_rows[
    ['nomor_usulan', 'komen_AL']
].drop_duplicates()

# Check whether each null row has a matching duplicate key by joining the null␣
↪rows with the duplicate combinations
check = null_rows.merge(
    duplicate_keys,
    on=['nomor_usulan', 'komen_AL'],
    how='inner'
)

# Print how many null rows are also duplicate rows
print(f"Matching rows: {len(check)}")
```

```
281
562
Matching rows: 281
```

[ ]:

```
display(
    duplicate_rows.sort_values(
        ['nomor_usulan', 'komen_AL']
    ).head(20)
)
```

[ ]:

```
# Remove rows where BOTH assessor columns are missing
df_al = df_al.drop(
    df_al[
        df_al['kode_asesor_AL'].isna() &
        df_al['tipe_asesor_AL'].isna()
    ].index
).reset_index(drop=True)
```

[ ]:

```
# Verify remaining null values
print(df_al.isna().sum())
```

```
nomor_usulan      0
komen_AL          0
```

```
kode_asesor_AL    0
tipe_asesor_AL    0
dtype: int64
```

[ ]:
```python
# Save updated dataframe
df_al.to_excel('/content/drive/MyDrive/komentar_AL.xlsx', index=False)

print(f"Remaining rows: {len(df_al)}")
print("Updated file saved to /content/drive/MyDrive/komentar_AL.xlsx")
```

```
Remaining rows: 281
Updated file saved to /content/drive/MyDrive/TA/komentar_AL.xlsx
```

2 Tahap Deteksi dan Penyamaran Identitas Institusi

Langkah ini dilakukan untuk menjaga kerahasiaan institusi yang dinilai, karena komentar asesor berbentuk narasi bebas yang kadang menyebut identitas institusi secara langsung maupun tidak langsung. Deteksi dilakukan melalui dua tahap, yaitu pencarian pola struktural menggunakan sebelas pola regular expression dan peninjauan token berbasis frekuensi untuk menangkap akronim maupun singkatan. Kedua tahap ini hanya menandai kandidat teks; seluruh hasil penandaan kemudian ditinjau dan disamarkan secara manual dengan placeholder generik seperti universitas x atau program studi x.

[ ]:
```
import pandas as pd
import numpy as np
import re
import string
from collections import Counter
```

[ ]:
```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

[ ]:
```
df_ak = pd.read_excel('/content/drive/MyDrive/komentar_AK.xlsx')
df_al = pd.read_excel('/content/drive/MyDrive/komentar_AL.xlsx')

for df in [df_ak, df_al]:
    drop_cols = [c for c in df.columns if c in ('level_0', 'index', 'Unnamed:␣
 ↪0')]
    df.drop(columns=drop_cols, inplace=True)
    df.reset_index(drop=True, inplace=True)

print(f'komentar_AK rows : {len(df_ak)}')
print(f'komentar_AL rows : {len(df_al)}')
print(f'\ndf_ak columns : {list(df_ak.columns)}')
print(f'df_al columns : {list(df_al.columns)}')
```

```
komentar_AK rows : 576
komentar_AL rows : 281

df_ak columns : ['nomor_usulan', 'komen_AK', 'no_asesor_AK']
df_al columns : ['nomor_usulan', 'komen_AL', 'kode_asesor_AL', 'tipe_asesor_AL']
```

[ ]:
```
def preprocess_text(text):
    if pd.isna(text):
        return np.nan
    text = str(text)
    text = text.lower()                                        # 1. lowercase
    text = re.sub(r'https?://\S+|www\.\S+', '', text)   # 2. remove URLs
    text = re.sub(r'\s+', ' ', text).strip()          # 3. normalise spaces
    return text
```

```
[ ]: df_ak['komentar_AK_lower'] = df_ak['komen_AK'].apply(preprocess_text)
     df_al['komentar_AL_lower'] = df_al['komen_AL'].apply(preprocess_text)

     print('Preprocessing done.')
     display(df_ak[['komen_AK', 'komentar_AK_lower']].head(3))
     display(df_al[['komen_AL', 'komentar_AL_lower']].head(3))
```

```
Preprocessing done.

                                            komen_AK  \
0  - Ketersediaan dokumen terkait:  * CP: tersedi…
1  Kelulusan Tepat Waktu: berdasarkan table 9.2 d…
2  1) Pemenuhan CPL untuk mata kuliah aspek sikap…

                                   komentar_AK_lower
0  - ketersediaan dokumen terkait: * cp: tersedia…
1  kelulusan tepat waktu: berdasarkan table 9.2 d…
2  1) pemenuhan cpl untuk mata kuliah aspek sikap…

                                            komen_AL  \
0  Keterlaksanaan kebijakan, standar, IKU, dan IK…
1  Pemenuhan luaran pendidikan mencakup: 1) Pemen…
2  Pemenuhan Capaian Pembelajaran Lulusan yang te…

                                   komentar_AL_lower
0  keterlaksanaan kebijakan, standar, iku, dan ik…
1  pemenuhan luaran pendidikan mencakup: 1) pemen…
2  pemenuhan capaian pembelajaran lulusan yang te…
```

```
[ ]: df_ak.to_excel('/content/drive/MyDrive/komentar_AK_lower.xlsx', index=False)
     df_al.to_excel('/content/drive/MyDrive/komentar_AL_lower.xlsx', index=False)
```

2.1 Structural Pattern Matches

```
[ ]: STRUCTURAL_PATTERNS = [
         ('universitas / universiti + next word',
          r'\b(?:universitas|universiti)\s+\w+'),
         ('program studi + next word',
          r'\bprogram\s+studi\s+\w+'),
         ('prodi + next word',
          r'\bprodi\s+\w+'),
         ('jurusan + next word',
          r'\bjurusan\s+\w+'),
         ('fakultas + next word',
          r'\bfakultas\s+\w+'),
         ('perguruan tinggi + next word',
          r'\bperguruan\s+tinggi\s+\w+'),
         ('institut / politeknik / akademi + next word',
          r'\b(?:institut|politeknik|poltek|akademi)\s+\w+'),
```

```
    ('sekolah tinggi + next word',
     r'\bsekolah\s+tinggi\s+\w+'),
    ('word + university',
     r'\b\w+\s+university\b'),
    ('himpunan + next word',
     r'\bhimpunan\s+\w+'),
    ('bem + next word',
     r'\bbem\s+\w+'),
]


def context_preview(text, match, window=6):
    words = text.split()
    pos = text.find(match)
    if pos == -1:
        return match
    word_idx = len(text[:pos].split())
    match_span = len(match.split())
    end = min(len(words), word_idx + match_span + window)
    snippet = ' '.join(words[word_idx:end])
    suffix = ' ...' if end < len(words) else ''
    return f'{snippet}{suffix}'


def scan_structural(df, text_col):
    rows = []
    for pos, (_, row) in enumerate(df.iterrows()):
        text = row[text_col]
        if pd.isna(text):
            continue
        for label, pattern in STRUCTURAL_PATTERNS:
            hits = re.findall(pattern, text)
            for hit in hits:
                rows.append({
                    'excel_row': pos + 2,
                    'pattern':   label,
                    'matched':   hit,
                    'context':   context_preview(text, hit),
                })
    return pd.DataFrame(rows) if rows else pd.DataFrame(
        columns=['excel_row', 'pattern', 'matched', 'context'])
```

[ ]:
```
hits_ak = scan_structural(df_ak, 'komentar_AK_lower')
print(f'AK - {len(hits_ak)} structural hit(s) across '
      f'{hits_ak["excel_row"].nunique() if len(hits_ak) else 0} row(s)')
display(hits_ak)
```

```
[ ]: hits_al = scan_structural(df_al, 'komentar_AL_lower')
print(f'AL - {len(hits_al)} structural hit(s) across '
      f'{hits_al["excel_row"].nunique() if len(hits_al) else 0} row(s)')
display(hits_al)
```

2.2 Words Starting with u: Frequency Table

```
[ ]: def u_word_freq(df, text_col):
    counter = Counter()
    row_map = {}  # word -> list of excel_rows
    for idx, text in df[text_col].dropna().items():
        seen_in_row = set()
        for raw_word in text.split():
            word = raw_word.strip(string.punctuation)
            if len(word) >= 2 and word.startswith('u'):
                counter[word] += 1
                if word not in seen_in_row:
                    row_map.setdefault(word, []).append(idx + 2)  # Excel row
                    seen_in_row.add(word)
    result = pd.DataFrame(counter.most_common(), columns=['word', 'count'])
    result['excel_rows'] = result.apply(
        lambda r: row_map.get(r['word'], []) if r['count'] < 20 else '',
        axis=1
    )
    return result
```

```
[ ]: freq_ak = u_word_freq(df_ak, 'komentar_AK_lower')
print(f'AK - {len(freq_ak)} unique u-words')
display(freq_ak)
```

```
AK - 88 unique u-words

                word  count excel_rows
0              untuk    560
1               umum     58
2               upps     55
3        universitas     50
4             uraian     33
..               ...    ...        ...
83        universiti      1      [485]
84            unusia      1      [505]
85               upp      1      [507]
86   upi-yptkpadang       1      [564]
87           umumnya      1      [567]

[88 rows x 3 columns]
```

```
[ ]: freq_al = u_word_freq(df_al, 'komentar_AL_lower')
print(f'AL - {len(freq_al)} unique u-words')
display(freq_al)
```

```
AL - 74 unique u-words

           word  count                excel_rows
0         untuk    269
1          umum     39
2   universitas     38
3          upps     20
4         usaha     11  [62, 83, 124, 142, 209]
..          …     …                        …
69           ub      1                     [237]
70   universiti      1                     [241]
71        unhan      1                     [256]
72      unisyss      1                     [263]
73          unm      1                     [273]

[74 rows x 3 columns]
```

3 Tahap Konstruksi Tesaurus - ACD (AK)

Tahap ini menjalankan ekstraksi sinonim, singkatan, dan padanan istilah untuk setiap aspek (*Aspect Category Detection* / ACD), menggunakan model LLM (`gemma4:latest`) yang dijalankan melalui Ollama dengan pendekatan *Subject-Predicate-Object* (SPO). Seluruh delapan aspek diproses dalam satu panggilan `ollama.chat()` per baris komentar. Hasil ekstraksi disimpan sebagai kandidat tesaurus domain yang akan melalui tahap pembersihan dan validasi pada tahap berikutnya.

> **Versi ini menggunakan data AK (Asesmen Kecukupan).** Untuk memproses data AL (Asesmen Lapangan), ubah `FILE_NAME`, `TEXT_COL`, `TIPE_KOMENTAR`, dan `DATASET_NAME` pada cell konfigurasi di bawah; baris yang perlu diubah ditandai dengan komentar `# ← ganti ke AL di sini`.

[ ]:
```
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call
drive.mount("/content/drive", force_remount=True).
```

3.1 Install Ollama + Start Server + Pull Model

[ ]:
```
import subprocess, threading, time

# 1. System deps + Ollama install
!sudo apt-get install -y zstd pciutils -q
!curl -fsSL https://ollama.com/install.sh | sh

# 2. Start server in background thread
def run_ollama_serve():
    subprocess.Popen(["ollama", "serve"])

thread = threading.Thread(target=run_ollama_serve)
thread.start()
time.sleep(5)
print("Ollama server started.")

# 3. Pull model - change MODEL_NAME below to switch models
MODEL_NAME = "gemma4:latest"   # ← change here if needed
!ollama pull {MODEL_NAME}
print(f"  {MODEL_NAME} ready.")
```

```
Reading package lists…
Building dependency tree…
Reading state information…
pciutils is already the newest version (1:3.7.0-6).
zstd is already the newest version (1.4.8+dfsg-3build1).
0 upgraded, 0 newly installed, 0 to remove and 53 not upgraded.
>>> Cleaning up old version at /usr/local/lib/ollama
>>> Installing ollama to /usr/local
```

```
>>> Downloading ollama-linux-amd64.tar.zst
######################################################################## 100.0%
>>> Adding ollama user to video group…
>>> Adding current user to ollama group…
>>> Creating ollama systemd service…
WARNING: systemd is not running
>>> NVIDIA GPU installed.
>>> The Ollama API is now available at 127.0.0.1:11434.
>>> Install complete. Run "ollama" from the command line.
Ollama server started.
  gemma4:latest ready.
```

[ ]:
```python
import sys, subprocess

PACKAGES = {
    "ollama":   "ollama",
    "pandas":   "pandas",
    "openpyxl": "openpyxl",
    "tqdm":     "tqdm",
}

for imp, pkg in PACKAGES.items():
    try:
        __import__(imp)
    except ImportError:
        subprocess.check_call([sys.executable, "-m", "pip", "install", pkg,␣
 ↪"-q"])

print("All packages OK.")
```

```
All packages OK.
```

[ ]:
```python
import os, re
import pandas as pd
import ollama
from datetime import datetime
from tqdm import tqdm

print("All imports OK.")
```

```
All imports OK.
```

[ ]:
```python
BASE_PATH     = "/content/drive/MyDrive"

FILE_NAME     = "komentar_AK_lower.xlsx"    # ← ganti ke AL di sini:␣
 ↪"komentar_AL_lower.xlsx"
TEXT_COL      = "komentar_AK_lower"         # ← ganti ke AL di sini:␣
 ↪"komentar_AL_lower"
```

```
TIPE_KOMENTAR = "AK"                        # ← ganti ke AL di sini: "AL"
MODEL_NAME    = "gemma4:latest"
DATASET_NAME  = "ACD_AK"                    # ← ganti ke AL di sini: "ACD_AL"

MODEL_FOLDER = MODEL_NAME.replace(":", "-").replace("/", "-")
OUTPUT_DIR   = os.path.join(BASE_PATH, "Thesaurus", "ACD_AK", MODEL_FOLDER)   #␣
 ↪← ganti "ACD_AK" ke "ACD_AL" di sini juga
LOGS_DIR     = os.path.join(OUTPUT_DIR, "logs")
OUTPUT_EXCEL = os.path.join(OUTPUT_DIR, "HASIL_ACD_AK_synonym.xlsx")         #␣
 ↪← ganti "AK" ke "AL" di sini juga

os.makedirs(LOGS_DIR, exist_ok=True)

ROW_START = 1  # ← e.g. 1   → start from row 1 of the Excel
ROW_END   = 576

print(f"Base path  : {BASE_PATH}")
print(f"Source file: {FILE_NAME}")
print(f"Output dir : {OUTPUT_DIR}")
print(f"Excel out  : {OUTPUT_EXCEL}")
print(f"Logs dir   : {LOGS_DIR}")
```

[ ]:
```
ASPEK_LIST = [
    "capaian pembelajaran lulusan",
    "indeks prestasi kumulatif",
    "kelulusan tepat waktu",
    "masa tunggu lulusan",
    "pelacakan dan perekaman data lulusan",
    "prestasi mahasiswa",
    "karya dengan hak kekayaan intelektual",
    "kesesuaian bidang kerja",
]

print(f"{len(ASPEK_LIST)} aspects configured.")
```

```
8 aspects configured.
```

[ ]:
```
import sys

try:
    models    = ollama.list()
    available = [m.model for m in models.models]
    print(f"Available models: {available}")
    if not any(MODEL_NAME in m for m in available):
        print(f"\n[ERROR] Model '{MODEL_NAME}' not found.")
        print(f"  Run:  !ollama pull {MODEL_NAME}")
        raise RuntimeError(f"Model not found: {MODEL_NAME}")
```

```python
    print(f"Model '{MODEL_NAME}' found. OK.")
except Exception as e:
    print(f"[ERROR] Cannot connect to Ollama: {e}")
    print("  Make sure Cell 1 (Ollama install) was run successfully.")
    raise
```

```
Available models: ['gemma4:latest']
Model 'gemma4:latest' found. OK.
```

[ ]:
```python
file_path = os.path.join(BASE_PATH, FILE_NAME)
df        = pd.read_excel(file_path)

if "nomor_usulan" not in df.columns:
    df["nomor_usulan"] = df.index.astype(str)
df["nomor_usulan"] = df["nomor_usulan"].astype(str).str.strip()
total_rows = len(df)

if ROW_START is not None or ROW_END is not None:
    _s = (ROW_START - 1) if ROW_START is not None else 0
    _e = ROW_END          if ROW_END   is not None else total_rows
    df = df.iloc[_s:_e].copy()
    print(f"Row window applied : rows {ROW_START or 1} - {ROW_END or␣
↪total_rows} "
          f"({len(df)} rows in window)")

all_results     = []
rows_to_process = df
remaining       = len(rows_to_process)

print(f"Dataset             : {FILE_NAME}")
print(f"Column              : {TEXT_COL}")
print(f"Total rows in file : {total_rows}")
print(f"Rows to process     : {remaining}")

if remaining == 0:
    print("\nNo rows to process.")
```

3.2 System Prompt

[ ]:
```python
SPO_SYSTEM = """## Task Role
Anda adalah seorang linguis yang berspesialisasi dalam leksikologi
dan identifikasi relasi ekuivalensi antar istilah, termasuk variasi
terminologi, singkatan, dan padanan yang muncul dalam laporan asesor
akreditasi pendidikan tinggi.

## Task Context
```

```
Anda membantu menyusun thesaurus domain-spesifik untuk mendukung Aspect-Based␣
↪Sentiment Analysis (ABSA) pada komentar penilaian akreditasi pendidikan␣
↪tinggi.

Dataset berisi komentar asesor yang tersimpan pada kolom komentar_AK.

Komentar dapat mengandung:
- variasi istilah
- singkatan
- padanan istilah aspek

## Task Objective
Bacalah komentar berikut dan identifikasi semua sinonim, padanan istilah,␣
↪singkatan, variasi penyebutan, serta rujukan eksplisit maupun implisit yang␣
↪mengacu pada istilah aspek yang diberikan.

## Instruction Rules

1. Gunakan struktur SPO (Subject\u2013Predicate\u2013Object) untuk␣
↪mengidentifikasi relasi sinonim.

2. Setiap entri HARUS mengikuti format:

Istilah Aspek | Frasa Kunci Penanda Relasi | Sinonim/Padanan

HANYA 1 Hasil Sinonim per row.

3. Hanya lakukan ekstraksi berdasarkan komentar yang diberikan.

4. Hanya identifikasi sinonim/padanan yang:
- muncul secara eksplisit/implicit pada komentar

5. Frasa kunci penanda relasi HANYA boleh:
- disebut sebagai
- disingkat sebagai
- dikenal sebagai
- merujuk pada
- mengacu pada
- merupakan istilah lain dari

6. Jangan membuat istilah baru yang tidak muncul pada komentar.

8. Jika istilah aspek pada input sudah berupa nama lengkap, pertahankan nama␣
↪lengkap tersebut pada output. Jangan ubah menjadi singkatan kecuali␣
↪singkatan itu muncul eksplisit di komentar sebagai hubungan singkatan.

9. Sertakan komentar asli sebagai bukti ekstraksi.
```

```
10. Pertahankan metadata:
- tipe komentar
- nomor_usulan

untuk menjaga traceability.
"""

print("SPO system prompt loaded.")
print(f"Length: {len(SPO_SYSTEM)} chars")
```

```
SPO system prompt loaded.
Length: 1775 chars
```

[ ]:
```
def build_messages_all(aspek_list, komentar, tipe_komentar, nomor_usulan):
    aspek_lines = "\n".join(f"- {a}" for a in aspek_list)

    user_content = f"""{komentar}

Tipe Komentar: {tipe_komentar}
Nomor Usulan: {nomor_usulan}

Aspek:
{aspek_lines}"""

    return [
        {"role": "system", "content": SPO_SYSTEM},
        {"role": "user",   "content": user_content},
    ]

print("Prompt builder ready.")
```

```
Prompt builder ready.
```

[ ]:
```
import threading

# Output cleaner
def clean_raw_output(raw: str) -> str:
    cleaned = re.sub(r"<think>.*?</think>", "", raw, flags=re.DOTALL)
    cleaned = re.sub(r"```[a-zA-Z]*", "", cleaned).replace("```", "")
    return cleaned.strip()


# Table parser
def parse_table_all(clean_text, aspek_list, tipe_komentar, nomor_usulan):
    rows      = []
    seen_keys = set()
```

```python
    for line in clean_text.splitlines():
        if "|" not in line:
            continue
        parts = [p.strip() for p in line.split("|")]
        parts = [p for p in parts if p]
        if len(parts) < 3:
            continue
        if all(re.fullmatch(r"[-:= ]+", p) for p in parts):
            continue
        if any(h in parts[0].lower() for h in ("istilah", "frasa", "aspek")):
            continue
        key = (parts[0].lower(), parts[1].lower(), parts[2].lower())
        if key in seen_keys:
            continue
        seen_keys.add(key)
        rows.append({
            "Istilah Aspek":               parts[0],
            "Frasa Kunci Penanda Relasi": parts[1],
            "Sinonim/Padanan":             parts[2],
            "Tipe Komentar":               tipe_komentar,
            "Nomor Usulan":                nomor_usulan,
        })

    # Fallback: ensure every aspect has at least one row
    found = {r["Istilah Aspek"].lower() for r in rows}
    for aspek in aspek_list:
        if aspek.lower() not in found:
            rows.append({
                "Istilah Aspek":               aspek,
                "Frasa Kunci Penanda Relasi": "Tidak ditemukan",
                "Sinonim/Padanan":             "Tidak ditemukan",
                "Tipe Komentar":               tipe_komentar,
                "Nomor Usulan":                nomor_usulan,
            })
    return rows


# Inference runner
def run_inference(messages):
    _done    = threading.Event()
    _result = {}

    def _worker():
        _result["response"] = ollama.chat(
            model=MODEL_NAME,
            messages=messages,
            options={
```

```
                "temperature": 0.0,
                "num_ctx":     8192,
                "num_predict": 4096,
                "num_keep":    0,
            }
        )
        _done.set()

    thread = threading.Thread(target=_worker, daemon=True)
    thread.start()

    with tqdm(
        desc="  Inferencing",
        unit="s",
        ncols=72,
        bar_format="{desc}: {elapsed} elapsed [{postfix}]",
        dynamic_ncols=False,
        leave=False,
        position=1,
    ) as ibar:
        while not _done.is_set():
            time.sleep(1)
            ibar.update(1)
            ibar.set_postfix_str("waiting for Ollama...")
        ibar.set_postfix_str("done [OK]")

    thread.join()
    response      = _result["response"]
    raw           = response.message.content
    thinking_text = getattr(response.message, "thinking", "") or ""
    return raw, thinking_text


# Log saver
def save_row_log(nomor_usulan, idx, komentar, raw, thinking_text, cleaned,␣
 ↪parsed_rows):
    log_path = os.path.join(LOGS_DIR, f"{nomor_usulan}_row{idx}.txt")
    with open(log_path, "w", encoding="utf-8") as f:
        f.write(f"{'=' * 70}\n")
        f.write(f"NOMOR USULAN  : {nomor_usulan}\n")
        f.write(f"DATASET       : {DATASET_NAME}  |  COLUMN: {TEXT_COL}\n")
        f.write(f"TIPE KOMENTAR : {TIPE_KOMENTAR}\n")
        f.write(f"TIMESTAMP     : {datetime.now().strftime('%Y-%m-%d %H:%M:
 ↪%S')}\n")
        f.write(f"{'=' * 70}\n\n")
        f.write("[KOMENTAR ASLI]\n")
        f.write(komentar + "\n\n")
```

```
        f.write("[THINKING - reasoning block]\n")
        f.write((thinking_text or "(empty)") + "\n\n")
        f.write("[RAW OUTPUT]\n")
        f.write((raw or "(empty)") + "\n\n")
        f.write("[CLEANED OUTPUT]\n")
        f.write((cleaned or "(empty)") + "\n\n")
        f.write("[REPR - exact bytes into parser]\n")
        f.write(repr(cleaned) + "\n\n")
        f.write(f"[PARSED ROWS: {len(parsed_rows)}]\n")
        for r in parsed_rows:
            f.write(
                f"  Aspek  : {r['Istilah Aspek']}\n"
                f"  Frasa  : {r['Frasa Kunci Penanda Relasi']}\n"
                f"  Sinonim: {r['Sinonim/Padanan']}\n\n"
            )


# Excel saver
def save_excel(results):
    pd.DataFrame(results).to_excel(OUTPUT_EXCEL, index=False)


print("All helper functions ready.")
```

```
All helper functions ready.
```

3.3 Run LLM

[ ]:
```
if remaining == 0:
    print("No rows to process. Check ROW_START / ROW_END in Cell 4.")
else:
    print(f"Starting - {remaining} row(s) to process")
    print(f"  Logs  → {LOGS_DIR}")
    print(f"  Excel → {OUTPUT_EXCEL}")
    print()

    last_processed = None

    try:
      with tqdm(
        total=remaining,
        desc="  Rows",
        unit="row",
        ncols=80,
        position=0,
    ) as pbar:

        for idx, row in rows_to_process.iterrows():
```

```
            nomor_usulan = str(row.get("nomor_usulan", idx))
            komentar     = str(row.get(TEXT_COL, "")).strip()

            # Empty / NaN row
            if not komentar or komentar.lower() == "nan":
                skipped = [{
                    "Istilah Aspek":              aspek,
                    "Frasa Kunci Penanda Relasi": "SKIPPED",
                    "Sinonim/Padanan":            "komentar kosong",
                    "Tipe Komentar":              TIPE_KOMENTAR,
                    "Nomor Usulan":               nomor_usulan,
                    "Dataset":                    DATASET_NAME,
                } for aspek in ASPEK_LIST]
                all_results.extend(skipped)
                save_excel(all_results)
                with open(os.path.join(LOGS_DIR, f"{nomor_usulan}_row{idx}.txt"), "w", encoding="utf-8") as f:
                    f.write(f"NOMOR USULAN: {nomor_usulan}\nSTATUS: SKIPPED - komentar kosong\n")
                pbar.set_postfix_str(f"{nomor_usulan} SKIPPED")
                pbar.update(1)
                continue

            # Build prompt + run inference
            pbar.set_postfix_str(f"{nomor_usulan} inferencing...")
            messages = build_messages_all(ASPEK_LIST, komentar, TIPE_KOMENTAR, nomor_usulan)

            try:
                raw, thinking_text = run_inference(messages)
            except Exception as e:
                tqdm.write(f"  [ERROR] {nomor_usulan}: {e}")
                error_rows = [{
                    "Istilah Aspek":              aspek,
                    "Frasa Kunci Penanda Relasi": "ERROR",
                    "Sinonim/Padanan":            str(e),
                    "Tipe Komentar":              TIPE_KOMENTAR,
                    "Nomor Usulan":               nomor_usulan,
                    "Dataset":                    DATASET_NAME,
                } for aspek in ASPEK_LIST]
                all_results.extend(error_rows)
                save_excel(all_results)
                with open(os.path.join(LOGS_DIR, f"{nomor_usulan}_row{idx}.txt"), "w", encoding="utf-8") as f:
                    f.write(f"NOMOR USULAN: {nomor_usulan}\nSTATUS: ERROR\n{e}\n")
                pbar.update(1)
```

```
                continue

            # Guard: model returned nothing
            if not raw or not raw.strip():
                tqdm.write(f"  [WARN] {nomor_usulan}: model returned empty output")
                if thinking_text:
                    tqdm.write(
                        f"         Thinking was {len(thinking_text)} chars - "
                        f"raise num_predict if this keeps happening"
                    )

            # Parse
            cleaned     = clean_raw_output(raw or "")
            parsed_rows = parse_table_all(cleaned, ASPEK_LIST, TIPE_KOMENTAR, nomor_usulan)
            for r in parsed_rows:
                r["Dataset"] = DATASET_NAME

            all_results.extend(parsed_rows)

            # Save log + Excel immediately after each row
            save_row_log(nomor_usulan, idx, komentar, raw or "", thinking_text, cleaned, parsed_rows)
            save_excel(all_results)

            last_processed = nomor_usulan
            pbar.set_postfix_str(f"{nomor_usulan} [OK] {len(parsed_rows)} rows")
            pbar.update(1)

    except KeyboardInterrupt:
        print(f"\n[WARNING]  Processing interrupted!")
        print(f"   Last row completed : {last_processed or 'none (no rows finished)'}")
        print(f"   Resume: re-run Cell 7 → Cell 11; already-saved rows are auto-skipped.")
    finally:
        if last_processed:
            print(f"\n  [OK] Last successfully saved row : {last_processed}")

    print()
    print("=" * 80)
    print("DONE!")
    print(f"  Total result rows : {len(all_results)}")
    print(f"  Excel             : {OUTPUT_EXCEL}")
    print(f"  Logs folder       : {LOGS_DIR}")
    print("=" * 80)
```

4 Tahap Konstruksi Leksikon Sentimen - ASC (AK)

Tahap ini menjalankan ekstraksi ekspresi sentimen untuk setiap aspek (*Aspect Sentiment Classification* / ASC), menggunakan model LLM (`gemma4:latest`) yang dijalankan melalui Ollama dengan pendekatan *Subject-Predicate-Object* (SPO). Untuk setiap aspek, model menentukan kelas sentimen, mengutip ekspresi indikator dari komentar sebagai bukti, dan menghasilkan bentuk generalisasi dari ekspresi tersebut agar dapat digunakan kembali pada komentar lain. Seluruh delapan aspek diproses dalam satu panggilan `ollama.chat()` per baris komentar. Hasil ekstraksi disimpan sebagai kandidat leksikon sentimen domain yang akan melalui tahap pembersihan dan validasi pada tahap berikutnya.

> **Versi ini menggunakan data AK (Asesmen Kecukupan).** Untuk memproses data AL (Asesmen Lapangan), ubah `FILE_NAME`, `TEXT_COL`, `TIPE_KOMENTAR`, dan `DATASET_NAME` pada cell konfigurasi di bawah; baris yang perlu diubah ditandai dengan komentar `# ← ganti ke AL di sini`.

[ ]:
```
from google.colab import drive
drive.mount('/content/drive')
```

4.1 Install Ollama + Start Server + Pull Model

[ ]:
```
import subprocess, threading, time

# 1. System deps + Ollama install
!sudo apt-get install -y zstd pciutils -q
!curl -fsSL https://ollama.com/install.sh | sh

# 2. Start server in background thread
def run_ollama_serve():
    subprocess.Popen(["ollama", "serve"])

thread = threading.Thread(target=run_ollama_serve)
thread.start()
time.sleep(5)
print("Ollama server started.")

# 3. Pull model - change MODEL_NAME below to switch models
MODEL_NAME = "gemma4:latest"    # ← change here if needed
!ollama pull {MODEL_NAME}
print(f"  {MODEL_NAME} ready.")
```

[ ]:
```
import sys, subprocess

PACKAGES = {
    "ollama":   "ollama",
    "pandas":   "pandas",
    "openpyxl": "openpyxl",
    "tqdm":     "tqdm",
```

```
}

for imp, pkg in PACKAGES.items():
    try:
        __import__(imp)
    except ImportError:
        subprocess.check_call([sys.executable, "-m", "pip", "install", pkg,␣
↪"-q"])

print("All packages OK.")
```

[ ]:
```
import os, re
import pandas as pd
import ollama
from datetime import datetime
from tqdm import tqdm

print("All imports OK.")
```

[ ]:
```
BASE_PATH     = "/content/drive/MyDrive"

FILE_NAME     = "komentar_AK_lower.xlsx"   # ← ganti ke AL di sini:␣
↪"komentar_AL_lower.xlsx"
TEXT_COL      = "komentar_AK_lower"        # ← ganti ke AL di sini:␣
↪"komentar_AL_lower"
TIPE_KOMENTAR = "AK"                       # ← ganti ke AL di sini: "AL"
MODEL_NAME    = "gemma4:latest"    # must match what was pulled in Cell 1
DATASET_NAME  = "ASC_AK"                   # ← ganti ke AL di sini: "ASC_AL"

# Output organised by dataset and model name
MODEL_FOLDER = MODEL_NAME.replace(":", "-").replace("/", "-")
OUTPUT_DIR   = os.path.join(BASE_PATH, "Thesaurus", "ASC_AK", MODEL_FOLDER)   #␣
↪← ganti "ASC_AK" ke "ASC_AL" di sini juga
LOGS_DIR     = os.path.join(OUTPUT_DIR, "logs")
OUTPUT_EXCEL = os.path.join(OUTPUT_DIR, "HASIL_ASC_AK_sentiment.xlsx")        #␣
↪← ganti "AK" ke "AL" di sini juga

os.makedirs(LOGS_DIR, exist_ok=True)

ROW_START = 1  # ← e.g. 1   → start from row 1 of the Excel
ROW_END   = 576

print(f"Base path  : {BASE_PATH}")
print(f"Source file: {FILE_NAME}")
print(f"Output dir : {OUTPUT_DIR}")
print(f"Excel out  : {OUTPUT_EXCEL}")
print(f"Logs dir   : {LOGS_DIR}")
```

[ ]:
```python
ASPEK_LIST = [
    "capaian pembelajaran lulusan",
    "indeks prestasi kumulatif",
    "kelulusan tepat waktu",
    "masa tunggu lulusan",
    "pelacakan dan perekaman data lulusan",
    "prestasi mahasiswa",
    "karya dengan hak kekayaan intelektual",
    "kesesuaian bidang kerja",
]

print(f"{len(ASPEK_LIST)} aspects configured.")
```

[ ]:
```python
import sys

try:
    models    = ollama.list()
    available = [m.model for m in models.models]
    print(f"Available models: {available}")
    if not any(MODEL_NAME in m for m in available):
        print(f"\n[ERROR] Model '{MODEL_NAME}' not found.")
        print(f"  Run:  !ollama pull {MODEL_NAME}")
        raise RuntimeError(f"Model not found: {MODEL_NAME}")
    print(f"Model '{MODEL_NAME}' found. OK.")
except Exception as e:
    print(f"[ERROR] Cannot connect to Ollama: {e}")
    print("  Make sure Cell 1 (Ollama install) was run successfully.")
    raise
```

[ ]:
```python
file_path = os.path.join(BASE_PATH, FILE_NAME)
df        = pd.read_excel(file_path)

if "nomor_usulan" not in df.columns:
    df["nomor_usulan"] = df.index.astype(str)
df["nomor_usulan"] = df["nomor_usulan"].astype(str).str.strip()
total_rows = len(df)

if ROW_START is not None or ROW_END is not None:
    _s = (ROW_START - 1) if ROW_START is not None else 0
    _e = ROW_END          if ROW_END   is not None else total_rows
    df = df.iloc[_s:_e].copy()
    print(f"Row window applied : rows {ROW_START or 1} - {ROW_END or␣
 ↪total_rows} "
          f"({len(df)} rows in window)")

# Resume: load existing Excel results if present
if os.path.exists(OUTPUT_EXCEL):
```

```
    existing_df  = pd.read_excel(OUTPUT_EXCEL)
    all_results  = existing_df.to_dict("records")
    print(f"Resumed            : {len(all_results)} rows loaded from existing␣
 ↪Excel")
else:
    all_results  = []
    print("No existing Excel found - starting fresh.")

rows_to_process = df
remaining       = len(rows_to_process)

print(f"Dataset            : {FILE_NAME}")
print(f"Column             : {TEXT_COL}")
print(f"Total rows in file : {total_rows}")
print(f"Rows to process    : {remaining}")

if remaining == 0:
    print("\nNo rows to process.")
```

4.2 System Prompt

[ ]:
```
SPO_SYSTEM = """## Task Role
Anda adalah seorang linguis yang berspesialisasi dalam analisis sentimen
dan identifikasi ekspresi evaluatif dalam laporan asesor akreditasi
pendidikan tinggi.

## Task Context
Anda membantu membangun leksikon sentimen domain-spesifik dari komentar
asesor akreditasi pendidikan tinggi untuk Aspect-Based Sentiment Analysis␣
 ↪(ABSA).
Tugas Anda adalah ekstraksi penanda linguistik, bukan melakukan ABSA itu␣
 ↪sendiri.

## Task Objective
Untuk setiap aspek yang diberikan, bacalah komentar dan tentukan:
1. Kelas sentimen aspek tersebut (positif / negatif / netral / tidak relevan).
2. Ekspresi pendek yang dikutip langsung dari komentar sebagai bukti sentimen.
3. Generalisasi linguistik dari ekspresi tersebut - tanpa angka, nama aspek,
   atau data spesifik - yang dapat berlaku pada komentar lain dengan konteks␣
 ↪berbeda.

Anda HARUS menentukan sentimen sendiri berdasarkan isi komentar.
Tidak ada label yang diberikan - klasifikasi adalah bagian dari tugas Anda.

## Output Format

WAJIB: Setiap baris memiliki TEPAT 5 kolom dipisahkan TEPAT 4 tanda |.
```

```
Tidak ada header. Tidak ada teks lain. Tidak ada penjelasan. Hanya baris data.

  Istilah Aspek | Kelas Sentimen | Frasa Kunci Penanda Sentimen | Ekspresi␣
 ↪Indikator Sentimen | Ekspresi Generalisasi

## Aturan Per Kolom

### Kolom 1 - Istilah Aspek
Salin nama aspek persis seperti diberikan. Satu baris per ekspresi evaluatif␣
 ↪yang ditemukan.

### Kolom 2 - Kelas Sentimen
Pilih TEPAT SATU. Tidak boleh kosong. Tidak boleh nilai lain:
  positif       → aspek diekspresikan secara baik dalam komentar
  negatif       → aspek dikritik atau dinyatakan kurang dalam komentar
  netral        → aspek disebut tanpa muatan evaluatif yang jelas
  tidak relevan → aspek tidak disebut sama sekali dalam komentar

### Kolom 3 - Frasa Kunci Penanda Sentimen
Pilih TEPAT SATU dari empat nilai berikut. Tidak boleh diubah, disingkat, atau␣
 ↪diparafrase:
  dinilai positif melalui frasa
  dinilai negatif melalui frasa
  diidentifikasi sebagai penanda netral melalui frasa
  Tidak Ditemukan

Panduan pemilihan:
  → Kelas Sentimen = positif       : gunakan "dinilai positif melalui frasa"
  → Kelas Sentimen = negatif       : gunakan "dinilai negatif melalui frasa"
  → Kelas Sentimen = netral        : gunakan "diidentifikasi sebagai penanda␣
 ↪netral melalui frasa"
  → Kelas Sentimen = tidak relevan : gunakan "Tidak Ditemukan"

### Kolom 4 - Ekspresi Indikator Sentimen
Kutipan PENDEK (1-7 kata) yang diambil LANGSUNG dari komentar sebagai bukti␣
 ↪sentimen.
  → Jika Kelas Sentimen = tidak relevan : isi "Tidak Ditemukan"
  → Jika tidak ada bukti yang dapat dikutip : isi "Tidak Ditemukan"
  → DILARANG membuat kutipan yang tidak muncul di komentar.

### Kolom 5 - Ekspresi Generalisasi
Ikuti PERSIS tiga langkah ini secara berurutan. Jangan lakukan hal lain.

  LANGKAH 1: Jika Kolom 4 = "Tidak Ditemukan" → tulis "Tidak Ditemukan".␣
 ↪SELESAI.
```

```
  LANGKAH 2: Periksa apakah Kolom 4 mengandung setidaknya satu dari:
             angka, digit, persentase (%), nama institusi, nama orang.
             → TIDAK mengandung satupun → salin Kolom 4 PERSIS kata per kata.␣
↪SELESAI.
             → MENGANDUNG salah satu    → lanjut ke Langkah 3.

  LANGKAH 3: Hapus semua angka, digit, persentase, nama institusi, nama orang.
             Ungkapkan sisa makna evaluatifnya dalam 2-5 kata TANPA menyebut
             angka atau data spesifik apapun.
             Contoh: "rata2 ipk > 3.35"      → "nilai akademik tinggi"
             Contoh: "prosentase lulus 100%" → "kelulusan sangat tinggi"
             Contoh: "kurang dari 4 bulan"   → "waktu tunggu singkat"

Pertahankan metadata (tipe komentar dan nomor_usulan) pada setiap baris.
"""

print("SPO system prompt loaded.")
print(f"Length: {len(SPO_SYSTEM)} chars")
```

[ ]:
```
def build_messages_all(aspek_list, komentar, tipe_komentar, nomor_usulan):
    aspek_lines = "\n".join(f"- {a}" for a in aspek_list)

    user_content = f"""{komentar}

Tipe Komentar: {tipe_komentar}
Nomor Usulan: {nomor_usulan}

Aspek:
{aspek_lines}"""

    return [
        {"role": "system", "content": SPO_SYSTEM},
        {"role": "user",   "content": user_content},
    ]

print("Prompt builder ready.")
```

[ ]:
```
import threading

def clean_raw_output(raw: str) -> str:
    cleaned = re.sub(r"<think>.*?</think>", "", raw, flags=re.DOTALL)
    cleaned = re.sub(r"```[a-zA-Z]*", "", cleaned).replace("```", "")
    return cleaned.strip()


def parse_table_all(clean_text, aspek_list, tipe_komentar, nomor_usulan):
    rows = []
```

```python
    seen = set()

    for line in clean_text.splitlines():
        if "|" not in line:
            continue
        parts = [p.strip() for p in line.strip().strip("|").split("|")]
        # Need at least aspek + sentimen
        if len(parts) < 2:
            continue
        # Skip separator and header lines
        if all(re.fullmatch(r"[-: ]+", p) for p in parts):
            continue
        if parts[0].lower() in ("istilah aspek", "istilah", "aspek"):
            continue
        # Pad missing columns rather than dropping the row
        while len(parts) < 5:
            parts.append("Tidak Ditemukan")
        key = tuple(p.lower() for p in parts[:5])
        if key in seen:
            continue
        seen.add(key)
        rows.append({
            "Istilah Aspek":                parts[0],
            "Kelas Sentimen":               parts[1],
            "Frasa Kunci Penanda Sentimen": parts[2],
            "Ekspresi Indikator Sentimen":  parts[3],
            "Ekspresi Generalisasi":        parts[4],
            "Tipe Komentar":                tipe_komentar,
            "Nomor Usulan":                 nomor_usulan,
        })

    # Ensure every aspect has at least one row
    found = {r["Istilah Aspek"].lower() for r in rows}
    for aspek in aspek_list:
        if aspek.lower() not in found:
            rows.append({
                "Istilah Aspek":                aspek,
                "Kelas Sentimen":               "tidak relevan",
                "Frasa Kunci Penanda Sentimen": "Tidak Ditemukan",
                "Ekspresi Indikator Sentimen":  "Tidak Ditemukan",
                "Ekspresi Generalisasi":        "Tidak Ditemukan",
                "Tipe Komentar":                tipe_komentar,
                "Nomor Usulan":                 nomor_usulan,
            })
    return rows
```

```python
def run_inference(messages):
    _done   = threading.Event()
    _result = {}

    def _worker():
        _result["response"] = ollama.chat(
            model=MODEL_NAME,
            messages=messages,
            options={"temperature": 0.0, "num_ctx": 8192, "num_predict": 4096}
        )
        _done.set()

    thread = threading.Thread(target=_worker, daemon=True)
    thread.start()

    with tqdm(
        desc="  Inferencing", unit="s", ncols=72,
        bar_format="{desc}: {elapsed} elapsed [{postfix}]",
        dynamic_ncols=False, leave=False, position=1,
    ) as ibar:
        while not _done.is_set():
            time.sleep(1)
            ibar.update(1)
            ibar.set_postfix_str("waiting for Ollama...")
        ibar.set_postfix_str("done [OK]")

    thread.join()
    raw           = _result["response"].message.content
    thinking_text = getattr(_result["response"].message, "thinking", "") or ""
    return raw, thinking_text


def save_row_log(nomor_usulan, idx, komentar, raw, thinking_text, cleaned,␣
↪parsed_rows):
    log_path = os.path.join(LOGS_DIR, f"{nomor_usulan}_row{idx}.txt")
    with open(log_path, "w", encoding="utf-8") as f:
        f.write(f"{'=' * 70}\n")
        f.write(f"NOMOR USULAN  : {nomor_usulan}\n")
        f.write(f"DATASET       : {DATASET_NAME}  |  COLUMN: {TEXT_COL}\n")
        f.write(f"TIPE KOMENTAR : {TIPE_KOMENTAR}\n")
        f.write(f"TIMESTAMP     : {datetime.now().strftime('%Y-%m-%d %H:%M:
↪%S')}\n")
        f.write(f"{'=' * 70}\n\n")
        f.write("[KOMENTAR ASLI]\n")
        f.write(komentar + "\n\n")
        f.write("[THINKING - reasoning block]\n")
        f.write((thinking_text or "(empty)") + "\n\n")
```

```python
        f.write("[RAW OUTPUT]\n")
        f.write((raw or "(empty)") + "\n\n")
        f.write("[CLEANED OUTPUT]\n")
        f.write((cleaned or "(empty)") + "\n\n")
        f.write("[REPR - exact bytes into parser]\n")
        f.write(repr(cleaned) + "\n\n")
        f.write(f"[PARSED ROWS: {len(parsed_rows)}]\n")
        for r in parsed_rows:
            f.write(
                f"  Aspek         : {r['Istilah Aspek']}\n"
                f"  Sentimen      : {r['Kelas Sentimen']}\n"
                f"  Frasa         : {r['Frasa Kunci Penanda Sentimen']}\n"
                f"  Ekspresi      : {r['Ekspresi Indikator Sentimen']}\n"
                f"  Generalisasi : {r['Ekspresi Generalisasi']}\n\n"
            )


def save_excel(results):
    pd.DataFrame(results).to_excel(OUTPUT_EXCEL, index=False)


print("All helper functions ready.")
```

4.3 Run LLM

```python
if remaining == 0:
    print("No rows to process. Check ROW_START / ROW_END in Cell 4.")
else:
    print(f"Starting - {remaining} row(s) to process")
    print(f"  Logs  → {LOGS_DIR}")
    print(f"  Excel → {OUTPUT_EXCEL}")
    print()

    last_processed = None   # tracks the most-recently completed nomor_usulan

    try:
      with tqdm(
        total=remaining,
        desc="  Rows",
        unit="row",
        ncols=80,
        position=0,
    ) as pbar:

        for idx, row in rows_to_process.iterrows():
            nomor_usulan = str(row.get("nomor_usulan", idx))
            komentar     = str(row.get(TEXT_COL, "")).strip()
```

```
            # Empty / NaN row
            if not komentar or komentar.lower() == "nan":
                skipped = [{
                    "Istilah Aspek":                  aspek,
                    "Kelas Sentimen":                 "tidak relevan",
                    "Frasa Kunci Penanda Sentimen":   "SKIPPED",
                    "Ekspresi Indikator Sentimen":    "komentar kosong",
                    "Ekspresi Generalisasi":          "komentar kosong",
                    "Tipe Komentar":                  TIPE_KOMENTAR,
                    "Nomor Usulan":                   nomor_usulan,
                    "Dataset":                        DATASET_NAME,
                } for aspek in ASPEK_LIST]
                all_results.extend(skipped)
                save_excel(all_results)
                with open(os.path.join(LOGS_DIR, f"{nomor_usulan}_row{idx}.
↪txt"), "w", encoding="utf-8") as f:
                    f.write(f"NOMOR USULAN: {nomor_usulan}\nSTATUS: SKIPPED -␣
↪komentar kosong\n")
                pbar.set_postfix_str(f"{nomor_usulan} SKIPPED")
                pbar.update(1)
                continue

            # Build prompt + run inference
            pbar.set_postfix_str(f"{nomor_usulan} inferencing...")
            messages = build_messages_all(ASPEK_LIST, komentar, TIPE_KOMENTAR,␣
↪nomor_usulan)

            try:
                raw, thinking_text = run_inference(messages)
            except Exception as e:
                tqdm.write(f"  [ERROR] {nomor_usulan}: {e}")
                error_rows = [{
                    "Istilah Aspek":                  aspek,
                    "Kelas Sentimen":                 "ERROR",
                    "Frasa Kunci Penanda Sentimen":   "ERROR",
                    "Ekspresi Indikator Sentimen":    str(e),
                    "Ekspresi Generalisasi":          "ERROR",
                    "Tipe Komentar":                  TIPE_KOMENTAR,
                    "Nomor Usulan":                   nomor_usulan,
                    "Dataset":                        DATASET_NAME,
                } for aspek in ASPEK_LIST]
                all_results.extend(error_rows)
                save_excel(all_results)
                with open(os.path.join(LOGS_DIR, f"{nomor_usulan}_row{idx}.
↪txt"), "w", encoding="utf-8") as f:
```

```
                    f.write(f"NOMOR USULAN: {nomor_usulan}\nSTATUS:␣
↪ERROR\n{e}\n")
                pbar.update(1)
                continue

            # Guard: model returned nothing
            if not raw or not raw.strip():
                tqdm.write(f"  [WARN] {nomor_usulan}: model returned empty␣
↪output")
                if thinking_text:
                    tqdm.write(
                        f"         Thinking was {len(thinking_text)} chars - "
                        f"raise num_predict if this keeps happening"
                    )

            # Parse
            cleaned     = clean_raw_output(raw or "")
            parsed_rows = parse_table_all(cleaned, ASPEK_LIST, TIPE_KOMENTAR,␣
↪nomor_usulan)
            for r in parsed_rows:
                r["Dataset"] = DATASET_NAME

            all_results.extend(parsed_rows)

            # Save log + Excel immediately after each row
            save_row_log(nomor_usulan, idx, komentar, raw or "", thinking_text,␣
↪cleaned, parsed_rows)
            save_excel(all_results)

            last_processed = nomor_usulan
            pbar.set_postfix_str(f"{nomor_usulan} [OK] {len(parsed_rows)} rows")
            pbar.update(1)

    except KeyboardInterrupt:
        print(f"\n[WARNING]  Processing interrupted!")
        print(f"   Last row completed : {last_processed or 'none (no rows␣
↪finished)'}")
        print(f"   Resume: re-run Cell 7 → Cell 11; already-saved rows are␣
↪auto-skipped.")
    finally:
        if last_processed:
            print(f"\n  [OK] Last successfully saved row : {last_processed}")

    print()
    print("=" * 80)
    print("DONE!")
```

```
    print(f"  Total result rows : {len(all_results)}")
    print(f"  Excel             : {OUTPUT_EXCEL}")
    print(f"  Logs folder       : {LOGS_DIR}")
    print("=" * 80)
```

5 Analisis & Kombinasi Tesaurus - ACD (AK + AL)

Tahap ini memproses tesaurus ACD_AK dan ACD_AL secara terpisah (lower-casing, quality control terpadu, penghapusan duplikat), menggabungkan keduanya menjadi satu tesaurus ACD, memeriksa duplikat lintas-sumber, lalu menyimpan hasil akhir ke Excel.

Quality control terpadu (`drop_invalid_rows`): menghapus baris dengan (1) Frasa Kunci Penanda Relasi tidak valid, dan/atau (2) Istilah Aspek di luar 8 aspek yang ditentukan, sehingga angka `Total baris dimuat` dan total tabel distribusi selalu konsisten.

[ ]:
```
import pandas as pd
import os
from google.colab import drive
drive.mount('/content/drive')

# Konfigurasi
BASE_DIR = "/content/drive/MyDrive/Thesaurus"

FILE_PATH_AK = f"{BASE_DIR}/ACD/gemma4-latest/HASIL_ACD_AK_synonym.xlsx"
FILE_PATH_AL = f"{BASE_DIR}/ACD/gemma4-latest/HASIL_ACD_AL_synonym.xlsx"

OUTPUT_DIR_COMBINED       = f"{BASE_DIR}/ACD/Distribution"
OUTPUT_DIR_COMBINED_CLEAN = f"{BASE_DIR}/ACD/Distribution_clean"

ASPEK_LIST = [
    "capaian pembelajaran lulusan",
    "indeks prestasi kumulatif",
    "kelulusan tepat waktu",
    "masa tunggu lulusan",
    "pelacakan dan perekaman data lulusan",
    "prestasi mahasiswa",
    "karya dengan hak kekayaan intelektual",
    "kesesuaian bidang kerja",
]

VALID_FRASA_RELASI = {
    "disebut sebagai",
    "disingkat sebagai",
    "dikenal sebagai",
    "merujuk pada",
    "mengacu pada",
    "merupakan istilah lain dari",
    "tidak ditemukan",
}

# Kolom teks yang di-lowercase di awal (sebelum analisis apa pun) agar
# perbandingan & deduplikasi antar-sumber (AK vs AL) konsisten.
```

```
TEXT_COLUMNS_TO_LOWER = ["Istilah Aspek", "Frasa Kunci Penanda Relasi",␣
↪"Sinonim/Padanan"]
DUP_KEYS = ["Istilah Aspek", "Sinonim/Padanan"]

os.makedirs(OUTPUT_DIR_COMBINED, exist_ok=True)
os.makedirs(OUTPUT_DIR_COMBINED_CLEAN, exist_ok=True)
print(f"[OK] File AK             : {FILE_PATH_AK}")
print(f"[OK] File AL             : {FILE_PATH_AL}")
print(f"[OK] Combined raw dir    : {OUTPUT_DIR_COMBINED}")
print(f"[OK] Combined clean dir : {OUTPUT_DIR_COMBINED_CLEAN}")
```

```
Mounted at /content/drive
[OK] File AK             : /content/drive/MyDrive/Thesaurus/ACD/gemma4-latest/
↪HASIL_ACD_AK_synonym.xlsx
[OK] File AL             : /content/drive/MyDrive/Thesaurus/ACD/gemma4-latest/
↪HASIL_ACD_AL_synonym.xlsx
[OK] Combined raw dir    : /content/drive/MyDrive/Thesaurus/ACD/Distribution
[OK] Combined clean dir :
/content/drive/MyDrive/Thesaurus/ACD/Distribution_clean
```

5.1 Fungsi Bantu

[ ]:

```
def lowercase_text_columns(df: pd.DataFrame, columns=TEXT_COLUMNS_TO_LOWER) ->␣
↪pd.DataFrame:
    """Lower-case (+ strip) kolom teks agar perbandingan & deduplikasi␣
↪konsisten."""
    df = df.copy()
    for col in columns:
        if col in df.columns:
            df[col] = df[col].astype(str).str.strip().str.lower()
    return df


def print_summary_table(summary_df: pd.DataFrame, title: str = "RINGKASAN"):
    """Cetak tabel ringkasan ASCII per aspek."""
    col_w = max(len(a) for a in summary_df["Aspek"]) + 4
    sep      = "=" * (col_w + 16 + 20 + 10)
    hdr_line = "-" * col_w + "+" + "-" * 12 + "+" + "-" * 18 + "+" + "-" * 10

    print(sep)
    print(f" {title}")
    print(sep)
    print(f" {'ASPEK':<{col_w}} | {'DITEMUKAN':>10} | {'TIDAK DITEMUKAN':>15} |␣
↪{'TOTAL':>7}")
    print(hdr_line)
    for _, row in summary_df.iterrows():
        print(
```

```
            f" {row['Aspek']:<{col_w}} | {row['Ditemukan']:>10,} "
            f"| {row['Tidak Ditemukan']:>15,} | {row['Total']:>7,}"
        )
    print(sep)
    totals = summary_df[["Ditemukan", "Tidak Ditemukan", "Total"]].sum()
    print(
        f" {'TOTAL':<{col_w}} | {totals['Ditemukan']:>10,} "
        f"| {totals['Tidak Ditemukan']:>15,} | {totals['Total']:>7,}"
    )
    print(sep)


def count_found(sub: pd.DataFrame):
    """Hitung baris berdasarkan nilai literal kolom Sinonim/Padanan."""
    not_found = sub["Sinonim/Padanan"].astype(str).str.strip().str.lower() ==␣
↪"tidak ditemukan"
    return (~not_found).sum(), not_found.sum()


def build_distribution(df: pd.DataFrame, aspek_list=ASPEK_LIST) -> pd.DataFrame:
    """Bangun dataframe ringkasan Ditemukan / Tidak Ditemukan / Total per aspek.
↪"""
    rows = []
    for aspek in aspek_list:
        sub = df[df["Istilah Aspek"].str.strip().str.lower() == aspek.lower()]
        ditemukan, tidak = count_found(sub)
        rows.append({"Aspek": aspek, "Ditemukan": ditemukan,
                     "Tidak Ditemukan": tidak, "Total": len(sub)})
    return pd.DataFrame(rows)


def drop_invalid_rows(df: pd.DataFrame, valid_frasa=VALID_FRASA_RELASI,
                      aspek_list=ASPEK_LIST, label=""):
    """Data quality control - satu pass, dua cek:
      1. Frasa Kunci Penanda Relasi tidak ada di VALID_FRASA_RELASI
      2. Istilah Aspek tidak ada di ASPEK_LIST
    Baris yang gagal salah satu (atau keduanya) dihapus sekaligus.
    """
    col_frasa   = df["Frasa Kunci Penanda Relasi"].astype(str).str.strip().str.
↪lower()
    col_aspek   = df["Istilah Aspek"].astype(str).str.strip().str.lower()
    aspek_lower = {a.lower() for a in aspek_list}

    mask_bad_frasa = ~col_frasa.isin(valid_frasa)
    mask_bad_aspek = ~col_aspek.isin(aspek_lower)
    mask_bad       = mask_bad_frasa | mask_bad_aspek
    df_bad         = df[mask_bad].copy()
```

```
    print("=" * 70)
    print(f" DATA QUALITY REPORT - {label}")
    print("=" * 70)
    print(f"  Total baris diperiksa            : {len(df):,}")
    print(f"  Total baris invalid              : {len(df_bad):,}")
    print(f"  - Frasa Kunci Penanda Relasi inv : {mask_bad_frasa.sum():,}")
    print(f"  - Istilah Aspek di luar daftar   : {mask_bad_aspek.sum():,}")
    print(f"  - Keduanya invalid               : {(mask_bad_frasa & mask_bad_aspek).sum():,}")
    print()
    print(f"  {'ASPEK':<45} | {'FRASA INV':>10} | {'TOTAL INV':>10}")
    print("  " + "-" * 70)
    for aspek in aspek_list:
        m = col_aspek == aspek.lower()
        f = (m & mask_bad_frasa).sum()
        t = (m & mask_bad).sum()
        print(f"  {aspek:<45} | {f:>10,} | {t:>10,}")
    # Rows with unknown aspek are already counted in the summary above;
    # no extra table row - the header line "Istilah Aspek di luar daftar" covers them.
    print("=" * 70)

    if len(df_bad) > 0:
        print(f"\nSample (up to 10) invalid rows:")
        print(df_bad[["Istilah Aspek", "Frasa Kunci Penanda Relasi",
                      "Sinonim/Padanan"]].head(10).to_string(index=True))
    else:
        print("\n[OK] Tidak ada baris invalid.")

    df_ok = df[~mask_bad].copy().reset_index(drop=True)
    print(f"\n[OK] Dihapus {len(df_bad):,} baris invalid. Tersisa: {len(df_ok):,} baris.")
    return df_ok


def report_duplicates(df: pd.DataFrame, aspek_list=ASPEK_LIST, dup_keys=DUP_KEYS, label=""):
    """Hitung & cetak jumlah duplikat per aspek (tanpa menghapus)."""
    print("=" * 70)
    print(f" LAPORAN DUPLIKAT PER ASPEK - {label}")
    print("=" * 70)
    total_dup = 0
    for aspek in aspek_list:
        sub   = df[df["Istilah Aspek"].str.strip().str.lower() == aspek.lower()]
        n_dup = sub.duplicated(subset=dup_keys, keep="first").sum()
        total_dup += n_dup
```

```
        print(f"  {aspek:<45} → {n_dup:>4} duplikat")
    print("=" * 70)
    print(f"  {'TOTAL':<45} → {total_dup:>4} duplikat")
    print("=" * 70)


def remove_duplicates(df: pd.DataFrame, aspek_list=ASPEK_LIST,␣
↪dup_keys=DUP_KEYS, label=""):
    """Hapus duplikat per aspek & cetak laporan sebelum/sesudah."""
    print("=" * 70)
    print(f" PENGHAPUSAN DUPLIKAT PER ASPEK - {label}")
    print("=" * 70)
    parts = []
    for aspek in aspek_list:
        mask      = df["Istilah Aspek"].str.strip().str.lower() == aspek.lower()
        sub       = df[mask].copy()
        before    = len(sub)
        sub_clean = sub.drop_duplicates(subset=dup_keys, keep="first")
        removed   = before - len(sub_clean)
        parts.append(sub_clean)
        print(f"  {aspek:<45} → dihapus {removed:>4} baris  "
              f"(sebelum: {before:>4}, sesudah: {len(sub_clean):>4})")
    df_clean = pd.concat(parts, ignore_index=True)
    total_removed = len(df) - len(df_clean)
    print("=" * 70)
    print(f"  {'TOTAL DIHAPUS':<45} → {total_removed:>4} baris")
    print(f"  {'TOTAL TERSISA':<45} → {len(df_clean):>4} baris")
    print("=" * 70)
    return df_clean


def save_per_aspek(df: pd.DataFrame, output_dir: str, aspek_list=ASPEK_LIST,␣
↪suffix=""):
    """Simpan dataframe per aspek menjadi file Excel terpisah."""
    print(f"Menyimpan file ke {output_dir} ...")
    for aspek in aspek_list:
        sub = df[df["Istilah Aspek"].str.strip().str.lower() == aspek.lower()].
↪copy()
        safe_name = aspek.replace(" ", "_").replace("/", "-")
        out_path  = os.path.join(output_dir, f"{safe_name}{suffix}.xlsx")
        sub.to_excel(out_path, index=False)
        print(f"  [OK] {aspek:<45} → {len(sub):>4} baris → {out_path}")
```

5.2 ACD_AK

[ ]:
```
# [AK] Load data
df_ak = pd.read_excel(FILE_PATH_AK)
print(f"Total baris dimuat : {len(df_ak):,}")
print(f"Kolom              : {list(df_ak.columns)}")
df_ak.head(3)
```

```
Total baris dimuat : 4,752
Kolom              : ['Istilah Aspek', 'Frasa Kunci Penanda Relasi',
'Sinonim/Padanan', 'Tipe Komentar', 'Nomor Usulan', 'Dataset']
```

[ ]:
```
                  Istilah Aspek Frasa Kunci Penanda Relasi  \
0  capaian pembelajaran lulusan          disingkat sebagai
1     indeks prestasi kumulatif          disingkat sebagai
2         kelulusan tepat waktu            disebut sebagai

                      Sinonim/Padanan Tipe Komentar             Nomor Usulan  \
0                                  cp            AK
1                                 ipk            AK
2  prosentase yang lulus tepat waktu            AK

  Dataset
0  ACD_AK
1  ACD_AK
2  ACD_AK
```

[ ]:
```
# [AK] Lower-case seluruh kolom teks
df_ak = lowercase_text_columns(df_ak)
```

[ ]:
```
# [AK] Data quality control: Frasa Kunci invalid + Istilah Aspek di luar daftar
df_ak = drop_invalid_rows(df_ak, label="ACD_AK")
```

```
======================================================================
 DATA QUALITY REPORT - ACD_AK
======================================================================
  Total baris diperiksa           : 4,752
  Total baris invalid             : 265
  - Frasa Kunci Penanda Relasi inv : 252
  - Istilah Aspek di luar daftar   : 14
  - Keduanya invalid              : 1

  ASPEK                                        |  FRASA INV |  TOTAL INV
  ---------------------------------------------------------------------
  capaian pembelajaran lulusan                 |         25 |         25
  indeks prestasi kumulatif                    |         13 |         13
  kelulusan tepat waktu                        |         31 |         31
  masa tunggu lulusan                          |         38 |         38
  pelacakan dan perekaman data lulusan         |         40 |         40
```

```
  prestasi mahasiswa                                |          41 |           41
  karya dengan hak kekayaan intelektual             |          22 |           22
  kesesuaian bidang kerja                           |          41 |           41
======================================================================

Sample (up to 10) invalid rows:
                            Istilah Aspek                   Frasa Kunci Penanda
Relasi                       Sinonim/Padanan
58                    kelulusan tepat waktu                              muncul
sebagai                   kelulusan tepat waktu
59                      masa tunggu lulusan                              muncul
sebagai                             masa tunggu
61                       prestasi mahasiswa                              muncul
sebagai                 akademik / non akademik
63                  kesesuaian bidang kerja                              muncul
sebagai                 kesesuaian bidang kerja
108  pelacakan dan perekaman data lulusan  (tidak ditemukan sinonim/variasi
eksplisit)                                     -
161           capaian pembelajaran lulusan                            dilakukan
melalui               audit mutu internal (ami)
165  pelacakan dan perekaman data lulusan                             diperoleh
melalui                             tracer study
168                      prestasi mahasiswa
meliputi  juara 2 turnamen futsal ypp cup 2017
219                     masa tunggu lulusan                  (tidak ada sinonim
eksplisit)         rata-rata masa tunggu lulusan
235                     masa tunggu lulusan
(n/a)                   rata-rata masa tunggu

[OK] Dihapus 265 baris invalid. Tersisa: 4,487 baris.
```

[ ]:
```
# [AK] Distribusi setelah quality control
dist_ak_filtered = build_distribution(df_ak)
print_summary_table(dist_ak_filtered, title="ACD_AK - SEBELUM PENGHAPUSAN␣
↪DUPLIKAT")
```

```
================================================================================
=======
 ACD_AK - SEBELUM PENGHAPUSAN DUPLIKAT
================================================================================
=======
 ASPEK                                      |  DITEMUKAN | TIDAK DITEMUKAN |
TOTAL
-----------------------------------------+------------+------------------+------
----
 capaian pembelajaran lulusan               |        440 |             147 |
587
 indeks prestasi kumulatif                  |        495 |              83 |
```

```
578
 kelulusan tepat waktu                      |        237 |            311 |
548
 masa tunggu lulusan                        |        297 |            245 |
542
 pelacakan dan perekaman data lulusan       |        285 |            269 |
554
 prestasi mahasiswa                         |        289 |            272 |
561
 karya dengan hak kekayaan intelektual      |        388 |            190 |
578
 kesesuaian bidang kerja                    |        275 |            264 |
539
================================================================================
=======
 TOTAL                                      |      2,706 |          1,781 |
4,487
================================================================================
=======
```

[ ]:
```python
# [AK] Cek duplikat per aspek
report_duplicates(df_ak, label="ACD_AK")
```

```
======================================================================
 LAPORAN DUPLIKAT PER ASPEK - ACD_AK
======================================================================
  capaian pembelajaran lulusan                 →  503 duplikat
  indeks prestasi kumulatif                    →  532 duplikat
  kelulusan tepat waktu                        →  427 duplikat
  masa tunggu lulusan                          →  433 duplikat
  pelacakan dan perekaman data lulusan         →  412 duplikat
  prestasi mahasiswa                           →  372 duplikat
  karya dengan hak kekayaan intelektual        →  538 duplikat
  kesesuaian bidang kerja                      →  371 duplikat
======================================================================
  TOTAL                                        → 3588 duplikat
======================================================================
```

[ ]:
```python
# [AK] Hapus duplikat per aspek
df_ak_clean = remove_duplicates(df_ak, label="ACD_AK")
```

```
======================================================================
 PENGHAPUSAN DUPLIKAT PER ASPEK - ACD_AK
======================================================================
  capaian pembelajaran lulusan                 → dihapus  503 baris  (sebelum:
587, sesudah:   84)
  indeks prestasi kumulatif                    → dihapus  532 baris  (sebelum:
578, sesudah:   46)
  kelulusan tepat waktu                        → dihapus  427 baris  (sebelum:
```

```
548, sesudah:  121)
  masa tunggu lulusan                        → dihapus  433 baris  (sebelum:
542, sesudah:  109)
  pelacakan dan perekaman data lulusan       → dihapus  412 baris  (sebelum:
554, sesudah:  142)
  prestasi mahasiswa                         → dihapus  372 baris  (sebelum:
561, sesudah:  189)
  karya dengan hak kekayaan intelektual      → dihapus  538 baris  (sebelum:
578, sesudah:   40)
  kesesuaian bidang kerja                    → dihapus  371 baris  (sebelum:
539, sesudah:  168)
======================================================================
  TOTAL DIHAPUS                              → 3588 baris
  TOTAL TERSISA                              →  899 baris
======================================================================
```

[ ]:
```
# [AK] Distribusi setelah penghapusan duplikat
dist_ak_clean = build_distribution(df_ak_clean)
print_summary_table(dist_ak_clean, title="ACD_AK - SESUDAH PENGHAPUSAN␣
↪DUPLIKAT")
```

```
================================================================================
=======
 ACD_AK - SESUDAH PENGHAPUSAN DUPLIKAT
================================================================================
=======
 ASPEK                                       |  DITEMUKAN | TIDAK DITEMUKAN |
TOTAL
-----------------------------------------+------------+-----------------+------
----
 capaian pembelajaran lulusan                |         83 |               1 |
84
 indeks prestasi kumulatif                   |         45 |               1 |
46
 kelulusan tepat waktu                       |        120 |               1 |
121
 masa tunggu lulusan                         |        108 |               1 |
109
 pelacakan dan perekaman data lulusan        |        141 |               1 |
142
 prestasi mahasiswa                          |        188 |               1 |
189
 karya dengan hak kekayaan intelektual       |         39 |               1 |
40
 kesesuaian bidang kerja                     |        167 |               1 |
168
================================================================================
=======
```

```
 TOTAL                                      |        891 |               8 |
899
================================================================================
=======
```

```
[ ]: # (cel ini sengaja dikosongkan setelah restrukturisasi blok AK)
```

5.3 ACD_AL

```
[ ]: # [AL] Load data
df_al = pd.read_excel(FILE_PATH_AL)
print(f"Total baris dimuat : {len(df_al):,}")
print(f"Kolom              : {list(df_al.columns)}")
df_al.head(3)
```

```
Total baris dimuat : 2,319
Kolom              : ['Istilah Aspek', 'Frasa Kunci Penanda Relasi',
'Sinonim/Padanan', 'Tipe Komentar', 'Nomor Usulan', 'Dataset']
```

```
[ ]:                            Istilah Aspek Frasa Kunci Penanda Relasi  \
 0          capaian pembelajaran lulusan            disebut sebagai
 1             indeks prestasi kumulatif          disingkat sebagai
 2  pelacakan dan perekaman data lulusan                merujuk pada

             Sinonim/Padanan Tipe Komentar              Nomor Usulan Dataset
 0                      cpl            AL     ACD_AL
 1                      ipk            AL     ACD_AL
 2  rekap hasil tracer study           AL     ACD_AL
```

```
[ ]: # [AL] Lower-case seluruh kolom teks
df_al = lowercase_text_columns(df_al)
```

```
[ ]: # [AL] Data quality control: Frasa Kunci invalid + Istilah Aspek di luar daftar
df_al = drop_invalid_rows(df_al, label="ACD_AL")
```

```
======================================================================
 DATA QUALITY REPORT - ACD_AL
======================================================================
  Total baris diperiksa            : 2,319
  Total baris invalid              : 157
  - Frasa Kunci Penanda Relasi inv : 152
  - Istilah Aspek di luar daftar   : 5
  - Keduanya invalid               : 0

  ASPEK                                    |  FRASA INV |  TOTAL INV
  ----------------------------------------------------------------------
  capaian pembelajaran lulusan             |         13 |         13
  indeks prestasi kumulatif                |          8 |          8
  kelulusan tepat waktu                    |         19 |         19
```

```
  masa tunggu lulusan                          |         16 |          16
  pelacakan dan perekaman data lulusan         |         24 |          24
  prestasi mahasiswa                           |         27 |          27
  karya dengan hak kekayaan intelektual        |         21 |          21
  kesesuaian bidang kerja                      |         24 |          24
======================================================================

Sample (up to 10) invalid rows:
                             Istilah Aspek          Frasa Kunci Penanda Relasi
Sinonim/Padanan
11                    kelulusan tepat waktu                                  -
rata-rata kelulusan tepat waktu
90             capaian pembelajaran lulusan                     dijelaskan atas
capaian prodi
114                   kelulusan tepat waktu                                nan
rata-rata kelulusan selama 4 tahun terakhir
115                     masa tunggu lulusan                                nan
rata-rata masa tunggu
118                      prestasi mahasiswa                                nan
mahasiswa berprestasi
121                 kesesuaian bidang kerja                                nan
kesesuaian bidang kerja
179                      prestasi mahasiswa  tidak ada sinonim/padanan eksplisit
-
181                 kesesuaian bidang kerja  tidak ada sinonim/padanan eksplisit
-
186   pelacakan dan perekaman data lulusan  *(tidak ditemukan sinonim/variasi)*
-
188  karya dengan hak kekayaan intelektual   *(tidak ditemukan dalam komentar)*
-

[OK] Dihapus 157 baris invalid. Tersisa: 2,162 baris.
```

[ ]:
```
# [AL] Distribusi setelah quality control
dist_al_filtered = build_distribution(df_al)
print_summary_table(dist_al_filtered, title="ACD_AL - SEBELUM PENGHAPUSAN␣
↪DUPLIKAT")
```

```
================================================================================
=======
 ACD_AL - SEBELUM PENGHAPUSAN DUPLIKAT
================================================================================
=======
 ASPEK                                 |  DITEMUKAN | TIDAK DITEMUKAN |
TOTAL
------------------------------------------+------------+------------------+------
----
 capaian pembelajaran lulusan          |        212 |              75 |
```

```
287
 indeks prestasi kumulatif                |       250 |             32 |
282
 kelulusan tepat waktu                    |       129 |            136 |
265
 masa tunggu lulusan                      |       154 |            112 |
266
 pelacakan dan perekaman data lulusan     |       160 |            104 |
264
 prestasi mahasiswa                       |       159 |            112 |
271
 karya dengan hak kekayaan intelektual    |       182 |             84 |
266
 kesesuaian bidang kerja                  |       145 |            116 |
261
================================================================================
=======
 TOTAL                                    |     1,391 |            771 |
2,162
================================================================================
=======
```

[ ]:
```
# [AL] Cek duplikat per aspek
report_duplicates(df_al, label="ACD_AL")
```

```
======================================================================
 LAPORAN DUPLIKAT PER ASPEK - ACD_AL
======================================================================
  capaian pembelajaran lulusan               →  241 duplikat
  indeks prestasi kumulatif                  →  256 duplikat
  kelulusan tepat waktu                      →  193 duplikat
  masa tunggu lulusan                        →  197 duplikat
  pelacakan dan perekaman data lulusan       →  173 duplikat
  prestasi mahasiswa                         →  166 duplikat
  karya dengan hak kekayaan intelektual      →  239 duplikat
  kesesuaian bidang kerja                    →  152 duplikat
======================================================================
  TOTAL                                      → 1617 duplikat
======================================================================
```

[ ]:
```
# [AL] Hapus duplikat per aspek
df_al_clean = remove_duplicates(df_al, label="ACD_AL")
```

```
======================================================================
 PENGHAPUSAN DUPLIKAT PER ASPEK - ACD_AL
======================================================================
  capaian pembelajaran lulusan               → dihapus  241 baris  (sebelum:
287, sesudah:   46)
  indeks prestasi kumulatif                  → dihapus  256 baris  (sebelum:
```

```
282, sesudah:   26)
  kelulusan tepat waktu                    → dihapus  193 baris  (sebelum:
265, sesudah:   72)
  masa tunggu lulusan                      → dihapus  197 baris  (sebelum:
266, sesudah:   69)
  pelacakan dan perekaman data lulusan     → dihapus  173 baris  (sebelum:
264, sesudah:   91)
  prestasi mahasiswa                       → dihapus  166 baris  (sebelum:
271, sesudah:  105)
  karya dengan hak kekayaan intelektual    → dihapus  239 baris  (sebelum:
266, sesudah:   27)
  kesesuaian bidang kerja                  → dihapus  152 baris  (sebelum:
261, sesudah:  109)
======================================================================
  TOTAL DIHAPUS                            → 1617 baris
  TOTAL TERSISA                            →  545 baris
======================================================================
```

[ ]:

```
# [AL] Distribusi setelah penghapusan duplikat
dist_al_clean = build_distribution(df_al_clean)
print_summary_table(dist_al_clean, title="ACD_AL - SESUDAH PENGHAPUSAN␣
↪DUPLIKAT")
```

```
================================================================================
=======
 ACD_AL - SESUDAH PENGHAPUSAN DUPLIKAT
================================================================================
=======
 ASPEK                                    |  DITEMUKAN | TIDAK DITEMUKAN |
TOTAL
-----------------------------------------+------------+------------------+------
----
 capaian pembelajaran lulusan             |         45 |                1 |
46
 indeks prestasi kumulatif                |         25 |                1 |
26
 kelulusan tepat waktu                    |         71 |                1 |
72
 masa tunggu lulusan                      |         68 |                1 |
69
 pelacakan dan perekaman data lulusan     |         90 |                1 |
91
 prestasi mahasiswa                       |        104 |                1 |
105
 karya dengan hak kekayaan intelektual    |         26 |                1 |
27
 kesesuaian bidang kerja                  |        108 |                1 |
109
```

```
================================================================================
=======
 TOTAL                                       |        537 |                8 |
545
================================================================================
=======
```

[ ]:
```
# (cel ini sengaja dikosongkan setelah restrukturisasi blok AL)
```

5.4 Kombinasi ACD_AK + ACD_AL → ACD

[ ]:
```
# Gabungkan thesaurus AK (bersih) + AL (bersih)
df_combined = pd.concat([df_ak_clean, df_al_clean], ignore_index=True)
print(f"Total baris gabungan: {len(df_combined):,} "
      f"(AK: {len(df_ak_clean):,} + AL: {len(df_al_clean):,})")

dist_combined_raw = build_distribution(df_combined)
print_summary_table(dist_combined_raw, title="ACD GABUNGAN - SETELAH KOMBINASI")
```

```
Total baris gabungan: 1,444 (AK: 899 + AL: 545)
================================================================================
=======
 ACD GABUNGAN - SETELAH KOMBINASI
================================================================================
=======
 ASPEK                                       |  DITEMUKAN | TIDAK DITEMUKAN |
TOTAL
-----------------------------------------+------------+------------------+------
----
 capaian pembelajaran lulusan                |        128 |                2 |
130
 indeks prestasi kumulatif                   |         70 |                2 |
72
 kelulusan tepat waktu                       |        191 |                2 |
193
 masa tunggu lulusan                         |        176 |                2 |
178
 pelacakan dan perekaman data lulusan        |        231 |                2 |
233
 prestasi mahasiswa                          |        292 |                2 |
294
 karya dengan hak kekayaan intelektual       |         65 |                2 |
67
 kesesuaian bidang kerja                     |        275 |                2 |
277
================================================================================
=======
 TOTAL                                       |      1,428 |               16 |
```

```
1,444
================================================================================
=======
```

[ ]:
```
# Simpan hasil gabungan ke folder ACD/Distribution
save_per_aspek(df_combined, OUTPUT_DIR_COMBINED)
```

```
Menyimpan file ke /content/drive/MyDrive/Thesaurus/ACD/Distribution …
  [OK] capaian pembelajaran lulusan                  →  130 baris → /content/
 ↪drive/
MyDrive/Thesaurus/ACD/Distribution/capaian_pembelajaran_lulusan.xlsx
  [OK] indeks prestasi kumulatif                     →   72 baris → /content/
 ↪drive/
MyDrive/Thesaurus/ACD/Distribution/indeks_prestasi_kumulatif.xlsx
  [OK] kelulusan tepat waktu                         →  193 baris →
/content/drive/MyDrive/Thesaurus/ACD/Distribution/kelulusan_tepat_waktu.xlsx
  [OK] masa tunggu lulusan                           →  178 baris →
/content/drive/MyDrive/Thesaurus/ACD/Distribution/masa_tunggu_lulusan.xlsx
  [OK] pelacakan dan perekaman data lulusan          →  233 baris → /content/
 ↪drive/
MyDrive/Thesaurus/ACD/Distribution/pelacakan_dan_perekaman_data_lulusan.xlsx
  [OK] prestasi mahasiswa                            →  294 baris →
/content/drive/MyDrive/Thesaurus/ACD/Distribution/prestasi_mahasiswa.xlsx
  [OK] karya dengan hak kekayaan intelektual         →   67 baris → /content/
 ↪drive/
MyDrive/Thesaurus/ACD/Distribution/karya_dengan_hak_kekayaan_intelektual.xlsx
  [OK] kesesuaian bidang kerja                       →  277 baris → /content/
 ↪drive/
MyDrive/Thesaurus/ACD/Distribution/kesesuaian_bidang_kerja.xlsx
```

[ ]:
```
# Cek duplikat lintas-sumber pada data gabungan (AK & AL bisa beririsan)
report_duplicates(df_combined, label="ACD GABUNGAN")
```

```
======================================================================
 LAPORAN DUPLIKAT PER ASPEK - ACD GABUNGAN
======================================================================
  capaian pembelajaran lulusan                 →   17 duplikat
  indeks prestasi kumulatif                    →   11 duplikat
  kelulusan tepat waktu                        →   41 duplikat
  masa tunggu lulusan                          →   42 duplikat
  pelacakan dan perekaman data lulusan         →   37 duplikat
  prestasi mahasiswa                           →   43 duplikat
  karya dengan hak kekayaan intelektual        →    7 duplikat
  kesesuaian bidang kerja                      →   52 duplikat
======================================================================
  TOTAL                                        →  250 duplikat
======================================================================
```

[ ]:
```python
# Hapus duplikat pada data gabungan & simpan hasil bersih
df_combined_clean = remove_duplicates(df_combined, label="ACD GABUNGAN")
save_per_aspek(df_combined_clean, OUTPUT_DIR_COMBINED_CLEAN, suffix="_clean")

dist_combined_clean = build_distribution(df_combined_clean)
print_summary_table(dist_combined_clean, title="ACD GABUNGAN - SESUDAH␣
↪PENGHAPUSAN DUPLIKAT")
```

```
======================================================================
 PENGHAPUSAN DUPLIKAT PER ASPEK - ACD GABUNGAN
======================================================================
  capaian pembelajaran lulusan              → dihapus   17 baris  (sebelum:
130, sesudah:  113)
  indeks prestasi kumulatif                 → dihapus   11 baris  (sebelum:
72, sesudah:   61)
  kelulusan tepat waktu                     → dihapus   41 baris  (sebelum:
193, sesudah:  152)
  masa tunggu lulusan                       → dihapus   42 baris  (sebelum:
178, sesudah:  136)
  pelacakan dan perekaman data lulusan      → dihapus   37 baris  (sebelum:
233, sesudah:  196)
  prestasi mahasiswa                        → dihapus   43 baris  (sebelum:
294, sesudah:  251)
  karya dengan hak kekayaan intelektual     → dihapus    7 baris  (sebelum:
67, sesudah:   60)
  kesesuaian bidang kerja                   → dihapus   52 baris  (sebelum:
277, sesudah:  225)
======================================================================
  TOTAL DIHAPUS                             →  250 baris
  TOTAL TERSISA                             → 1194 baris
======================================================================
Menyimpan file ke /content/drive/MyDrive/Thesaurus/ACD/Distribution_clean
…
  [OK] capaian pembelajaran lulusan              →  113 baris → /content/
↪drive/
MyDrive/Thesaurus/ACD/Distribution_clean/capaian_pembelajaran_lulusan_clean.x
lsx
  [OK] indeks prestasi kumulatif                 →   61 baris → /content/
↪drive/
MyDrive/Thesaurus/ACD/Distribution_clean/indeks_prestasi_kumulatif_clean.xlsx
  [OK] kelulusan tepat waktu                     →  152 baris → /content/
↪drive/
MyDrive/Thesaurus/ACD/Distribution_clean/kelulusan_tepat_waktu_clean.xlsx
  [OK] masa tunggu lulusan                       →  136 baris → /content/
↪drive/
MyDrive/Thesaurus/ACD/Distribution_clean/masa_tunggu_lulusan_clean.xlsx
```

```
  [OK] pelacakan dan perekaman data lulusan          →  196 baris → /content/
 ↪drive/
MyDrive/Thesaurus/ACD/Distribution_clean/pelacakan_dan_perekaman_data_lulusan
_clean.xlsx
  [OK] prestasi mahasiswa                            →  251 baris → /content/
 ↪drive/
MyDrive/Thesaurus/ACD/Distribution_clean/prestasi_mahasiswa_clean.xlsx
  [OK] karya dengan hak kekayaan intelektual         →   60 baris → /content/
 ↪drive/
MyDrive/Thesaurus/ACD/Distribution_clean/karya_dengan_hak_kekayaan_intelektua
l_clean.xlsx
  [OK] kesesuaian bidang kerja                       →  225 baris → /content/
 ↪drive/
MyDrive/Thesaurus/ACD/Distribution_clean/kesesuaian_bidang_kerja_clean.xlsx
================================================================================
=======
 ACD GABUNGAN - SESUDAH PENGHAPUSAN DUPLIKAT
================================================================================
=======
 ASPEK                                       |  DITEMUKAN | TIDAK DITEMUKAN |
TOTAL
------------------------------------------+------------+------------------+------
----
 capaian pembelajaran lulusan                |        112 |               1 |
113
 indeks prestasi kumulatif                   |         60 |               1 |
61
 kelulusan tepat waktu                       |        151 |               1 |
152
 masa tunggu lulusan                         |        135 |               1 |
136
 pelacakan dan perekaman data lulusan        |        195 |               1 |
196
 prestasi mahasiswa                          |        250 |               1 |
251
 karya dengan hak kekayaan intelektual       |         59 |               1 |
60
 kesesuaian bidang kerja                     |        224 |               1 |
225
================================================================================
=======
 TOTAL                                       |      1,186 |               8 |
1,194
================================================================================
=======
```

5.5 Setelah Manual Validasi

```
[ ]: # [OPSIONAL] Muat ulang thesaurus ACD final dari folder & cek distribusi
def load_combined_from_folder(folder: str, aspek_list=ASPEK_LIST,␣
 ↪suffix="_clean"):
    parts = []
    for aspek in aspek_list:
        safe_name = aspek.replace(" ", "_").replace("/", "-")
        path = os.path.join(folder, f"{safe_name}{suffix}.xlsx")
        parts.append(pd.read_excel(path))
    return pd.concat(parts, ignore_index=True)

df_verify = load_combined_from_folder(OUTPUT_DIR_COMBINED_CLEAN)
print(f"Total baris dimuat dari folder : {len(df_verify):,}")

dist_verify = build_distribution(df_verify)
print_summary_table(dist_verify, title="VERIFIKASI - ACD/Distribution_clean")
```

```
Total baris dimuat dari folder : 736
================================================================================
=======
 VERIFIKASI - ACD/Distribution_clean
================================================================================
=======
 ASPEK                                    |  DITEMUKAN | TIDAK DITEMUKAN |
TOTAL
-----------------------------------------+------------+-----------------+------
----
 capaian pembelajaran lulusan             |         82 |               0 |
82
 indeks prestasi kumulatif                |         44 |               0 |
44
 kelulusan tepat waktu                    |        108 |               0 |
108
 masa tunggu lulusan                      |         73 |               0 |
73
 pelacakan dan perekaman data lulusan     |        120 |               0 |
120
 prestasi mahasiswa                       |        156 |               0 |
156
 karya dengan hak kekayaan intelektual    |         45 |               0 |
45
 kesesuaian bidang kerja                  |        108 |               0 |
108
================================================================================
=======
 TOTAL                                    |        736 |               0 |
736
```

==
=======

6 Analisis & Kombinasi Sentimen - ASC (AK + AL)

Tahap ini memproses hasil ekstraksi sentimen ASC_AK dan ASC_AL secara terpisah (lowercasing, quality control terpadu, penghapusan duplikat), menggabungkan keduanya menjadi satu dataset ASC, memeriksa duplikat lintas-sumber, lalu menyimpan hasil akhir ke Excel.

Quality control terpadu (`drop_invalid_rows`): menghapus baris dengan (1) Kelas Sentimen tidak valid, (2) Frasa Kunci Penanda Sentimen tidak valid, dan/atau (3) Istilah Aspek di luar 8 aspek yang ditentukan, sehingga angka `Total baris dimuat` dan total tabel distribusi selalu konsisten.

[ ]:
```
import pandas as pd
import os
from google.colab import drive
drive.mount('/content/drive')

# Konfigurasi
BASE_DIR = "/content/drive/MyDrive/Thesaurus"

FILE_PATH_AK = f"{BASE_DIR}/ASC_AK/gemma4-latest/HASIL_ASC_AK_sentiment.xlsx"
FILE_PATH_AL = f"{BASE_DIR}/ASC_AL/gemma4-latest/HASIL_ASC_AL_sentiment.xlsx"

OUTPUT_DIR_COMBINED       = f"{BASE_DIR}/ASC/Distribution"
OUTPUT_DIR_COMBINED_CLEAN = f"{BASE_DIR}/ASC/Distribution_clean"

ASPEK_LIST = [
    "capaian pembelajaran lulusan",
    "indeks prestasi kumulatif",
    "kelulusan tepat waktu",
    "masa tunggu lulusan",
    "pelacakan dan perekaman data lulusan",
    "prestasi mahasiswa",
    "karya dengan hak kekayaan intelektual",
    "kesesuaian bidang kerja",
]

VALID_SENTIMEN = {"positif", "negatif", "netral", "tidak relevan"}
VALID_FRASA    = {
    "dinilai positif melalui frasa",
    "dinilai negatif melalui frasa",
    "diidentifikasi sebagai penanda netral melalui frasa",
    "tidak ditemukan",
}

# Kolom teks yang di-lowercase di awal (sebelum analisis apa pun) agar
# perbandingan & deduplikasi antar-sumber (AK vs AL) konsisten.
TEXT_COLUMNS_TO_LOWER = [
    "Istilah Aspek",
```

```
    "Kelas Sentimen",
    "Frasa Kunci Penanda Sentimen",
    "Ekspresi Indikator Sentimen",
    "Ekspresi Generalisasi",
]

# Duplikat ditentukan berdasarkan Istilah Aspek + Ekspresi Generalisasi
DUP_KEYS = ["Istilah Aspek", "Ekspresi Generalisasi"]

os.makedirs(OUTPUT_DIR_COMBINED, exist_ok=True)
os.makedirs(OUTPUT_DIR_COMBINED_CLEAN, exist_ok=True)
print(f"[OK] File AK            : {FILE_PATH_AK}")
print(f"[OK] File AL            : {FILE_PATH_AL}")
print(f"[OK] Combined raw dir   : {OUTPUT_DIR_COMBINED}")
print(f"[OK] Combined clean dir : {OUTPUT_DIR_COMBINED_CLEAN}")
```

```
Mounted at /content/drive
[OK] File AK            : /content/drive/MyDrive/Thesaurus/ASC_AK/gemma4-
latest/HASIL_ASC_AK_sentiment.xlsx
[OK] File AL            : /content/drive/MyDrive/Thesaurus/ASC_AL/gemma4-
latest/HASIL_ASC_AL_sentiment.xlsx
[OK] Combined raw dir   : /content/drive/MyDrive/Thesaurus/ASC/Distribution
[OK] Combined clean dir :
/content/drive/MyDrive/Thesaurus/ASC/Distribution_clean
```

6.1 Fungsi Bantu

[ ]:
```
def lowercase_text_columns(df: pd.DataFrame, columns=TEXT_COLUMNS_TO_LOWER) ->␣
 ↪pd.DataFrame:
    """Lower-case (+ strip) kolom teks agar perbandingan & deduplikasi␣
 ↪konsisten."""
    df = df.copy()
    for col in columns:
        if col in df.columns:
            df[col] = df[col].astype(str).str.strip().str.lower()
    return df


def count_sentimen(sub: pd.DataFrame):
    """Hitung distribusi Kelas Sentimen per aspek."""
    s = sub["Kelas Sentimen"].astype(str).str.strip().str.lower()
    return {
        "positif":       (s == "positif").sum(),
        "negatif":       (s == "negatif").sum(),
        "netral":        (s == "netral").sum(),
        "tidak relevan": (s == "tidak relevan").sum(),
    }
```

```python
def print_summary_table(summary_df: pd.DataFrame, title: str = "RINGKASAN"):
    """Cetak tabel ringkasan ASCII distribusi sentimen per aspek."""
    col_w = max(len(a) for a in summary_df["Aspek"]) + 4
    sep   = "=" * (col_w + 14 + 10 + 10 + 10 + 18 + 10)
    hdr   = "-" * col_w + "+" + "-" * 10 + "+" + "-" * 10 + "+" + "-" * 10 + "+" + "-" * 16 + "+" + "-" * 10
    print(sep)
    print(f" {title}")
    print(sep)
    print(f" {'ASPEK':<{col_w}} | {'POSITIF':>8} | {'NEGATIF':>8} | {'NETRAL':>8} | {'TIDAK RELEVAN':>13} | {'TOTAL':>7}")
    print(hdr)
    for _, row in summary_df.iterrows():
        print(
            f" {row['Aspek']:<{col_w}} "
            f"| {row['positif']:>8,} "
            f"| {row['negatif']:>8,} "
            f"| {row['netral']:>8,} "
            f"| {row['tidak relevan']:>13,} "
            f"| {row['Total']:>7,}"
        )
    print(sep)
    totals = summary_df[["positif", "negatif", "netral", "tidak relevan", "Total"]].sum()
    print(
        f" {'TOTAL':<{col_w}} "
        f"| {totals['positif']:>8,} "
        f"| {totals['negatif']:>8,} "
        f"| {totals['netral']:>8,} "
        f"| {totals['tidak relevan']:>13,} "
        f"| {totals['Total']:>7,}"
    )
    print(sep)


def build_distribution(df: pd.DataFrame, aspek_list=ASPEK_LIST) -> pd.DataFrame:
    """Bangun dataframe ringkasan distribusi sentimen per aspek."""
    rows = []
    for aspek in aspek_list:
        sub    = df[df["Istilah Aspek"].str.strip().str.lower() == aspek.lower()]
        counts = count_sentimen(sub)
        rows.append({"Aspek": aspek, **counts, "Total": len(sub)})
    return pd.DataFrame(rows)
```

```python
def drop_invalid_rows(df: pd.DataFrame, valid_sentimen=VALID_SENTIMEN, valid_frasa=VALID_FRASA,
                      aspek_list=ASPEK_LIST, label=""):
    """Data quality control - satu pass, tiga cek:
      1. Kelas Sentimen tidak ada di VALID_SENTIMEN
      2. Frasa Kunci Penanda Sentimen tidak ada di VALID_FRASA
      3. Istilah Aspek tidak ada di ASPEK_LIST
    Baris yang gagal salah satu (atau lebih) dihapus sekaligus.
    """
    col_sentimen = df["Kelas Sentimen"].astype(str).str.strip().str.lower()
    col_frasa    = df["Frasa Kunci Penanda Sentimen"].astype(str).str.strip().str.lower()
    col_aspek    = df["Istilah Aspek"].astype(str).str.strip().str.lower()
    aspek_lower  = {a.lower() for a in aspek_list}

    mask_bad_sentimen = ~col_sentimen.isin(valid_sentimen)
    mask_bad_frasa    = ~col_frasa.isin(valid_frasa)
    mask_bad_aspek    = ~col_aspek.isin(aspek_lower)
    mask_bad          = mask_bad_sentimen | mask_bad_frasa | mask_bad_aspek
    df_bad            = df[mask_bad].copy()

    print("=" * 70)
    print(f" DATA QUALITY REPORT - {label}")
    print("=" * 70)
    print(f"  Total baris diperiksa            : {len(df):,}")
    print(f"  Total baris invalid              : {len(df_bad):,}")
    print(f"  - Kelas Sentimen invalid         : {mask_bad_sentimen.sum():,}")
    print(f"  - Frasa Kunci invalid            : {mask_bad_frasa.sum():,}")
    print(f"  - Istilah Aspek di luar daftar   : {mask_bad_aspek.sum():,}")
    print()
    print(f"  {'ASPEK':<45} | {'SENT INV':>9} | {'FRASA INV':>10} | {'TOTAL INV':>10}")
    print("  " + "-" * 83)
    for aspek in aspek_list:
        m = col_aspek == aspek.lower()
        s = (m & mask_bad_sentimen).sum()
        f = (m & mask_bad_frasa).sum()
        t = (m & mask_bad).sum()
        print(f"  {aspek:<45} | {s:>9,} | {f:>10,} | {t:>10,}")
    print("=" * 70)

    if len(df_bad) > 0:
        print(f"\nSample (up to 10) invalid rows:")
        print(df_bad[["Istilah Aspek", "Kelas Sentimen",
                      "Frasa Kunci Penanda Sentimen",
```

```
                        "Ekspresi Indikator Sentimen"]].head(10).
↪to_string(index=True))
    else:
        print("\n[OK] Tidak ada baris invalid.")

    df_ok = df[~mask_bad].copy().reset_index(drop=True)
    print(f"\n[OK] Dihapus {len(df_bad):,} baris invalid. Tersisa: {len(df_ok):
↪,} baris.")
    return df_ok


def report_duplicates(df: pd.DataFrame, aspek_list=ASPEK_LIST,␣
↪dup_keys=DUP_KEYS, label=""):
    """Hitung & cetak jumlah duplikat per aspek (tanpa menghapus)."""
    print("=" * 70)
    print(f" LAPORAN DUPLIKAT PER ASPEK - {label}")
    print("=" * 70)
    total_dup = 0
    for aspek in aspek_list:
        sub   = df[df["Istilah Aspek"].str.strip().str.lower() == aspek.lower()]
        n_dup = sub.duplicated(subset=dup_keys, keep="first").sum()
        total_dup += n_dup
        print(f"  {aspek:<45} → {n_dup:>4} duplikat")
    print("=" * 70)
    print(f"  {'TOTAL':<45} → {total_dup:>4} duplikat")
    print("=" * 70)


def remove_duplicates(df: pd.DataFrame, aspek_list=ASPEK_LIST,␣
↪dup_keys=DUP_KEYS, label=""):
    """Hapus duplikat per aspek & cetak laporan sebelum/sesudah."""
    print("=" * 70)
    print(f" PENGHAPUSAN DUPLIKAT PER ASPEK - {label}")
    print("=" * 70)
    parts = []
    for aspek in aspek_list:
        mask      = df["Istilah Aspek"].str.strip().str.lower() == aspek.lower()
        sub       = df[mask].copy()
        before    = len(sub)
        sub_clean = sub.drop_duplicates(subset=dup_keys, keep="first")
        removed   = before - len(sub_clean)
        parts.append(sub_clean)
        print(f"  {aspek:<45} → dihapus {removed:>4} baris  "
              f"(sebelum: {before:>4}, sesudah: {len(sub_clean):>4})")
    df_clean = pd.concat(parts, ignore_index=True)
    total_removed = len(df) - len(df_clean)
    print("=" * 70)
```

```
    print(f"  {'TOTAL DIHAPUS':<45} → {total_removed:>4} baris")
    print(f"  {'TOTAL TERSISA':<45} → {len(df_clean):>4} baris")
    print("=" * 70)
    return df_clean


def save_per_aspek(df: pd.DataFrame, output_dir: str, aspek_list=ASPEK_LIST,␣
 ↪suffix=""):
    """Simpan dataframe per aspek menjadi file Excel terpisah."""
    print(f"Menyimpan file ke {output_dir} ...")
    for aspek in aspek_list:
        sub = df[df["Istilah Aspek"].str.strip().str.lower() == aspek.lower()].
 ↪copy()
        safe_name = aspek.replace(" ", "_").replace("/", "-")
        out_path  = os.path.join(output_dir, f"{safe_name}{suffix}.xlsx")
        sub.to_excel(out_path, index=False)
        print(f"  [OK] {aspek:<45} → {len(sub):>4} baris → {out_path}")
```

6.2 ASC_AK

[ ]:
```
# [AK] Load data
df_ak = pd.read_excel(FILE_PATH_AK)
print(f"Total baris dimuat : {len(df_ak):,}")
print(f"Kolom              : {list(df_ak.columns)}")
df_ak.head(3)
```

```
Total baris dimuat : 4,658
Kolom              : ['Istilah Aspek', 'Kelas Sentimen', 'Frasa Kunci Penanda
Sentimen', 'Ekspresi Indikator Sentimen', 'Ekspresi Generalisasi', 'Tipe
Komentar', 'Nomor Usulan', 'Dataset']
```

[ ]:
```
                   Istilah Aspek Kelas Sentimen   Frasa Kunci Penanda Sentimen  \
0  capaian pembelajaran lulusan        positif  dinilai positif melalui frasa
1     indeks prestasi kumulatif        positif  dinilai positif melalui frasa
2         kelulusan tepat waktu        positif  dinilai positif melalui frasa

               Ekspresi Indikator Sentimen    Ekspresi Generalisasi  \
0            tersedia (di bagian suplemen)     ketersediaan dokumen
1                                ipk>3.35    nilai akademik tinggi
2  prosentase yang lulus tepat waktu 100%  kelulusan sangat tinggi

  Tipe Komentar             Nomor Usulan Dataset
0            AK     ASC_AK
1            AK     ASC_AK
2            AK     ASC_AK
```

[ ]:
```
# [AK] Lower-case seluruh kolom teks
df_ak = lowercase_text_columns(df_ak)
```

[ ]:
```
# [AK] Data quality control: Sentimen invalid, Frasa invalid, Aspek di luar␣
↪daftar
df_ak = drop_invalid_rows(df_ak, label="ASC_AK")
```

```
======================================================================
 DATA QUALITY REPORT - ASC_AK
======================================================================
  Total baris diperiksa          : 4,658
  Total baris invalid            : 189
  - Kelas Sentimen invalid       : 167
  - Frasa Kunci invalid          : 161
  - Istilah Aspek di luar daftar  : 46

  ASPEK                                     |  SENT INV |  FRASA INV |
TOTAL INV
  ---------------------------------------------------------------------------
-----
  capaian pembelajaran lulusan              |        10 |         10 |
10
  indeks prestasi kumulatif                 |        18 |         19 |
19
  kelulusan tepat waktu                     |        17 |         18 |
18
  masa tunggu lulusan                       |        19 |         18 |
20
  pelacakan dan perekaman data lulusan      |        18 |         19 |
20
  prestasi mahasiswa                        |        18 |         17 |
19
  karya dengan hak kekayaan intelektual     |        14 |         14 |
17
  kesesuaian bidang kerja                   |        19 |         19 |
20
======================================================================

Sample (up to 10) invalid rows:
                             Istilah Aspek
Kelas Sentimen                                               Frasa Kunci Penanda
Sentimen                 Ekspresi Indikator Sentimen
201              indeks prestasi kumulatif
positif                                           ipk terendah 2.77 dan tertinggi
3.90             rentang nilai akademik yang baik
202                  kelulusan tepat waktu
netral  data kelulusan tepat waktu, yang untuk tahun 2017/2018 sebanyak 10
orang.                 jumlah kelulusan tepat waktu
```

```
204   pelacakan dan perekaman data lulusan
positif                                      data hasil survei ke pengguna
lulusan                hasil survei dari dunia kerja
205                     prestasi mahasiswa
positif                            juara 3 kompetisi tingkat nasional smart
inotek    prestasi akademik dan non-akademik tinggi
206  karya dengan hak kekayaan intelektual
positif                        mendapat hki dari hasil penelitian berjumlah 7
buah  jumlah hak kekayaan intelektual yang banyak
207                  kesesuaian bidang kerja
positif                                       perusahaan yang menjadi tempat
kerja         penyerapan lulusan di dunia industri
352           keterlaksanaan pemenuhan cpl
netral                        diidentifikasi sebagai penanda netral melalui
frasa                 keterlaksanaan pemenuhan cpl
361                            tidak relevan
tidak ditemukan                                                        tidak
ditemukan                              tidak ditemukan
362                                positif                         dinilai
positif melalui frasa                                        telah terlaksana
dengan sangat baik             pelaksanaan akademik sangat baik
363                                 netral  diidentifikasi sebagai penanda
netral melalui frasa                                   rata-rata kelulusan
mahasiswa tepat waktu                tingkat kelulusan tepat waktu

[OK] Dihapus 189 baris invalid. Tersisa: 4,469 baris.
```

[ ]:
```python
# [AK] Distribusi setelah quality control
dist_ak_filtered = build_distribution(df_ak)
print_summary_table(dist_ak_filtered, title="ASC_AK - SEBELUM PENGHAPUSAN␣
 ↪DUPLIKAT")
```

```
================================================================================
=================================
 ASC_AK - SEBELUM PENGHAPUSAN DUPLIKAT
================================================================================
=================================
 ASPEK                                     |  POSITIF |  NEGATIF |   NETRAL |
TIDAK RELEVAN |   TOTAL
-----------------------------------------+----------+----------+----------+-----
-----------+----------
 capaian pembelajaran lulusan              |      239 |      126 |       90 |
113 |     568
 indeks prestasi kumulatif                 |      238 |       59 |      185 |
76 |     558
 kelulusan tepat waktu                     |      138 |      128 |      206 |
86 |     558
 masa tunggu lulusan                       |      162 |       84 |      193 |
```

```
117 |      556
 pelacakan dan perekaman data lulusan      |       166 |       135 |       145 |
111 |      557
 prestasi mahasiswa                        |       279 |        98 |        71 |
109 |      557
 karya dengan hak kekayaan intelektual     |       159 |       121 |        80 |
200 |      560
 kesesuaian bidang kerja                   |       225 |        81 |       126 |
123 |      555
================================================================================
=================================
 TOTAL                                     |     1,606 |       832 |     1,096 |
935 |    4,469
================================================================================
=================================
```

[ ]:
```
# [AK] Cek duplikat per aspek
report_duplicates(df_ak, label="ASC_AK")
```

```
======================================================================
 LAPORAN DUPLIKAT PER ASPEK - ASC_AK
======================================================================
  capaian pembelajaran lulusan                 → 177 duplikat
  indeks prestasi kumulatif                    → 317 duplikat
  kelulusan tepat waktu                        → 232 duplikat
  masa tunggu lulusan                          → 275 duplikat
  pelacakan dan perekaman data lulusan         → 184 duplikat
  prestasi mahasiswa                           → 188 duplikat
  karya dengan hak kekayaan intelektual        → 260 duplikat
  kesesuaian bidang kerja                      → 238 duplikat
======================================================================
  TOTAL                                        → 1871 duplikat
======================================================================
```

[ ]:
```
# [AK] Hapus duplikat per aspek
df_ak_clean = remove_duplicates(df_ak, label="ASC_AK")
```

```
======================================================================
 PENGHAPUSAN DUPLIKAT PER ASPEK - ASC_AK
======================================================================
  capaian pembelajaran lulusan                 → dihapus  177 baris  (sebelum:
568, sesudah:  391)
  indeks prestasi kumulatif                    → dihapus  317 baris  (sebelum:
558, sesudah:  241)
  kelulusan tepat waktu                        → dihapus  232 baris  (sebelum:
558, sesudah:  326)
  masa tunggu lulusan                          → dihapus  275 baris  (sebelum:
556, sesudah:  281)
  pelacakan dan perekaman data lulusan         → dihapus  184 baris  (sebelum:
```

```
557, sesudah:  373)
  prestasi mahasiswa                         → dihapus  188 baris  (sebelum:
557, sesudah:  369)
  karya dengan hak kekayaan intelektual      → dihapus  260 baris  (sebelum:
560, sesudah:  300)
  kesesuaian bidang kerja                    → dihapus  238 baris  (sebelum:
555, sesudah:  317)
======================================================================
  TOTAL DIHAPUS                              → 1871 baris
  TOTAL TERSISA                              → 2598 baris
======================================================================
```

[ ]:
```
# [AK] Distribusi setelah penghapusan duplikat
dist_ak_clean = build_distribution(df_ak_clean)
print_summary_table(dist_ak_clean, title="ASC_AK - SESUDAH PENGHAPUSAN␣
 ↪DUPLIKAT")
```

```
================================================================================
=================================
 ASC_AK - SESUDAH PENGHAPUSAN DUPLIKAT
================================================================================
=================================
 ASPEK                                      |  POSITIF |  NEGATIF |   NETRAL |
TIDAK RELEVAN |   TOTAL
-----------------------------------------+----------+----------+----------+-----
-----------+----------
 capaian pembelajaran lulusan               |      210 |      114 |       66 |
1 |     391
 indeks prestasi kumulatif                  |      113 |       50 |       77 |
1 |     241
 kelulusan tepat waktu                      |       97 |      104 |      124 |
1 |     326
 masa tunggu lulusan                        |       92 |       74 |      114 |
1 |     281
 pelacakan dan perekaman data lulusan       |      142 |      124 |      106 |
1 |     373
 prestasi mahasiswa                         |      229 |       95 |       44 |
1 |     369
 karya dengan hak kekayaan intelektual      |      118 |      117 |       63 |
2 |     300
 kesesuaian bidang kerja                    |      164 |       73 |       79 |
1 |     317
================================================================================
=================================
 TOTAL                                      |    1,165 |      751 |      673 |
9 |   2,598
================================================================================
=================================
```

6.3 ASC_AL

[ ]:
```
# [AL] Load data
df_al = pd.read_excel(FILE_PATH_AL)
print(f"Total baris dimuat : {len(df_al):,}")
print(f"Kolom              : {list(df_al.columns)}")
df_al.head(3)
```

```
Total baris dimuat : 2,277
Kolom              : ['Istilah Aspek', 'Kelas Sentimen', 'Frasa Kunci Penanda
Sentimen', 'Ekspresi Indikator Sentimen', 'Ekspresi Generalisasi', 'Tipe
Komentar', 'Nomor Usulan', 'Dataset']
```

[ ]:
```
                  Istilah Aspek Kelas Sentimen  \
0  capaian pembelajaran lulusan        positif
1     indeks prestasi kumulatif         netral
2         kelulusan tepat waktu         netral

                        Frasa Kunci Penanda Sentimen  \
0                       dinilai positif melalui frasa
1  diidentifikasi sebagai penanda netral melalui …
2  diidentifikasi sebagai penanda netral melalui …

                         Ekspresi Indikator Sentimen  \
0  "terlaksananya pemenuhan capaian pembelajaran …
1               "instrumen penilaian rata-rata ipk"
2                  "laporan pbm - laporan yudisium"

                         Ekspresi Generalisasi Tipe Komentar  \
0  pemenuhan luaran dan capaian pembelajaran            AL
1       ketersediaan instrumen penilaian IPK            AL
2   dokumentasi status kelulusan tepat waktu            AL

              Nomor Usulan Dataset
0    ASC_AL
1    ASC_AL
2    ASC_AL
```

[ ]:
```
# [AL] Lower-case seluruh kolom teks
df_al = lowercase_text_columns(df_al)
```

[ ]:
```
# [AL] Data quality control: Sentimen invalid, Frasa invalid, Aspek di luar␣
↪daftar
df_al = drop_invalid_rows(df_al, label="ASC_AL")
```

```
======================================================================
 DATA QUALITY REPORT - ASC_AL
======================================================================
  Total baris diperiksa            : 2,277
```

```
  Total baris invalid             : 107
  - Kelas Sentimen invalid        : 96
  - Frasa Kunci invalid           : 96
  - Istilah Aspek di luar daftar  : 26

  ASPEK                                     |  SENT INV |  FRASA INV |
TOTAL INV
  --------------------------------------------------------------------------------
-----
  capaian pembelajaran lulusan              |         6 |          8 |
8
  indeks prestasi kumulatif                 |        11 |         10 |
12
  kelulusan tepat waktu                     |         9 |          8 |
9
  masa tunggu lulusan                       |        11 |         11 |
12
  pelacakan dan perekaman data lulusan      |        10 |         11 |
11
  prestasi mahasiswa                        |         9 |          9 |
10
  karya dengan hak kekayaan intelektual     |         9 |          8 |
9
  kesesuaian bidang kerja                   |         9 |          9 |
10
======================================================================

Sample (up to 10) invalid rows:
                              Istilah Aspek
Kelas Sentimen                                    Frasa Kunci Penanda Sentimen
Ekspresi Indikator Sentimen
224  pemenuhan capaian pembelajaran lulusan
positif                                   dinilai positif melalui frasa
disertai bukti yang sahih dan sangat lengkap
458              indeks prestasi kumulatif                        dinilai
positif melalui frasa                                 "sudah memenuhi ikt pnj
2022"                                     sudah memenuhi ikt pnj 2022
460                    masa tunggu lulusan                        dinilai
positif melalui frasa                            "waktu tunggu masa bekerja cukup
baik"                               waktu tunggu masa bekerja cukup baik
461    pelacakan dan perekaman data lulusan                       dinilai
positif melalui frasa                         "data keberhasilan lulusan program
studi"                            data keberhasilan lulusan program studi
462                     prestasi mahasiswa                        dinilai
positif melalui frasa                         "mendapatkan prestasi di bidang
akademik"                          mendapatkan prestasi di bidang akademik
463   karya dengan hak kekayaan intelektual  diidentifikasi sebagai penanda
netral melalui frasa                                  "hki dosen: hki
```

```
mahasiswa:"                                        hki dosen: hki mahasiswa:
464                 kesesuaian bidang kerja                        dinilai
negatif melalui frasa                                                  "makin
menurun"                                                     makin menurun
521           capaian pembelajaran lulusan                          dinilai
negatif melalui frasa  belum tersedia informasi mengenai instrumen hasil
pemenuhan cpl  belum tersedia informasi mengenai instrumen hasil pemenuhan cpl
522                indeks prestasi kumulatif                        dinilai
positif melalui frasa                                            ipk sudah
melampaui iku                                          ipk sudah melampaui iku
523                  kelulusan tepat waktu  diidentifikasi sebagai penanda
netral melalui frasa                             kelulusan tepat waktu = 52
mahasiswa                             kelulusan tepat waktu = 52 mahasiswa

[OK] Dihapus 107 baris invalid. Tersisa: 2,170 baris.
```

[ ]:
```python
# [AL] Distribusi setelah quality control
dist_al_filtered = build_distribution(df_al)
print_summary_table(dist_al_filtered, title="ASC_AL - SEBELUM PENGHAPUSAN␣
 ↪DUPLIKAT")
```

```
================================================================================
=================================
 ASC_AL - SEBELUM PENGHAPUSAN DUPLIKAT
================================================================================
=================================
 ASPEK                                      |  POSITIF |  NEGATIF |   NETRAL |
TIDAK RELEVAN |   TOTAL
-----------------------------------------+----------+----------+----------+-----
-----------+----------
 capaian pembelajaran lulusan               |      140 |       49 |       33 |
52 |     274
 indeks prestasi kumulatif                  |      129 |       21 |       93 |
26 |     269
 kelulusan tepat waktu                      |       80 |       59 |      104 |
29 |     272
 masa tunggu lulusan                        |       89 |       30 |       99 |
51 |     269
 pelacakan dan perekaman data lulusan       |      103 |       41 |       82 |
45 |     271
 prestasi mahasiswa                         |      187 |       22 |       17 |
45 |     271
 karya dengan hak kekayaan intelektual      |      115 |       28 |       38 |
92 |     273
 kesesuaian bidang kerja                    |      137 |       26 |       67 |
41 |     271
================================================================================
=================================
```

```
 TOTAL                                        |       980 |       276 |       533 |
381 |   2,170
================================================================================
=================================
```

[ ]:
```
# [AL] Cek duplikat per aspek
report_duplicates(df_al, label="ASC_AL")
```

```
======================================================================
 LAPORAN DUPLIKAT PER ASPEK - ASC_AL
======================================================================
  capaian pembelajaran lulusan                 →   74 duplikat
  indeks prestasi kumulatif                    →  134 duplikat
  kelulusan tepat waktu                        →   94 duplikat
  masa tunggu lulusan                          →  114 duplikat
  pelacakan dan perekaman data lulusan         →   68 duplikat
  prestasi mahasiswa                           →   74 duplikat
  karya dengan hak kekayaan intelektual        →  121 duplikat
  kesesuaian bidang kerja                      →   85 duplikat
======================================================================
  TOTAL                                        →  764 duplikat
======================================================================
```

[ ]:
```
# [AL] Hapus duplikat per aspek
df_al_clean = remove_duplicates(df_al, label="ASC_AL")
```

```
======================================================================
 PENGHAPUSAN DUPLIKAT PER ASPEK - ASC_AL
======================================================================
  capaian pembelajaran lulusan                 → dihapus   74 baris  (sebelum:
274, sesudah:  200)
  indeks prestasi kumulatif                    → dihapus  134 baris  (sebelum:
269, sesudah:  135)
  kelulusan tepat waktu                        → dihapus   94 baris  (sebelum:
272, sesudah:  178)
  masa tunggu lulusan                          → dihapus  114 baris  (sebelum:
269, sesudah:  155)
  pelacakan dan perekaman data lulusan         → dihapus   68 baris  (sebelum:
271, sesudah:  203)
  prestasi mahasiswa                           → dihapus   74 baris  (sebelum:
271, sesudah:  197)
  karya dengan hak kekayaan intelektual        → dihapus  121 baris  (sebelum:
273, sesudah:  152)
  kesesuaian bidang kerja                      → dihapus   85 baris  (sebelum:
271, sesudah:  186)
======================================================================
  TOTAL DIHAPUS                                →  764 baris
  TOTAL TERSISA                                → 1406 baris
======================================================================
```

[ ]:
```
# [AL] Distribusi setelah penghapusan duplikat
dist_al_clean = build_distribution(df_al_clean)
print_summary_table(dist_al_clean, title="ASC_AL - SESUDAH PENGHAPUSAN␣
↪DUPLIKAT")
```

```
================================================================================
=================================
 ASC_AL - SESUDAH PENGHAPUSAN DUPLIKAT
================================================================================
=================================
 ASPEK                                      |  POSITIF |  NEGATIF |   NETRAL |
TIDAK RELEVAN |   TOTAL
-----------------------------------------+----------+----------+----------+-----
-----------+----------
 capaian pembelajaran lulusan               |      124 |       48 |       27 |
1 |      200
 indeks prestasi kumulatif                  |       62 |       20 |       52 |
1 |      135
 kelulusan tepat waktu                      |       55 |       51 |       71 |
1 |      178
 masa tunggu lulusan                        |       58 |       28 |       68 |
1 |      155
 pelacakan dan perekaman data lulusan       |       90 |       39 |       73 |
1 |      203
 prestasi mahasiswa                         |      164 |       22 |       10 |
1 |      197
 karya dengan hak kekayaan intelektual      |      101 |       26 |       24 |
1 |      152
 kesesuaian bidang kerja                    |      112 |       24 |       49 |
1 |      186
================================================================================
=================================
 TOTAL                                      |      766 |      258 |      374 |
8 |    1,406
================================================================================
=================================
```

[ ]:
```
# (cel ini sengaja dikosongkan setelah restrukturisasi blok AL)
```

6.4 Kombinasi ASC_AK + ASC_AL → ASC

[ ]:
```python
# Gabungkan data sentimen AK (bersih) + AL (bersih)
df_combined = pd.concat([df_ak_clean, df_al_clean], ignore_index=True)
print(f"Total baris gabungan: {len(df_combined):,} "
      f"(AK: {len(df_ak_clean):,} + AL: {len(df_al_clean):,})")

dist_combined_raw = build_distribution(df_combined)
print_summary_table(dist_combined_raw, title="ASC GABUNGAN - SETELAH KOMBINASI")
```

```
Total baris gabungan: 4,004 (AK: 2,598 + AL: 1,406)
================================================================================
=================================
 ASC GABUNGAN - SETELAH KOMBINASI
================================================================================
=================================
 ASPEK                                    |  POSITIF |  NEGATIF |   NETRAL |
TIDAK RELEVAN |   TOTAL
-----------------------------------------+----------+----------+----------+-----
-----------+----------
 capaian pembelajaran lulusan             |      334 |      162 |       93 |
2 |      591
 indeks prestasi kumulatif                |      175 |       70 |      129 |
2 |      376
 kelulusan tepat waktu                    |      152 |      155 |      195 |
2 |      504
 masa tunggu lulusan                      |      150 |      102 |      182 |
2 |      436
 pelacakan dan perekaman data lulusan     |      232 |      163 |      179 |
2 |      576
 prestasi mahasiswa                       |      393 |      117 |       54 |
2 |      566
 karya dengan hak kekayaan intelektual    |      219 |      143 |       87 |
3 |      452
 kesesuaian bidang kerja                  |      276 |       97 |      128 |
2 |      503
================================================================================
=================================
 TOTAL                                    |    1,931 |    1,009 |    1,047 |
17 |    4,004
================================================================================
=================================
```

[ ]:
```python
# Simpan hasil gabungan ke folder ASC/Distribution
save_per_aspek(df_combined, OUTPUT_DIR_COMBINED)
```

```
Menyimpan file ke /content/drive/MyDrive/Thesaurus/ASC/Distribution …
  [OK] capaian pembelajaran lulusan                  →  591 baris → /content/
↪drive/
MyDrive/Thesaurus/ASC/Distribution/capaian_pembelajaran_lulusan.xlsx
```

```
  [OK] indeks prestasi kumulatif                     → 376 baris → /content/
 ↪drive/
MyDrive/Thesaurus/ASC/Distribution/indeks_prestasi_kumulatif.xlsx
  [OK] kelulusan tepat waktu                         → 504 baris →
/content/drive/MyDrive/Thesaurus/ASC/Distribution/kelulusan_tepat_waktu.xlsx
  [OK] masa tunggu lulusan                           → 436 baris →
/content/drive/MyDrive/Thesaurus/ASC/Distribution/masa_tunggu_lulusan.xlsx
  [OK] pelacakan dan perekaman data lulusan          → 576 baris → /content/
 ↪drive/
MyDrive/Thesaurus/ASC/Distribution/pelacakan_dan_perekaman_data_lulusan.xlsx
  [OK] prestasi mahasiswa                            → 566 baris →
/content/drive/MyDrive/Thesaurus/ASC/Distribution/prestasi_mahasiswa.xlsx
  [OK] karya dengan hak kekayaan intelektual         → 452 baris → /content/
 ↪drive/
MyDrive/Thesaurus/ASC/Distribution/karya_dengan_hak_kekayaan_intelektual.xlsx
  [OK] kesesuaian bidang kerja                       → 503 baris → /content/
 ↪drive/
MyDrive/Thesaurus/ASC/Distribution/kesesuaian_bidang_kerja.xlsx
```

[ ]:
```
# Cek duplikat lintas-sumber pada data gabungan (AK & AL bisa beririsan)
report_duplicates(df_combined, label="ASC GABUNGAN")
```

```
======================================================================
 LAPORAN DUPLIKAT PER ASPEK - ASC GABUNGAN
======================================================================
  capaian pembelajaran lulusan                 →   35 duplikat
  indeks prestasi kumulatif                    →   45 duplikat
  kelulusan tepat waktu                        →   56 duplikat
  masa tunggu lulusan                          →   54 duplikat
  pelacakan dan perekaman data lulusan         →   40 duplikat
  prestasi mahasiswa                           →   37 duplikat
  karya dengan hak kekayaan intelektual        →   27 duplikat
  kesesuaian bidang kerja                      →   45 duplikat
======================================================================
  TOTAL                                        →  339 duplikat
======================================================================
```

[ ]:
```
# Hapus duplikat pada data gabungan & simpan hasil bersih
df_combined_clean = remove_duplicates(df_combined, label="ASC GABUNGAN")
save_per_aspek(df_combined_clean, OUTPUT_DIR_COMBINED_CLEAN, suffix="_clean")

dist_combined_clean = build_distribution(df_combined_clean)
print_summary_table(dist_combined_clean, title="ASC GABUNGAN - SESUDAH␣
 ↪PENGHAPUSAN DUPLIKAT")
```

```
======================================================================
 PENGHAPUSAN DUPLIKAT PER ASPEK - ASC GABUNGAN
======================================================================
```

```
  capaian pembelajaran lulusan                  → dihapus   35 baris  (sebelum:
591, sesudah:  556)
  indeks prestasi kumulatif                     → dihapus   45 baris  (sebelum:
376, sesudah:  331)
  kelulusan tepat waktu                         → dihapus   56 baris  (sebelum:
504, sesudah:  448)
  masa tunggu lulusan                           → dihapus   54 baris  (sebelum:
436, sesudah:  382)
  pelacakan dan perekaman data lulusan          → dihapus   40 baris  (sebelum:
576, sesudah:  536)
  prestasi mahasiswa                            → dihapus   37 baris  (sebelum:
566, sesudah:  529)
  karya dengan hak kekayaan intelektual         → dihapus   27 baris  (sebelum:
452, sesudah:  425)
  kesesuaian bidang kerja                       → dihapus   45 baris  (sebelum:
503, sesudah:  458)
=====================================================================
  TOTAL DIHAPUS                                 →  339 baris
  TOTAL TERSISA                                 → 3665 baris
=====================================================================
Menyimpan file ke /content/drive/MyDrive/Thesaurus/ASC/Distribution_clean
…
  [OK] capaian pembelajaran lulusan                  →  556 baris → /content/
 ↪drive/
MyDrive/Thesaurus/ASC/Distribution_clean/capaian_pembelajaran_lulusan_clean.x
lsx
  [OK] indeks prestasi kumulatif                     →  331 baris → /content/
 ↪drive/
MyDrive/Thesaurus/ASC/Distribution_clean/indeks_prestasi_kumulatif_clean.xlsx
  [OK] kelulusan tepat waktu                         →  448 baris → /content/
 ↪drive/
MyDrive/Thesaurus/ASC/Distribution_clean/kelulusan_tepat_waktu_clean.xlsx
  [OK] masa tunggu lulusan                           →  382 baris → /content/
 ↪drive/
MyDrive/Thesaurus/ASC/Distribution_clean/masa_tunggu_lulusan_clean.xlsx
  [OK] pelacakan dan perekaman data lulusan          →  536 baris → /content/
 ↪drive/
MyDrive/Thesaurus/ASC/Distribution_clean/pelacakan_dan_perekaman_data_lulusan
_clean.xlsx
  [OK] prestasi mahasiswa                            →  529 baris → /content/
 ↪drive/
MyDrive/Thesaurus/ASC/Distribution_clean/prestasi_mahasiswa_clean.xlsx
  [OK] karya dengan hak kekayaan intelektual         →  425 baris → /content/
 ↪drive/
MyDrive/Thesaurus/ASC/Distribution_clean/karya_dengan_hak_kekayaan_intelektua
l_clean.xlsx
```

```
  [OK] kesesuaian bidang kerja                        →  458 baris → /content/
 ↪drive/
MyDrive/Thesaurus/ASC/Distribution_clean/kesesuaian_bidang_kerja_clean.xlsx
================================================================================
=================================
 ASC GABUNGAN - SESUDAH PENGHAPUSAN DUPLIKAT
================================================================================
=================================
 ASPEK                                      |  POSITIF |  NEGATIF |   NETRAL |
TIDAK RELEVAN |   TOTAL
-----------------------------------------+----------+----------+----------+-----
-----------+----------
 capaian pembelajaran lulusan               |      310 |      158 |       87 |
1 |      556
 indeks prestasi kumulatif                  |      152 |       66 |      112 |
1 |      331
 kelulusan tepat waktu                      |      135 |      140 |      172 |
1 |      448
 masa tunggu lulusan                        |      125 |       97 |      159 |
1 |      382
 pelacakan dan perekaman data lulusan       |      210 |      159 |      166 |
1 |      536
 prestasi mahasiswa                         |      366 |      112 |       50 |
1 |      529
 karya dengan hak kekayaan intelektual      |      201 |      140 |       82 |
2 |      425
 kesesuaian bidang kerja                    |      247 |       96 |      114 |
1 |      458
================================================================================
=================================
 TOTAL                                      |    1,746 |      968 |      942 |
9 |    3,665
================================================================================
=================================
```

6.5 Sesi Manual

[ ]:
```python
# [OPSIONAL] Muat ulang data ASC final dari folder & cek distribusi
def load_combined_from_folder(folder: str, aspek_list=ASPEK_LIST,␣
 ↪suffix="_filtered"):
    parts = []
    for aspek in aspek_list:
        safe_name = aspek.replace(" ", "_").replace("/", "-")
        path = os.path.join(folder, f"{safe_name}{suffix}.xlsx")
        parts.append(pd.read_excel(path))
    return pd.concat(parts, ignore_index=True)
```

```python
df_verify = load_combined_from_folder(OUTPUT_DIR_COMBINED_CLEAN)
print(f"Total baris dimuat dari folder : {len(df_verify):,}")

dist_verify = build_distribution(df_verify)
print_summary_table(dist_verify, title="VERIFIKASI - ASC/Distribution_clean")
```

```
Total baris dimuat dari folder : 1,996
================================================================================
=================================
 VERIFIKASI - ASC/Distribution_clean
================================================================================
=================================
 ASPEK                                      |  POSITIF |  NEGATIF |   NETRAL |
TIDAK RELEVAN |   TOTAL
-----------------------------------------+-----------+-----------+-----------+-----
-----------+----------
 capaian pembelajaran lulusan               |      208 |      119 |       52 |
0 |      379
 indeks prestasi kumulatif                  |       56 |       36 |       47 |
0 |      139
 kelulusan tepat waktu                      |       85 |       97 |       87 |
0 |      269
 masa tunggu lulusan                        |       58 |       76 |       54 |
0 |      188
 pelacakan dan perekaman data lulusan       |      108 |      112 |       95 |
0 |      315
 prestasi mahasiswa                         |      155 |       62 |       34 |
0 |      251
 karya dengan hak kekayaan intelektual      |       91 |       71 |       43 |
0 |      205
 kesesuaian bidang kerja                    |      131 |       54 |       65 |
0 |      250
================================================================================
=================================
 TOTAL                                      |      892 |      627 |      477 |
0 |    1,996
================================================================================
=================================
```

7 Tahap Few-Shot Fine-Tuning - Baseline (IndoBERT)

Tahap ini melatih model IndoBERT (`indobenchmark/indobert-base-p2`) untuk ACD dan ASC di bawah kondisi data terbatas (few-shot), tanpa augmentasi apa pun - hanya mengandalkan class-weighted cross-entropy loss untuk menangani ketidakseimbangan kelas. Data dibagi 80/20 secara multilabel-stratified dengan minimal dua contoh uji per kelas per aspek, lalu diambil hingga lima belas komentar berlabel per kelas aspek untuk pelatihan. Kedelapan aspek dilatih melalui head klasifikasi terpisah di atas encoder IndoBERT yang sama. Hasil (prediksi, metrik precision/recall/F1, dan confusion matrix) disimpan untuk dibandingkan dengan konfigurasi tesaurus pada tahap berikutnya.

7.1 Install & Import

[ ]:
```
!pip install -q transformers datasets scikit-learn pandas openpyxl matplotlib␣
↪seaborn tqdm iterative-stratification
```

[ ]:
```
import os, re, random, warnings, gc
import numpy as np
import pandas as pd
from copy import deepcopy
from tqdm.auto import tqdm

import torch
import torch.nn as nn
from torch.optim import AdamW
from torch.utils.data import Dataset, DataLoader

from transformers import (
    BertTokenizer, BertModel,
    set_seed,
)
from sklearn.utils.class_weight import compute_class_weight
from sklearn.metrics import (
    precision_recall_fscore_support, f1_score,
    confusion_matrix, accuracy_score,
)
import matplotlib.pyplot as plt
import seaborn as sns

warnings.filterwarnings('ignore')
sns.set_style('whitegrid')
print('Libraries loaded.')
```

```
Libraries loaded.
```

7.2 Configuration

```
[ ]: MODEL_CHECKPOINT = 'indobenchmark/indobert-base-p2'

# Hyperparameters
LEARNING_RATE = 2e-5
BATCH_SIZE    = 8
DROPOUT       = 0.1
EPOCHS        = 5
WEIGHT_DECAY  = 0.3
MAX_SEQ_LEN   = 512
SEED          = 42

# Few-shot sampling & split
N_PER_CLASS = 15
TEST_FRAC   = 0.20
MIN_TEST_PER_CLASS = 2   # enforce >= this many test examples per class, per␣
 ↪aspect
STRATIFY    = True       # multilabel-stratified split (keeps per-aspect class␣
 ↪mix)

# Labels
ACD_LABEL_MAP   = {'relevan': 1, 'tidak relevan': 0}
ACD_LABEL_NAMES = ['tidak_relevan', 'relevan']

ASC_LABEL_MAP   = {'positif': 0, 'negatif': 1, 'netral': 2}
ASC_LABEL_NAMES = ['positif', 'negatif', 'netral']

# Aspect columns (8 fixed aspects, '(manual)' gold columns)
ASPECT_COLUMNS = [
    'capaian pembelajaran lulusan (manual)',
    'indeks prestasi kumulatif (manual)',
    'kelulusan tepat waktu (manual)',
    'masa tunggu lulusan (manual)',
    'pelacakan dan perekaman data lulusan (manual)',
    'prestasi mahasiswa (manual)',
    'karya dengan hak kekayaan intelektual (manual) ',
    'kesesuaian bidang kerja (manual)',
]
NUM_ASPECTS   = len(ASPECT_COLUMNS)
ASPECT_SHORTS = [c.replace(' (manual)', '').strip() for c in ASPECT_COLUMNS]


# Paths (Colab / Google Drive)
BASE_PATH = '/content/drive/MyDrive'

ACD_EXPERIMENTS = [
```

```
    {'name': 'ACD_AK', 'file': 'Hasil Ground Truth_ACD AK_clean.xlsx', 'task':␣
 ↪'ACD', 'text_col': 'komentar_AK'},
    {'name': 'ACD_AL', 'file': 'Hasil Ground Truth_ACD AL_clean.xlsx', 'task':␣
 ↪'ACD', 'text_col': 'komentar_AL'},
]
ASC_EXPERIMENTS = [
    {'name': 'ASC_AK', 'file': 'Hasil Ground Truth_ASC AK_clean.xlsx', 'task':␣
 ↪'ASC', 'text_col': 'komentar_AK'},
    {'name': 'ASC_AL', 'file': 'Hasil Ground Truth_ASC AL_clean.xlsx', 'task':␣
 ↪'ASC', 'text_col': 'komentar_AL'},
]

print(f'Model        : {MODEL_CHECKPOINT}')
print(f'LR {LEARNING_RATE} | batch {BATCH_SIZE} | dropout {DROPOUT} | epochs␣
 ↪{EPOCHS} | max_len {MAX_SEQ_LEN}')
print(f'Few-shot     : {N_PER_CLASS} per (aspect,class), NO augmentation')
print(f'Split        : {int((1-TEST_FRAC)*100)}/{int(TEST_FRAC*100)}, NO␣
 ↪validation | min {MIN_TEST_PER_CLASS} test/class/aspect')
```

```
Model        : indobenchmark/indobert-base-p2
LR 2e-05 | batch 8 | dropout 0.1 | epochs 5 | max_len 512
Few-shot     : 15 per (aspect,class), NO augmentation
Split        : 80/20, NO validation | min 2 test/class/aspect
```

7.3 Device & Reproducibility

[ ]:
```
def set_all_seeds(seed):
    random.seed(seed); np.random.seed(seed)
    torch.manual_seed(seed); torch.cuda.manual_seed_all(seed)
    set_seed(seed)

set_all_seeds(SEED)
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print('Device:', device)
if torch.cuda.is_available():
    print('  GPU:', torch.cuda.get_device_name(0))
```

```
Device: cuda
  GPU: NVIDIA L4
```

7.4 Mount Google Drive

[ ]:
```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

7.5 Tokenizer

```
[ ]: tokenizer = BertTokenizer.from_pretrained(MODEL_CHECKPOINT)
     print('Tokenizer loaded:', MODEL_CHECKPOINT, '| vocab', tokenizer.vocab_size)
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a
HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated
requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and
faster downloads.

tokenizer_config.json:   0%|          | 0.00/2.00 [00:00<?, ?B/s]

vocab.txt:   0%|          | 0.00/229k [00:00<?, ?B/s]

special_tokens_map.json:   0%|          | 0.00/112 [00:00<?, ?B/s]

Tokenizer loaded: indobenchmark/indobert-base-p2 | vocab 30521
```

7.6 Preprocessing (singkatan + typo normalization)

```
[ ]: import re
     # Singkatan + typo normalization dictionaries (provided by user).
     # NOTE: typo_dict2 arrived truncated at the end ("...,\"amik\": \"amik\","} );
     # the corrupted trailing fragment was dropped and the dict closed cleanly.

     replace_weird_words1 = {
     "bbrp":"beberapa","bhw":"bahwa","bk":"bidang_keahlian","bl":"bulan","blm":
      ↪"belum","bln":"bulan",
     "bnsp":"badan_nasional_sertifikasi_profesi","bpc":
      ↪"bidang_pengembang_kompetensi","bpjs":
      ↪"badan_penyelenggara_jaminan_sosial","brp":"berapa",
     "cmpk":"capaian_pembelajaran_mata_kuliah","cp":"capaian_pembelajaran","cs":
      ↪"computer_science","csd":"center_of_student_development","dcs":
      ↪"diploma_computer_science","dg":"dengan","dgn":"dengan","dll":
      ↪"dan_lain_lain","dlm":"dalam","dr":"dari","dsb":"dan_sebagainya",
     "dtpr":"dosen_tetap_penghitung_rasio","dtprmhs":
      ↪"dosen_tetap_penghitung_rasio_mahasiswa","fcns":
      ↪"foresec_certified_network_security",
     "fgd":"forum_group_discussion","ft":"fakultas_teknik","gbr":"gambar",
     "hk":"hak_kekayaan_intelektual","hkl":"hak_kekayaan_intelektual","hlm":
      ↪"halaman","jl":"jumlah","jml":"jumlah","jwd":"junior_web_developer",
     "kbk":"kurikulum_berbasis_kompetensi","kbm":"kegiatan_belajar_mengajar","kklp":
      ↪"kuliah_kerja_lapangan_pengganti","kkn":"kuliah_kerja_nyata","kp":
      ↪"kerja_praktek","kpt":"konsultan_pendidikan_tinggi","krn":"karena",
     "ktw":"kelulusan_tepat_waktu","lck":"laporan_capaian_kinerja","lkps":
      ↪"laporan_kinerja_program_studi","lm":"layanan_mahasiswa","lpdp":
      ↪"lembaga_pengelola_dana_pendidikan","lpk":"indeks_prestasi_kumulatif",
```

```
"lppm":"lembaga_penelitian_dan_pengabdian_masyarakat","lsp":
 ↪"lembaga_sertifikasi_profesi","lspk":
 ↪"lembaga_sertifikasi_profesi_komputer","lt":"information_technology","lttq":
 ↪"lembaga_tahfizh_dan_tilawah_alquran",
"mbkm":"merdeka_belajar_kampus_merdeka","mhn":"mohon","mhs":"mahasiswa","mhsw":
 ↪"mahasiswa","mk":"mata_kuliah","mkbm":"merdeka_belajar_kampus_merdeka","ms":
 ↪"mahasiswa","ntb":"nusa_tenggara_barat",
"pd":"pada","pk":"prestasi_kumulatif","pkm":
 ↪"program_kreativitas_mahasiswa","pkmk":
 ↪"program_kreativitas_mahasiswi_kewirausahaan","pkmkw":
 ↪"program_kreativitas_mahasiswi_kewirausahaan","pkmm":
 ↪"program_kreativitas_mahasiswi_pengabdian_masyarakat",
"pkmpm":"program_kreativitas_mahasiswi_pengabdian_masyarakat","pl":
 ↪"pembelajaran_lanjutan","pmmb":"program_magang_mahasiswa_bersertifikat",
"pp":"pimpinan_pusat","ppm":"pusat_penjaminan_mutu","ps":"program_studi","psbb":
 ↪"pembatasan_sosial_berskala_besar",
"pspsk":"pusat_studi_pendidikan_dan_kebijakan","ptnpts":
 ↪"perguruan_tinggi_negeri_perguruan_tinggi_swasta","ptpn":
 ↪"perkebunan_tanaman_nusantara","rpl":"rekayasa_perangkat_lunak","rpm":
 ↪"rencana_pembelajaran_mingguan","rpp":"rencana_pelaksanaan_pembelajaran",
"rps":"rencana_pembelajaran_semester","sbb":"sebagai_berikut","sbg":
 ↪"sebagai","sbgmn":"sebagaimana","scr":"secara","sd":"sampai_dengan",
"sdh":"sudah","sdlc":"system_development_life_cycle","sdm":
 ↪"sumber_daya_manusia","shg":"sehingga","sk":"surat_keputusan",
"sks":"satuan_kredit_semester","sn":"standar_nasional","snpt":
 ↪"standar_nasional_pendidikan_tinggi","spp":
 ↪"surat_perjanjian_pemanfaatan","str":"sarjana_terapan","stt":
 ↪"sekolah_tinggi_teknologi","tdk":"tidak","tgl":"tanggal","th":"tahun","thn":
 ↪"tahun",
"thd":"terhadap","tkj":"teknik_komputer_dan_jaringan","tkk":
 ↪"teknik_komputer_kontrol","tkt":"tingkat","trk":
 ↪"teknologi_rekayasa_komputer",
"trkj":"teknologi_rekayasa_komputer","ts":"tracer_study","tsb":
 ↪"tersebut","tsbt":"tersebut","ttg":"tentang",
"vm":"visi_misi","vmts":"visi_misi_tujuan_dan_strategi","wt":
 ↪"waktu_tunggu","prodi":"program_studi","ik":"ilmu_komputer","al":
 ↪"asesmen_lapangan","ak":"asesmen_kecukupan","rtm":"papat_tinjauan_manajemen",
"cdc":"career_development_center","spm":"sistem_penjaminan_mutu",
"phb":"program_hibah_binaan","pt":"perguruan_tinggi",
"dtps":"dosen_tetap_program_studi","lrs":"laporan_rencana_studi","lpks":
 ↪"lembaga_pengembangan_keterampilan_dan_sertifikasi",
"smk":"sekolah_menengah_kejuruan","dst":"dan_seterusnya","smp":"sampai","lpm":
 ↪"lembaga_penjaminan_mutu",
"std":"standar","bpm":"badan_penjaminan_mutu","bhs":"bahasa",
"kkmk":"kriteria_kinerja_manajemen_kualitas","pkl":
 ↪"rpaktek_kerja_lapangan","psms":"program_studi_manajemen_sistem",
```

```
"kk":"kode_klasifikasi","rtl":"rencana_tindak_lanjut","smt":"semester",
"pbl":"problem_based_learning","rpsm":"rencana_pembelajaran_semester",
"lp":"lembaga_penjaminan","lpd":"laporan_penelitian_dosen","pppm":
 ↪"pengembangan_pendidikan_mahasiswa",
"bppm":"badan_penelitian_pengabdian_masyarakat","lpppm":
 ↪"lembaga_penelitian_pengabdian_masyarakat","rpjp":
 ↪"rencana_pembangunan_jangka_panjang",
"bpp":"biaya_penyelenggaraan_pendidikan","st":"sarjana_teknik","sp":
 ↪"surat_peringatan",
"dpp":"dosen_pendamping_program","upps":"unit_pengelola_program_studi","pts":
 ↪"perguruan_tinggi_swasta","ptn":"perguruan_tinggi_negeri","pr":
 ↪"penelitian_riset","ttd":"tanda_tangan","pllm":
 ↪"lembaga_penelitian_dan_pengabdian_masyarakat",
"rp":"rencana_pembelajaran","drpm":"direktorat_riset_pengabdian_masyarakat",
"bph":"badan_pengurus_harian","ttp":"tetap","llppm":
 ↪"lembaga_penelitian_pengabdian_masyarakat",
"cfp":"certified_financial_planner","lppmbppm":
 ↪"lembaga_penelitian_pengabdian_masyarakat_badan_penjaminan_mutu_pendidikan",
"spk":"surat_perjanjian_kerja","pm":"penjaminan_mutu","dptr":
 ↪"dosen_tetap_penghitung_rasio","kpd":"kepada","dtpt":
 ↪"dosen_tetap_perguruan_tinggi","bgm":"bagaimana",
"dtp":"dosen_tetap_program","kts":"kurikulum_tingkat_satuan",
"jg":"juga","dgx":"dengan","rpjmn":
 ↪"rencana_pembangunan_jangka_menengah_nasional","ppkm":
 ↪"program_pengembangan_keterampilan_mahasiswa",
"dprpm":"direktorat_penelitian_pengabdian_masyarakat","bg":"bagaimana","spmpt":
 ↪"sistem_penjaminan_mutu_pendidikan_tinggi","pnbp":
 ↪"penerimaan_negara_bukan_pajak",
"wkt":"waktu","csr":"corporate_social_responsibility",
"kkp":"kualitas_kinerja_program","pkmdtpr":
 ↪"program_kreativitas_mahasiswa_dosen_tetap_penghitung_rasio",
"lppp":"lembaga_penelitian_pengabdian_masyarakat",
"dtr":"dosen_tetap_penghitung_rasio","ln":"luar_negeri","tmps":
 ↪"tim_penjaminan_mutu_program_studi","tpmj":"tim_penjaminan_mutu_jurusan",
"pmk":"penjaminan_mutu_kinerja","bkn":"bukan","pkms":
 ↪"program_kreativitas_mahasiswa_sosial","skripsita":"skripsi_tugas_akhir",
"pstp":"program_studi_teknik_pertambangan","ppkmdm":
 ↪"program_pengembangan_keterampilan_mahasiswa_dan_dosen",
"plp":"program_latihan_profesi","skl":"surat_keterangan_lulus",
"cplcpmk":
 ↪"capaian_pembelajaran_lulusan_dan_capaian_pembelajaran_mata_kuliah","cpmk":
 ↪"capaian_pembelajaran_mata_kuliah","rkps":
 ↪"rencana_kegiatan_pembelajaran_semester",
"sbgmn":"sebagaimana","jmlh":"jumlah","sms":"semester","lplppm":
 ↪"lembaga_penelitian_pengabdian_masyarakat","pstg":
 ↪"program_studi_teknik_geologi","pstk":"program_studi_teknik_kimia",
```

```
"skp":"surat_kerja_praktek","pblpbm":
 ↪"problem_based_learning_project_based_learning","prn":
 ↪"program_riset_nasional","ppg":"program_profesi_guru","psbm":
 ↪"program_studi_bisnis_dan_manajemen",
"lpj":"laporan_pertanggungjawaban","kkt":"kualitas_kinerja_tim",
"pjj":"pendidikan_jarak_jauh","pmp":"pemetaan_mutu_pendidikan","bpmp":
 ↪"balai_penjaminan_mutu_pendidikan","bppmp":
 ↪"balai_penjaminan_mutu_pendidikan","bdp":"balai_guru_penggerak",
"bkd":"beban_kerja_dosen","bkk":"bursa_kerja_khusus","bmw":
 ↪"belajar_melanjut_wirausaha","bnpb":
 ↪"badan_nasional_penanggulangan_bencana","bnpp":
 ↪"badan_nasional_pengelola_perbatasan","bnpt":
 ↪"badan_nasional_penanggulangan_terorisme",
"bp":"bimbingan_penyuluhan","bpk":"badan_pengurus_keuangan","bpl":
 ↪"badang_pengelola_latihan","bssn":"badan_siber_dan_sandi_negara",
"cpns":"calon_pegawai_negeri_sipil","crf":
 ↪"center_for_research_and_facilitation","crm":
 ↪"sistem_manajemen_hubungan_pelanggan","csrpkbl":
 ↪"corporate_social_responsibility_program_kemitraan_bina_lingkungan","pkbl":
 ↪"program_kemitraan_bina_lingkungan",
"ct":"computational_thinking","ctf":"career_and_technology_foundations","cv":
 ↪"curriculum_vitae",
"ddtpr":"dosen_tetap_penghitung_rasio","dkk":"dan_kawan_kawan","dn":"dan","dng":
 ↪"dengan","dpkm":"direktorat_pengabdian_masyarakat","dpl":
 ↪"dosen_pembimbing_lapangan","dpm":"dewan_perwakilan_mahasiswa","dppm":
 ↪"direktorat_penelitian_pengabdian_masyarakat",
"dpr":"laporan_kemajuan_gelar","dpt":"daftar_pemilih_tetap",
"hki":"hak_kekayaan_intelektual","hku":"hak_kekayaan_usaha",
"iku":"indikator_kinerja_utama","ikt":"indikator_kinerja_tambahan","ikuikt":
 ↪"indikator_kinerja_utama_indikator_kinerja_tambahan","cbr":
 ↪"community_based_research","cbt":"computer_based_test","cl":
 ↪"collaborative_learning",
}

replace_weird_words2 = {
"tpg":"tunjangan_profesi_guru","dtprmhs":
 ↪"dosen_tetap_penghitung_rasio_mahasiswa","mhs":"mahasiswa",
"upps":"unit_pengelola_program_studi","spmi":
 ↪"sistem_penjaminan_mutu_internal","renstra":"rencana_strategis",
"menemu":"menemukan","isbn":"international_standard_book_number","prosentase":
 ↪"presentase","yg":"yang","haki":"hak_atas_kekayaan_intelektual",
"led":"laporan_evaluasi_diri","danatau":"dan_atau","peneltian":
 ↪"penelitian","utk":"untuk","ta":"tugas_akhir","dharma":"tridharma","amik":
 ↪"akademi_manajemen_informatika_dan_komputer",
```

```
"pgri":"persatuan_guru_republik_indonesia","umkm":
 ↪"usaha_mikro_kecil_menengah","abdimas":"pengabdian_masyarakat","uppsps":
 ↪"unit_pengelola_program_studi",
"berdasar":"berdasarkan","ti":"teknologi_informasi","dtprmahasiswa":
 ↪"dosen_tetap_penghitung_rasio_mahasiswa","ppepp":
 ↪"penetapan_pelaksanaan_evaluasi_pengendalian_dan_peningkatan",
"si":"sitem_informasi","kaprodi":"kepala_program_studi","dipa":
 ↪"daftar_isian_pelaksanaan_anggaran","itb":"institut_teknologi_bandung",
"kebijakanstandar":"kebijakan_standar","renop":
 ↪"rencana_operasional","lemabagaunit":"lembaga_unit","akakom":
 ↪"akademi_aplikasi_komputer",
"permendikbud":"peraturan_menteri_pendidikan",
"standard":"standar","simlitabmas":
 ↪"sistem_informasi_penelitian_dan_pengabdian_masyarakat","pob":
 ↪"pedoman_operasional_baku",
"penelitan":"penelitian","fti":"fakultas_teknologi_industri","puslit":
 ↪"pusat_penelitian","ri":"republik_indonesia","dosenmahasiswa":
 ↪"dosen_mahasiswa",
"kkni":"kerangka_kualifikasi_nasional_indonesia",
"kkn":"kuliah_kerja_nyata","pkmgft":
 ↪"program_kreativitas_mahasiswa_gagasan_futuristik_tertulis","pkmgfk":
 ↪"program_kreativitas_mahasiswa_gagasan_futuristik_konstruktif",
}

replace_weird_words3 = {
"prodi":"program_studi","ik":"ilmu_komputer","ps":"program_studi","al":
 ↪"asesmen_lapangan","ak":"asesmen_kecukupan","pkm":
 ↪"program_kreativitas_mahasiswa","rtm":"papat_tinjauan_manajemen",
"ppm":"pengabdian_dan_pemberdayaan_masyarakat","cp":"capaian_pembelajaran",
"lkps":"laporan_kinerja_program_studi",
"lrs":"laporan_rencana_studi","kp":"kerja_praktek","lpks":
 ↪"lembaga_pengembangan_keterampilan_dan_sertifikasi",
"smk":"sekolah_menengah_kejuruan","dst":"dan_seterusnya","smp":"sampai","pk":
 ↪"penilaian_kinerja","lpm":"lembaga_penjaminan_mutu",
"kkmk":"kriteria_kinerja_manajemen_kualitas","pkl":
 ↪"rpaktek_kerja_lapangan","psms":"program_studi_manajemen_sistem",
"kk":"kode_klasifikasi","tkk":"tenaga_kependidikan_khusus",
"vmts":"visi_misi_tujuan_dan_sasaran","kbk":
 ↪"kurikulum_berbasis_kompetensi","pbl":"problem_based_learning","rpsm":
 ↪"rencana_pembelajaran_semester","ft":"fakultas_teknik","ttg":"tentang",
"snpt":"standar_nasional_perguruan_tinggi","lp":"lembaga_penjaminan","lpd":
 ↪"laporan_penelitian_dosen","pppm":"pengembangan_pendidikan_mahasiswa",
"rpjp":"rencana_pembangunan_jangka_panjang",
"st":"sarjana_teknik","sp":"surat_peringatan",
"dpp":"dosen_pendamping_program","pr":"penelitian_riset","ttd":
 ↪"tanda_tangan","pllm":"lembaga_penelitian_dan_pengabdian_masyarakat",
```

```
"rp":"rencana_pembelajaran","drpm":"direktorat_riset_pengabdian_masyarakat",
"ttp":"tetap","llppm":"lembaga_penelitian_pengabdian_masyarakat",
"cfp":"certified_financial_planner","lppmbppm":
 ↪"lembaga_penelitian_pengabdian_masyarakat_badan_penjaminan_mutu_pendidikan",
"spk":"surat_perjanjian_kerja","pm":"penjaminan_mutu","dptr":
 ↪"dosen_tetap_penghitung_rasio","kpd":"kepada","dtpt":
 ↪"dosen_tetap_perguruan_tinggi","bgm":"bagaimana",
"dtp":"dosen_tetap_program","kts":"kurikulum_tingkat_satuan",
"jg":"juga","dgx":"dengan","rpjmn":
 ↪"rencana_pembangunan_jangka_menengah_nasional","ppkm":
 ↪"program_pengembangan_keterampilan_mahasiswa",
"dprpm":"direktorat_penelitian_pengabdian_masyarakat","bg":"bagaimana","spmpt":
 ↪"sistem_penjaminan_mutu_pendidikan_tinggi","pnbp":
 ↪"penerimaan_negara_bukan_pajak",
"wkt":"waktu","csr":"corporate_social_responsibility","fgd":
 ↪"focus_group_discussion",
"kkp":"kualitas_kinerja_program","pkmdtpr":
 ↪"program_kreativitas_mahasiswa_dosen_tetap_penghitung_rasio",
"lppp":"lembaga_penelitian_pengabdian_masyarakat","rpp":
 ↪"rencana_pembelajaran_program",
"dtr":"dosen_tetap_penghitung_rasio","ln":"luar_negeri","tmps":
 ↪"tim_penjaminan_mutu_program_studi","tpmj":"tim_penjaminan_mutu_jurusan",
"pmk":"penjaminan_mutu_kinerja","bkn":"bukan","pkms":
 ↪"program_kreativitas_mahasiswa_sosial",
"pstp":"program_studi_teknik_pertambangan","ppkmdm":
 ↪"program_pengembangan_keterampilan_mahasiswa_dan_dosen",
"plp":"program_latihan_profesi","bnsp":
 ↪"badan_nasional_sertifikasi_profesi","skl":"surat_keterangan_lulus",
"cplcpmk":
 ↪"capaian_pembelajaran_lulusan_dan_capaian_pembelajaran_mata_kuliah","cpmk":
 ↪"capaian_pembelajaran_mata_kuliah","rkps":
 ↪"rencana_kegiatan_pembelajaran_semester","mhsw":"mahasiswa",
"jl":"jalan","sbgmn":"sebagaimana","jmlh":"jumlah","sms":"semester","lplppm":
 ↪"lembaga_penelitian_pengabdian_masyarakat","pstg":
 ↪"program_studi_teknik_geologi","pstk":"program_studi_teknik_kimia",
"kpt":"kualitas_pendidikan_tinggi","skp":"surat_kerja_praktek","rata2":
 ↪"rata_rata","mk":"mata_kuliah","pblpbm":
 ↪"problem_based_learning_project_based_learning","prn":
 ↪"program_riset_nasional","ppg":"program_profesi_guru","psbm":
 ↪"program_studi_bisnis_dan_manajemen",
"lpj":"laporan_pertanggungjawaban","dpts":"dosen_tetap_program_studi","kkt":
 ↪"kualitas_kinerja_tim",
"pjj":"pendidikan_jarak_jauh","pmp":"pemetaan_mutu_pendidikan","ts1":
 ↪"tracer_study_1","ts2":"tracer_study_2","ts3":"tracer_study_3","ts4":
 ↪"tracer_study_4","ts5":"tracer_study_5","ts6":"tracer_study_6",
}
```

```
typo_dict1 = {
"undri":"undur_diri","detil":"detail","luarancapaian":
 ↪"luaran_capaian","sebasar":"sebesar","komptetensi":"kompetensi",
"luluipk":"lulus_indeks_prestasi_kumulatif","perkulihan":"perkuliahan","dirjen":
 ↪"direktorat_jenderal","belmawa":"pembelajaran_dan_kemahasiswaan",
"systemaplikasi":"system_aplikasi","deskripsidatanarasi":
 ↪"deskripsi_data_narasi","informasidata":"informasi_data",
"egovernment":"government","ipkprestasi":
 ↪"indeks_prestasi_kumulatif_prestasi","perkaman":"perekaman","ydapat":"dapat",
"mahasiswat":"mahasiswa","lulululusan":"lulusan","senasif":
 ↪"seminar_nasional_informasi",
"prestai":"prestasi","standart":"standar","kessusuaian":"kesesuaian","kwerja":
 ↪"kerja","mendaoat":"mendapat",
"maksimun":"maksimum","berbading":"berbanding","kebuman":
 ↪"kebumen","diperolehlah":"diperoleh","tigatahunan":"tiga_tahunan",
"namum":"namun","detailriil":"detail_riil","dtprdosen":
 ↪"dosen_tetap_penghitung_rasio_dosen","klarfikiasi":"klarifikasi",
"kepuasanan":"kepuasan","fluaktif":"fluktuatif","ketidakkonsitenan":
 ↪"ketidakkonsistenan","simkatmawa":"sistem_pemeringkatan_kemahasiswaan",
"ekplisit":"eksplisit","terlenuhi":"terpenuhi","diihat":
 ↪"dilihat","pujiansebesar":"pujian_sebesar","pddikti":
 ↪"pangkalan_data_pendidikan_tinggi",
"atunggu":"tunggu","masy":"masyarakat","ditumkan":"ditemukan","reseponden":
 ↪"responden","kesesuaiaan":"kesesuaian",
"denagn":"dengan","berturutan":"berurutan","mnempuh":"menempuh","emenuhan":
 ↪"pemenuhan","mahaswasiswa":"mahasiswa",
"melampui":"melampaui","dilaporakan":"dilaporkan","laporaan":"laporan","elama":
 ↪"lama","mskor":"skor","erlacak":"terlacak",
"lulusanlulusan":"lulusan_lulusan","daapat":"dapat","reepositori":
 ↪"repositori","bidanginfokom":"bidang_infokom",
"jumlh":"jumlah","menyarakan":"menyarankan","polinas":
 ↪"politeknik_informatika_nasional","inggirs":"inggris",
"datajumlah":"data_jumlah","aktf":"aktif","kementrian":
 ↪"kementerian","eratcukup":"erat_cukup","pprodi":"program_studi",
"sertiikat":"sertifikat","terakahir":"terakhir","berkerjaberwirausaha":
 ↪"bekerja_berwirausaha",
"lokalwilayahberwirausaha":"lokal_wilayah_berusaha","depkominfo":
 ↪"departemen_komunikasi_dan_informatika",
"linktracer":"link_tracer","dtidak":"tidak","berhuubng":"berhubung","gkerja":
 ↪"kerja","subcpmk":"sub_capaian_pembelajaran_mata_kuliah",
"saaat":"saat","danstr":"dan_sarjana_terapan","hkiteknologi":
 ↪"hak_kekayaan_intelektual_teknologi",
"gunaprodukbuku":"guna_produk_buku","kurikulumlink":
 ↪"kurikulum_link","ketersedian":"ketersediaan",
```

```
"sikappengetahuanketerampilan":"sikap_pengetahuan_keterampilan","penilainnya":
 ↪"penilaiannya","lulusantabel":"lulusan_tabel",
"kerampilan":"keterampilan","tugaskuis":"tugas_kuis","pda":"pada","ket":
 ↪"keterangan","healthkathon":"health_kathon",
"defenisi":"definisi","perekeman":"perekaman","oprestasi":"prestasi","jkarya":
 ↪"karya","academyc":"academic",
"dibandikan":"dibandingkan","telahbelum":"telah_belum","sem":"semester","kkni":
 ↪"kerangka_kualifikasi_nasional_indonesia",
"diharapka":"diharapkan","penuruanan":"penurunan","renatan":"rentan","anatar":
 ↪"antar","dilaksanakantarcer":"dilaksanakan_tracer",
"dasboard":"dashboard","dalammkurun":"dalam_kurun","kelulusah":
 ↪"kelulusan","dtingkat":"tingkat","sekprodi":"sekretaris_program_studi",
"sertau":"serta","unipa":"universitas_papua","ppemenuhan":
 ↪"pemenuhan","ditelusur":"ditelusuri",
"sebelumnyasetiap":"sebelumnya_setiap","diberikansedangkan":
 ↪"diberikan_sedangkan","bbrapa":"beberapa",
"mahasiswabelum":"mahasiswa_belum","rekruitmen":"rekrutmen","hakidibawah":
 ↪"hak_kekayaan_intelektual_di_bawah",
"menghasillkan":"menghasilkan","disosialisasiakn":
 ↪"disosialisasikan","nasionalpada":"nasional_pada","infromatika":
 ↪"informatika",
"ratrata":"rata_rata","dtpsmahasiswa":
 ↪"dosen_tetap_program_studi_mahasiswa","tahunberdasarkan":"tahun_berdasarkan",
"eventevent":"event_event","erdasarkan":"berdasarkan","nyapelaksanaan":
 ↪"nya_pelaksanaan","temanteman":"teman_teman",
"studyunikama":"study_unikama","skore":"skor","menjacapi":
 ↪"mencapai","acehsumatera":"aceh_sumatera",
"internasiona":"internasional","tercpai":"tercapai","renstra":
 ↪"rencana_strategis",
"renstrarenoprkat":
 ↪"rencana_strategis_rencana_operasional_rencana_kerja_dan_anggaran_tahunan",
"kebermanfaatanaspek":"kebermanfaatan_aspek","tunggunnya":
 ↪"tunggunya","silahkan":"silakan","oresec":"foresec",
"mutinasionalinternasional":"multinasional_internasional","tercapaikesesuain":
 ↪"tercapai_kesesuaian","pelacakanperekaman":
 ↪"pelacakan_perekaman","matakuliah":"mata_kuliah",
"stimik":"sekolah_tinggi_ilmu_manajemen_informatika_dan_komputer","sepnop":
 ↪"sepuluh_nopember","ikuikt":
 ↪"indikator_kinerja_utama_indikator_kinerja_tambahan",
"pelbelajaran":"pembelajaran","dijelaskantetapi":"dijelaskan_tetapi","tingat":
 ↪"tingkat","katagari":"kategori",
"keberadaanya":"keberadaannya","adalahdiperoleh":"adalah_diperoleh","pertaa":
 ↪"pertama","dikatagorikan":"dikatagorikan",
"utuk":"untuk","dsetkan":"disertakan","standeriku":
 ↪"standar_indikator_kinerja_utama","prosentase":"persentase",
```

```
"nasionalsedangkan":"nasional_sedangkan","tekait":"terkait","crosscek":"cross_check","kiurang":"kurang",
"kompetisilomba":"kompetisi_lomba","acc":"accept","diuukur":"diukur","keterlaksannan":"keterlaksanaan","kemabali":"kembali",
"kesesuaiakn":"kesesuaian","psnya":"program_studi","utaraperingkat":"utara_peringkat","gbkperingkat":"gbk_peringkat","bahw":"bahwa","rerarta":"rerata",
"mendeskripsi":"mendeskripsikan","padalink":"pada_link","tercapaian":"ketercapaian","pa":"pembimbing_akademik",
"regionallokal":"regional_lokal","intektual":"intelektual","pdki":"pangkalan_data_kekayaan_intelektual","butirbutir":"butir_butir",
"penejelasan":"penjelasan","ilkom":"ilmu_komputer","padangsidimpuan":"padang_sidimpuan","didata":"di_data","kehlian":"keahlian","lulusn":"lulusan",
"finalisdan":"finalis_dan","usni":"universitas_satya_negara_indonesia","ekternal":"eksternal","khususkarena":"khusus_karena","runnerup":"runner_up",
"edupreuner":"edupreneur","sebagainyatidak":"sebagainya_tidak","led":"laporan_evaluasi_diri","ledlkps":"laporan_evaluasi_diri_laporan_kinerja_program_studi",
"meghasilkan":"menghasilkan","pemenuhana":"pemenuhan","sehingg":"sehingga","kekesahihan":"kesahihan","skalan":"skala","ditetapkansemua":"ditetapkan_semua",
"ckelulusan":"kelulusan","sebanyal":"sebanyak","drivecek":"drive_cek","lolakwilayah":"lolak_wilayah","adakelulusan":"ada_kelulusan","proditeknologi":"program_studi_teknologi",
"terpehuni":"terpenuhi","terpenhi":"terpenuhi","mahassiwa":"mahasiswa","tungg":"tunggu","lokalnasionalinternasional":"lokal_nasional_internasional","piagamsertifikat":"piagam_sertifikat",
"kelulusa":"kelulusan","dokumem":"dokumen","ratarataipk":"rata_rata_indeks_prestasi_kumulatif","berhki":"yang_memiliki_hak_kekayaan_intelektual","upi":"universitas_pendidikan_indonesia",
"univ":"universitas","akhiir":"akhir","capaiantabel":"capaian_tabel","peilaian":"penilaian","pekerjaanketerlaksanaan":"pekerjaan_keterlaksanaan","juarapesertapanitia":"juara_peserta_panitia",
"aklademiknon":"akademik_non","kaademik":"akademik","keseuain":"kesesuaian","kalrfikasi":"klarifikasi","kkniobeskkni":"kerangka_kualifikasi_nasional_indonesia_outcome_based_education_standar_kompetensi_kerja_n
"samapai":"sampai","kesesuaikan":"kesesuaian","tautantautan":"tautan_tautan","engisi":"mengisi","akreditas":"akreditasi","pembelajran":"pembelajaran","dikur":"diukur","presntasi":"presentasi","unuk":"untuk","sepeti":"seperti",
"pelaksaan":"pelaksanaan","pendidika":"pendidikan","thun":"tahun","lylusan":"lulusan","keterlaksanaanya":"keterlaksanaannya",
```

```
"pihakpihak":"pihak_pihak","unitama":
 ↪"universitas_teknologi_akba_makassar","memilik":"memiliki","unsika":
 ↪"universitas_singaperbangsa_karawang",
"darisama":"dari_sama","terkahir":"terakhir","studyratarata":
 ↪"study_rata_rata","almuni":"alumni","angakatan":
 ↪"angkatan","terlacakterdapat":"terlacak_terdapat",
"aspekapa":"aspek_apa","pertanyaaan":"pertanyaan","infokomps":
 ↪"infokom_program_studi","mengujur":"mengukur","miyaitu":"mi_yaitu",
"dilaksananakan":"dilaksanakan","tuingkat":"tingkat","nasonal":
 ↪"nasional","presteasi":"prestasi",
"tetang":"tentang","terjad":"terjadi","pemurunan":"penurunan","menadi":
 ↪"menjadi","peremkaman":"perekaman","rendahkurang":"rendah_kurang","perekama":
 ↪"perekaman","llusan":"lulusan","lulusanpada":"lulusan_pada","siginifikan":
 ↪"signifikan","karenkan":"karenakan","unesa":
 ↪"universitas_negeri_surabaya","perkerjaan":"pekerjaan","pemutahiran":
 ↪"pemutakhiran",
"dilakukankonfirmasi":"dilakukan_konfirmasi","mahsa":"masa","rataata":
 ↪"rata_rata",
"pelaksnaan":"pelaksanaan","tidaada":"tidak_ada","askpek":"aspek","tentan":
 ↪"tentang","kemanpuan":"kemampuan","bulanadapun":"bulan_adapun","ulusan":
 ↪"lulusan","infokomkepuasan":"infokom_kepuasan","wilayahnasional":
 ↪"wilayah_nasional",
"nformatika":"informatika","daengan":"dengan","dilaksakan":
 ↪"dilaksanakan","kuisoner":"kuesioner","adala":"adalah","masayang":
 ↪"masa_yang","pemerintahdalam":"pemerintah_dalam","amsa":"masa","tudi":
 ↪"studi","mendpatkan":"mendapatkan","lkti":
 ↪"lomba_karya_tulis_ilmiah","psychoferia":"psychofrenia","sigra":
 ↪"sistem_terintegrasi","kalrifikasin":"klarifikasi","terupsat":
 ↪"terpusat","diaksesbroken":"diakses_broken","keilmuankeahlian":
 ↪"keilmuan_keahlian",
"dilingkungan":"di_lingkungan","dimatakuliah":"di_mata_kuliah","tracertubacid":
 ↪"tracer_ub_ac_id","rataarat":"rata_rata",
"dilakukanperlu":"dilakukan_perlu","emndapat":"mendapat","mendapar":
 ↪"mendapat","tpi":"tapi","diidentifkasi":"diidentifikasi","pad":"pada","oba":
 ↪"outcome_based_assessment","acmis":
 ↪"academic_management_information_system","almni":"alumni","wirausaa":
 ↪"wirausaha","mekanime":"mekanisme",
"pemenuhancapaianpembelajaranlulusancplratarataipkprestasi":
 ↪"pemenuhan_capaian_pembelajaran_lulusan_rata_rata_indeks_prestasi_kumulatif_prestasi","reke
 ↪"rekayasa","komputerbekerja":"komputer_bekerja","akademiknonakademik":
 ↪"akademik_non_akademik","upp":"unit_pengelola_program_studi",
"keterlaksaan":"keterlaksanaan","idikator":"indikator","penetatpan":
 ↪"penetapan","disajika":"disajikan","memperolah":"memperoleh","nadhlatul":
 ↪"nahdlatul","tachnopreneur":"technopreneur","shasil":"hasil","waktuvisual":
 ↪"waktu","bersma":"bersama","isuisu":"isu_isu","semeter":
 ↪"semester","sedangkah":"sedangkan","peenuhan":"pemenuhan",
```

```
"careersitsacidtracer":"careersits_ac_id_tracer","standariku":
 ↪"standar_indikator_kinerja_utama","pelakanaan":"pelaksanaan","taraget":
 ↪"target","melaksankan":"melaksanakan","salama":"selama","kesesuaianya":
 ↪"kesesuaiannya","kesesusian":"kesesuaian",
"icts":
 ↪"international_conference_on_information_and_communication_technology_and_systems","sharepe
 ↪"sharepedia_id",
"kompetisikompetisi":"kompetisi_kompetisi","wa":"whatsapp","ebanyak":
 ↪"sebanyak","tsnya":"tracer_study_nya",
"lipihki":
 ↪"lembaga_inovasi_penulisan_ilmiah_dan_hak_kekayaan_intelektual","standarikuikt":
 ↪"standar_indikator_kinerja_utama_indikator_kinerja_tambahan","negaranegara":
 ↪"negara_negara","bidangbidang":"bidang_bidang","teknologiteknologi":
 ↪"teknologi_teknologi","kegiatankegiatan":"kegiatan_kegiatan",
}

typo_dict2 = {
"di konfirmasi":"dikonfirmasi","di validasi":"divalidasi","ter indentifikasi":
 ↪"terindentifikasi","ter sematkan":"tersematkan","pdik":
 ↪"program_studi_pendidikan","ilmu iahnya":"ilmiahnya","psmtif":
 ↪"program_studi_magister_teknik_informatika","perdosentahun":
 ↪"per_dosen_tahun",
"lacak":"lacakan","di tandai":"ditandai","ter pusat":"terpusat","per baikan":
 ↪"perbaikan","ber kualitas":"berkualitas","mkom":"magister_komputer","fmipa":
 ↪"fakultas_matematika_dan_ilmu_pengetahuan_alam","intelligence":
 ↪"kecerdasan","puslit":"pusat_penelitian","unmuh":
 ↪"universitas_muhammadiyah","pbi":
 ↪"pendidikan_bahasa_inggris","hilirisasipenerapan":
 ↪"hilirisasi_penerapan","matakuliah":"mata_kuliah",
"unsiq":"universitas_sains_al_quran","uppsprodi":
 ↪"unit_pengelola_program_studi","keseuaian":"kesesuaian","kebijakankebijakan":
 ↪"kebijakan_kebijakan","indicator":"indikator","procedure":
 ↪"prosedur","programprogram":"program_program","ketrampilan":
 ↪"keterampilan","pengmas":"pengabdian_masyarakat",
"ugn":"universitas_graha_nusantara","mtelah":"telah","kepahahaman":
 ↪"kepahaman","meningkatkanmengoptimalisasi":
 ↪"meningkatkan_mengoptimalisasi","noninfokom":"non_infokom","bindonesia":
 ↪"bahasa_indonesia",
"mi":"manajemen_informatika","renop":"rencana_operasi","sndikti":
 ↪"standar_nasional_pendidikan_tinggi",
}

DICTS = [replace_weird_words1, replace_weird_words2, replace_weird_words3,␣
 ↪typo_dict1, typo_dict2]
```

```python
# Merge dicts, drop empty / self-mapping entries
DICTS = [replace_weird_words1, replace_weird_words2, replace_weird_words3, typo_dict1, typo_dict2]
NORM_MAP = {}
for _d in DICTS:
    for _k, _v in _d.items():
        _k = _k.strip().lower(); _v = _v.strip()
        if _k and _v and _k != _v:
            NORM_MAP[_k] = _v

# Longest keys first so multi-word phrases match before their parts.
_NORM_KEYS = sorted(NORM_MAP.keys(), key=len, reverse=True)
_NORM_RE = re.compile(r'\b(' + '|'.join(re.escape(k) for k in _NORM_KEYS) + r')\b', re.IGNORECASE)
print(f'Normalization map: {len(NORM_MAP)} entries ({sum(1 for k in NORM_MAP if " " in k)} multi-word).')

def normalize_singkatan_typo(text):
    return _NORM_RE.sub(lambda m: NORM_MAP[m.group(0).lower()], text)

def clean_text(text):
    if pd.isna(text) or text == '':
        return ''
    text = str(text)
    text = re.sub(r'http\S+|www\S+|https\S+', '', text, flags=re.MULTILINE)
    text = re.sub(r'@\w+', '', text)
    text = _emoji.sub('', text)
    text = ' '.join(text.split())
    return text.strip()

def preprocess_pipeline(text):
    """clean -> lowercase -> singkatan/typo normalization (the provided dictionaries)."""
    if pd.isna(text) or text == '':
        return ''
    return normalize_singkatan_typo(clean_text(str(text)).lower()).strip()

# quick check
_ex = 'IPK mhs prodi TI sdh sesuai dgn standart, namum detil ts belum lengkap.'
print('Example:')
print('  in :', _ex)
print('  out:', preprocess_pipeline(_ex))
```

```
Normalization map: 701 entries (9 multi-word).
Example:
  in : IPK mhs prodi TI sdh sesuai dgn standart, namum detil ts belum lengkap.
  out: ipk mahasiswa program_studi teknologi_informasi sudah sesuai dengan
```

standar, namun detail tracer_study belum lengkap.

7.7 Load & Label

[ ]:
```
def load_and_preprocess(file_name, task, text_col):
    df = pd.read_excel(f'{BASE_PATH}/{file_name}')            # read the␣
 ↪labelled Excel file
    print(f'Loaded {file_name}  shape={df.shape}')            # show how many␣
 ↪rows/columns came in
    missing = [c for c in [text_col] + ASPECT_COLUMNS if c not in df.columns]
    if missing:                                               # stop early if a␣
 ↪needed column is absent
        raise ValueError(f'Missing columns: {missing}')
    df = df.dropna(subset=[text_col]).reset_index(drop=True) # drop rows with␣
 ↪no comment text
    df['comment_original'] = df[text_col].astype(str)         # keep a copy of␣
 ↪the raw text (used later for augmentation)
    df[text_col] = df['comment_original'].map(preprocess_pipeline)   # clean/
 ↪normalise the text we actually train on
    df = df[df[text_col].str.strip() != ''].reset_index(drop=True)   # drop rows␣
 ↪that became empty after cleaning
    lab_map = ACD_LABEL_MAP if task == 'ACD' else ASC_LABEL_MAP      # pick the␣
 ↪text->number label map for this task
    for col in ASPECT_COLUMNS:                                # convert each␣
 ↪aspect's word label to a number
        df[col] = df[col].astype(str).str.strip().str.lower().map(lab_map)
    if task == 'ACD':                                         # ACD: every␣
 ↪aspect must have a 0/1 label
        df = df.dropna(subset=ASPECT_COLUMNS).reset_index(drop=True)
        df[ASPECT_COLUMNS] = df[ASPECT_COLUMNS].astype(int)
    return df.reset_index(drop=True)                          # hand back the␣
 ↪cleaned, numeric-labelled table
```

7.8 Split (80/20), min-2-per-class guard, and few-shot sampling

[ ]:
```
# Turn the label columns into a plain matrix so the stratified splitter can␣
 ↪read them.
def _label_matrix(df, task):
    rows = []
    for _, r in df.iterrows():                     # go row by row (one comment␣
 ↪at a time)
        vec = []
        for col in ASPECT_COLUMNS:                 # walk all 8 aspects
            v = r[col]
            if task == 'ACD':
```

```
                vec.append(int(v))                # ACD: one 0/1 value per␣
↪aspect
            else:
                oh = [0, 0, 0]                    # ASC: one-hot for positif/
↪negatif/netral
                if not (isinstance(v, float) and np.isnan(v)):
                    oh[int(v)] = 1                # mark the class that is␣
↪present
                vec.extend(oh)
        rows.append(vec)
    return np.array(rows)                         # matrix of shape [comments,␣
↪labels]

# Split into 80% train / 20% test while keeping each aspect's class mix roughly␣
↪the same in both halves.
def stratified_split(df, task, test_size, seed):
    from sklearn.model_selection import train_test_split
    if not STRATIFY:                              # if stratification is turned␣
↪off, do a plain random split
        return train_test_split(df, test_size=test_size, random_state=seed,␣
↪shuffle=True)
    try:
        from iterstrat.ml_stratifiers import MultilabelStratifiedShuffleSplit
        Y = _label_matrix(df, task)               # label matrix the splitter␣
↪needs
        msss = MultilabelStratifiedShuffleSplit(n_splits=1,␣
↪test_size=test_size, random_state=seed)
        big, small = next(msss.split(df.values, Y))   # big = train indices,␣
↪small = test indices
        return df.iloc[big].reset_index(drop=True), df.iloc[small].
↪reset_index(drop=True)
    except Exception as e:                         # if the library is missing,␣
↪fall back to a plain split
        print(f'  [stratify] fallback to random split ({type(e).__name__})')
        return train_test_split(df, test_size=test_size, random_state=seed,␣
↪shuffle=True)

# Guarantee the test set has at least min_count examples of every class, moving␣
↪rows from train if needed.
def enforce_min_test_per_class(train_df, test_df, task, min_count, seed):
    '''Move rows train->test until each (aspect,class) has >= min_count in test␣
↪(where data allows).'''
    rng = random.Random(seed)                      # local RNG so the moves are␣
↪reproducible
    label_vals = [0, 1] if task == 'ACD' else [0, 1, 2]    # the class values to␣
↪check
```

```
    train_df = train_df.reset_index(drop=True); test_df = test_df.
↪reset_index(drop=True)
    moved = 0
    for col in ASPECT_COLUMNS:                              # for every aspect...
        for lv in label_vals:                               # ...and every class of␣
↪that aspect
            have = int((test_df[col] == lv).sum())  # how many are already in␣
↪the test set
            if have >= min_count:                           # enough already -> skip
                continue
            need = min_count - have                         # how many more we must␣
↪move over
            cand = train_df.index[train_df[col] == lv].tolist()  # training␣
↪rows that have this class
            rng.shuffle(cand)                               # shuffle so the choice␣
↪isn't order-dependent
            take = cand[:need]                              # take just enough
            if take:
                test_df  = pd.concat([test_df, train_df.loc[take]],␣
↪ignore_index=True)  # add to test
                train_df = train_df.drop(index=take).reset_index(drop=True)     ␣
↪         # remove from train
                moved += len(take)
    if moved:
        print(f'  [min-{min_count}] moved {moved} row(s) train->test to satisfy␣
↪per-class minimum')
    for col in ASPECT_COLUMNS:                              # warn about any class␣
↪still short (no data left to move)
        for lv in label_vals:
            if int((test_df[col] == lv).sum()) < min_count:
                nm = (ACD_LABEL_NAMES if task=='ACD' else ASC_LABEL_NAMES)[lv]
                print(f'  [warn] {col.strip()} / {nm}: only␣
↪{int((test_df[col]==lv).sum())} in test (data-limited)')
    return train_df.reset_index(drop=True), test_df.reset_index(drop=True)

# THE FEW-SHOT SAMPLER.
# For each (aspect, class) cell it draws up to n_per_class REAL comments, then␣
↪UNIONS all the draws.
# Because a comment carries a label for every aspect, one comment can satisfy␣
↪several cells at once,
# so the final fit set has each comment ONCE and the frequent classes end up␣
↪above n_per_class.
def few_shot_fit(train_df, task, n_per_class, seed):
    '''Union of up to n_per_class real comments per (aspect,class). No␣
↪augmentation.'''
    rng = random.Random(seed)                               # reproducible sampling
```

```
    pool = train_df.reset_index(drop=True)          # the training rows we␣
↪sample from
    label_vals = [0, 1] if task == 'ACD' else [0, 1, 2]
    keep = set()                                     # a SET -> automatically␣
↪de-duplicates comments
    report = []
    for col, short in zip(ASPECT_COLUMNS, ASPECT_SHORTS):   # loop over the 8␣
↪aspects
        for lv in label_vals:                            # loop over each␣
↪class of that aspect
            if task == 'ACD':
                idxs = pool.index[pool[col] == lv].tolist()          # rows␣
↪with this ACD class
            else:
                idxs = pool.index[pool[col] == float(lv)].tolist()   # rows␣
↪with this ASC class
            if not idxs:                             # none exist -> nothing␣
↪to sample
                continue
            sel = rng.sample(idxs, n_per_class) if len(idxs) >= n_per_class␣
↪else idxs  # 15, or all if fewer
            keep.update(sel)                         # add these comments to␣
↪the fit set (union)
            nm = (ACD_LABEL_NAMES if task=='ACD' else ASC_LABEL_NAMES)[lv]
            report.append(f'  {short[:28]:<28} | {nm:<13}: {len(sel):>2}/
↪{n_per_class} real')
    fit_df = pool.loc[sorted(keep)].reset_index(drop=True)  # build the final␣
↪training table from the kept rows
    print(f'  Few-shot fit set: {len(fit_df)} unique comments (target␣
↪{n_per_class}/class)')
    for r in report:                                 # print how many real␣
↪examples each cell contributed
        print(r)
    return fit_df
```

7.9 Dataset

```
[ ]: class ABSAMultiHeadDataset(Dataset):
         '''One comment -> input_ids/mask + label vector of length NUM_ASPECTS.
            ASC absent-aspect (NaN) -> -100 so it is ignored by that head's loss.'''
         def __init__(self, df, text_col, task, tokenizer, max_len):
             self.texts = df[text_col].astype(str).tolist()   # the cleaned comment␣
↪texts
             self.task = task
             labels = []
```

```
        for _, r in df.iterrows():                          # build one label row␣
↪per comment
            row = []
            for col in ASPECT_COLUMNS:                      # one entry per␣
↪aspect (8 total)
                v = r[col]
                if isinstance(v, float) and np.isnan(v):
                    row.append(-100)                        # -100 = "no label␣
↪here" -> loss will skip this head
                else:
                    row.append(int(v))                      # otherwise the class␣
↪index
            labels.append(row)
        self.labels = np.array(labels, dtype=np.int64)
        self.tokenizer = tokenizer; self.max_len = max_len
    def __len__(self):
        return len(self.texts)                              # how many comments␣
↪in this dataset
    def __getitem__(self, i):
        enc = self.tokenizer(self.texts[i], add_special_tokens=True,␣
↪max_length=self.max_len,
                             padding='max_length', truncation=True,␣
↪return_tensors='pt')  # tokenise one comment
        return {'input_ids': enc['input_ids'].flatten(),          # token IDs␣
↪the model reads
                'attention_mask': enc['attention_mask'].flatten(),# 1 for real␣
↪tokens, 0 for padding
                'labels': torch.tensor(self.labels[i], dtype=torch.long)}  #␣
↪the 8-aspect label vector
```

7.10 Model — shared IndoBERT + per-aspect heads

[ ]:
```
class IndoBERT_MultiHead_ABSA(nn.Module):
    def __init__(self, num_aspects, num_labels_per_head, dropout_rate=0.3):
        super().__init__()
        self.bert = BertModel.from_pretrained(MODEL_CHECKPOINT)   # the shared␣
↪IndoBERT encoder
        h = self.bert.config.hidden_size                          # size of␣
↪BERT's output vector (768)
        self.dropout = nn.Dropout(dropout_rate)                   #␣
↪regularisation before the heads
        # ONE small linear classifier per aspect (8 heads). Each head decides␣
↪for its own aspect only.
        self.heads = nn.ModuleList([nn.Linear(h, num_labels_per_head) for _ in␣
↪range(num_aspects)])
    def forward(self, input_ids, attention_mask):
```

```
        pooled = self.bert(input_ids=input_ids, attention_mask=attention_mask).
↪pooler_output  # encode the comment ONCE
        pooled = self.dropout(pooled)                               # apply␣
↪dropout to that shared vector
        return [head(pooled) for head in self.heads]                # each head␣
↪reads the SAME vector -> 8 sets of logits
```

7.11 Per-aspect class weights (from the few-shot fit set)

```
[ ]: def compute_head_weights(fit_df, task):
         '''Balanced class weights per aspect head, computed on available (non -100)␣
↪labels.'''
         n_lab = 2 if task == 'ACD' else 3                          # 2 classes for ACD, 3␣
↪for ASC
         weights = []
         for col in ASPECT_COLUMNS:                                  # compute a weight␣
↪vector for each aspect head
             y = fit_df[col].dropna().astype(int).values             # the labels actually␣
↪present for this aspect
             present = np.unique(y)                                  # which classes appear␣
↪at all
             w = np.ones(n_lab, dtype=np.float32)                    # default weight 1.0␣
↪for every class
             if len(present) >= 2:                                   # only balance if at␣
↪least 2 classes are present
                 cw = compute_class_weight('balanced', classes=present, y=y)  #␣
↪rarer class -> bigger weight
                 for c, val in zip(present, cw):
                     w[c] = val                                      # store the computed␣
↪weight in its class slot
             weights.append(torch.tensor(w, dtype=torch.float32).to(device))  # move␣
↪to GPU/CPU as a tensor
         return weights                                              # one weight vector per␣
↪aspect, used in the loss
```

7.12 Training (weighted CE, no validation, no early stopping)

```
[ ]: def train_model(fit_df, task, text_col, seed):
         set_all_seeds(seed)                                    # reset all RNGs so␣
↪training is reproducible
         n_lab = 2 if task == 'ACD' else 3                      # classes per head: 2␣
↪(ACD) or 3 (ASC)
         ds = ABSAMultiHeadDataset(fit_df, text_col, task, tokenizer, MAX_SEQ_LEN) ␣
↪# wrap the fit set
         loader = DataLoader(ds, batch_size=BATCH_SIZE, shuffle=True)              ␣
↪# feed it in shuffled batches
```

```python
    model = IndoBERT_MultiHead_ABSA(NUM_ASPECTS, n_lab, DROPOUT).to(device)    # build the 8-head model
    head_weights = compute_head_weights(fit_df, task)                          # per-aspect class weights
    # One weighted cross-entropy loss per aspect. ignore_index=-100 -> absent labels don't contribute.
    loss_fns = [nn.CrossEntropyLoss(weight=head_weights[a], ignore_index=-100)
                for a in range(NUM_ASPECTS)]

    optimizer = AdamW(model.parameters(), lr=LEARNING_RATE, weight_decay=WEIGHT_DECAY)  # flat LR, no scheduler

    history = []
    for epoch in range(EPOCHS):                          # train for a fixed number of epochs (5)
        model.train(); run = 0.0                         # training mode; running loss accumulator
        pbar = tqdm(loader, desc=f'  epoch {epoch+1}/{EPOCHS}', leave=False)
        for batch in pbar:                               # one mini-batch of comments at a time
            ids  = batch['input_ids'].to(device)         # token IDs -> GPU
            mask = batch['attention_mask'].to(device)# attention mask -> GPU
            lab  = batch['labels'].to(device)            # true labels, shape [batch, 8]
            logits_list = model(ids, mask)               # forward pass -> 8 heads' logits
            # total loss = sum of each head's loss (each head compares its column of labels)
            loss = sum(loss_fns[a](logits_list[a], lab[:, a]) for a in range(NUM_ASPECTS))
            optimizer.zero_grad(); loss.backward()       # clear old gradients, back-propagate
            optimizer.step()                             # update the weights
            run += loss.item(); pbar.set_postfix(loss=loss.item())
        ep_loss = run / len(loader)                      # average loss this epoch
        history.append(ep_loss)
        print(f'  epoch {epoch+1}: train_loss={ep_loss:.4f}')
    return model, history                                # trained model + loss curve
```

7.13 Prediction

[ ]:
```
@torch.no_grad()                                          # no gradients needed when␣
 ↪only predicting
def predict(model, df, text_col, task):
    model.eval()                                          # evaluation mode (turns␣
 ↪off dropout)
    ds = ABSAMultiHeadDataset(df, text_col, task, tokenizer, MAX_SEQ_LEN)
    loader = DataLoader(ds, batch_size=BATCH_SIZE, shuffle=False)  # keep order␣
 ↪so preds line up with rows
    preds, labels = [], []
    for batch in loader:
        ids  = batch['input_ids'].to(device)
        mask = batch['attention_mask'].to(device)
        logits_list = model(ids, mask)                    # 8 heads' logits for this␣
 ↪batch
        # for each head take the highest-scoring class -> one prediction per␣
 ↪aspect -> [batch, 8]
        p = torch.stack([torch.argmax(l, dim=1) for l in logits_list], dim=1)
        preds.append(p.cpu().numpy())                     # collect predictions
        labels.append(batch['labels'].numpy())            # collect the true labels
    return np.vstack(preds), np.vstack(labels)            # stack all batches into␣
 ↪full arrays
```

7.14 Metrics — per-aspect P/R/F1, macro-F1 + weighted-F1, confusion matrices

[ ]:
```
def _valid(true, pred, task):
    if task == 'ASC':
        m = true != -100
        return true[m], pred[m]
    return true, pred

def print_metrics(preds, labels, task, name):
    label_names = ACD_LABEL_NAMES if task == 'ACD' else ASC_LABEL_NAMES
    idxs = list(range(len(label_names)))
    print(f'\n{"="*64}\n  {name}\n{"="*64}')
    macro_list, weighted_list = [], []
    for a, asp in enumerate(ASPECT_SHORTS):
        true, pred = _valid(labels[:, a], preds[:, a], task)
        if len(true) == 0:
            print(f'\nAspect: {asp}  (no labels)'); continue
        p, r, f, _ = precision_recall_fscore_support(true, pred, labels=idxs,␣
 ↪average=None, zero_division=0)
        mf1 = f1_score(true, pred, average='macro',    zero_division=0)
        wf1 = f1_score(true, pred, average='weighted', zero_division=0)
        macro_list.append(mf1); weighted_list.append(wf1)
```

```
        print(f'\nAspect: {asp}')
        for i, ln in enumerate(label_names):
            print(f'  {ln:<22} P={p[i]:.3f}  R={r[i]:.3f}  F1={f[i]:.3f}')
        print(f'  Macro F1 {mf1:.4f}  |  Weighted F1 {wf1:.4f}')
    mean_macro = float(np.mean(macro_list)) if macro_list else float("nan")
    mean_weight = float(np.mean(weighted_list)) if weighted_list else␣
↪float("nan")
    print(f'\n  MEAN Macro F1: {mean_macro:.4f}  |  MEAN Weighted F1:␣
↪{mean_weight:.4f}')
    return {'mean_macro_f1': mean_macro, 'mean_weighted_f1': mean_weight,
            'per_aspect_macro': dict(zip(ASPECT_SHORTS, macro_list)),
            'per_aspect_weighted': dict(zip(ASPECT_SHORTS, weighted_list))}

def plot_cms(preds, labels, task, name):
    label_names = ACD_LABEL_NAMES if task == 'ACD' else ASC_LABEL_NAMES
    idxs = list(range(len(label_names)))
    fig, axes = plt.subplots(2, 4, figsize=(24, 12))
    for a, (asp, ax) in enumerate(zip(ASPECT_SHORTS, axes.flatten())):
        true, pred = _valid(labels[:, a], preds[:, a], task)
        if len(true) == 0:
            ax.set_title(f'{asp}\n(no data)', fontsize=9); ax.axis('off');␣
↪continue
        cm = confusion_matrix(true, pred, labels=idxs)
        sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
                    xticklabels=label_names, yticklabels=label_names, ax=ax,
                    cbar=True, linewidths=0.5, linecolor='white')
        ax.set_title(asp, fontsize=9); ax.set_xlabel('Pred', fontsize=8); ax.
↪set_ylabel('True', fontsize=8)
        ax.tick_params(labelsize=7)
    plt.suptitle(f'Confusion Matrices - {name}', fontsize=13, y=1.01)
    plt.tight_layout()
    fname = f'{BASE_PATH}/results/baseline/cm_{name}.png'; os.makedirs(os.path.
↪dirname(fname), exist_ok=True); plt.savefig(fname, dpi=150,␣
↪bbox_inches='tight'); plt.show()
    print('Saved ->', fname)

def plot_training(history, name):
    plt.figure(figsize=(7, 4))
    plt.plot(range(1, len(history) + 1), history, marker='o', linewidth=2,␣
↪color='#2b6cb0')
    plt.title(f'Training Loss - {name}', fontsize=12)
    plt.xlabel('Epoch'); plt.ylabel('Summed weighted CE loss')
    plt.xticks(range(1, len(history) + 1)); plt.grid(True, alpha=0.3)
    fname = f'{BASE_PATH}/results/baseline/training_{name}.png'
    os.makedirs(os.path.dirname(fname), exist_ok=True)
```

```
    plt.tight_layout(); plt.savefig(fname, dpi=150, bbox_inches='tight'); plt.
↪show()
    print('Saved ->', fname)
```

7.15 Save predictions to Excel

[ ]:
```
def save_predictions_excel(test_df, preds, labels, text_col, task, name):
    names = ACD_LABEL_NAMES if task == 'ACD' else ASC_LABEL_NAMES
    nm = lambda v: '' if int(v) == -100 else names[int(v)]
    rows = []
    for i in range(len(test_df)):
        r = {'comment': str(test_df.iloc[i][text_col])}
        for a, asp in enumerate(ASPECT_SHORTS):
            r[f'{asp} | true'] = nm(labels[i, a])
            r[f'{asp} | pred'] = nm(preds[i, a])
        rows.append(r)
    out = pd.DataFrame(rows)
    os.makedirs(f'{BASE_PATH}/results/baseline', exist_ok=True)
    path = f'{BASE_PATH}/results/baseline/predictions_{name}_weighted_baseline.
↪xlsx'
    out.to_excel(path, index=False)
    print(f'  Saved {len(out)} test comments + predictions -> {path}')
    return out
```

7.16 Runner

[ ]:
```
def run_experiment(exp):
    name, task, text_col = exp['name'], exp['task'], exp['text_col']
    print(f'\n{"#"*66}\n  {name}  [{task} / weighted-class baseline / 80-20 /␣
↪few-shot {N_PER_CLASS}]\n{"#"*66}')

    df = load_and_preprocess(exp['file'], task, text_col)          # load +␣
↪clean + numeric labels

    train_df, test_df = stratified_split(df, task, TEST_FRAC, SEED)  # 80/20␣
↪stratified split
    train_df, test_df = enforce_min_test_per_class(train_df, test_df, task,␣
↪MIN_TEST_PER_CLASS, SEED)  # guarantee test coverage
    print(f'  Split: total={len(df)} | train={len(train_df)} |␣
↪test={len(test_df)}  (NO validation)')

    # Save the EXACT split so the tahap tesaurus trains/tests on the SAME data
    # This is what makes the two experiments a fair comparison: identical train␣
↪and test rows.
    os.makedirs(f'{BASE_PATH}/results/baseline/splits', exist_ok=True)
    keep_cols = ['comment_original', text_col] + ASPECT_COLUMNS
```

```
    train_df[keep_cols].to_excel(f'{BASE_PATH}/results/baseline/splits/
↪{name}_train.xlsx', index=False)
    test_df[keep_cols].to_excel(f'{BASE_PATH}/results/baseline/splits/
↪{name}_test.xlsx', index=False)
    print(f'  Saved splits -> {BASE_PATH}/results/baseline/splits/{name}_train.
↪xlsx  &  {name}_test.xlsx')

    fit_df = few_shot_fit(train_df, task, N_PER_CLASS, SEED)       # pick the␣
↪few-shot training comments

    model, history = train_model(fit_df, task, text_col, SEED)    # fine-tune␣
↪IndoBERT on them

    print('\n  Evaluating on held-out TEST set...')
    preds, labels = predict(model, test_df, text_col, task)       # predict on␣
↪the untouched test set
    metrics = print_metrics(preds, labels, task, name)            # P/R/F1,␣
↪macro-F1, weighted-F1
    plot_cms(preds, labels, task, name)                           # confusion␣
↪matrices
    plot_training(history, name)                                  # loss curve
    save_predictions_excel(test_df, preds, labels, text_col, task, name)  #␣
↪dump predictions to Excel

    del model; gc.collect()                                       # free memory␣
↪before the next experiment
    if torch.cuda.is_available(): torch.cuda.empty_cache()
    return {'metrics': metrics, 'preds': preds, 'labels': labels,
            'n_train': len(train_df), 'n_fit': len(fit_df), 'n_test':␣
↪len(test_df)}
```

7.17 Run all four experiments (ACD AK/AL, ASC AK/AL)

[ ]:
```
ALL_RESULTS = {}
for exp in ACD_EXPERIMENTS + ASC_EXPERIMENTS:
    ALL_RESULTS[exp['name']] = run_experiment(exp)

print('\nDone.  Mean macro-F1 / weighted-F1:')
for k, v in ALL_RESULTS.items():
    m = v['metrics']
    print(f'  {k:<8}  macro={m["mean_macro_f1"]:.4f} ␣
↪weighted={m["mean_weighted_f1"]:.4f}')
```

```
##################################################################
  ACD_AK  [ACD / weighted-class baseline / 80-20 / few-shot 15]
##################################################################
```

```
Loaded Hasil Ground Truth_ACD AK_clean.xlsx  shape=(200, 12)
  [min-2] moved 1 row(s) train->test to satisfy per-class minimum
  Split: total=200 | train=159 | test=41  (NO validation)
  Saved splits ->
/content/drive/MyDrive/results/baseline/splits/ACD_AK_train.xlsx  &
ACD_AK_test.xlsx
  Few-shot fit set: 119 unique comments (target 15/class)
  capaian pembelajaran lulusan | tidak_relevan:  2/15 real
  capaian pembelajaran lulusan | relevan      : 15/15 real
  indeks prestasi kumulatif    | tidak_relevan: 15/15 real
  indeks prestasi kumulatif    | relevan      : 15/15 real
  kelulusan tepat waktu        | tidak_relevan: 15/15 real
  kelulusan tepat waktu        | relevan      : 15/15 real
  masa tunggu lulusan          | tidak_relevan: 15/15 real
  masa tunggu lulusan          | relevan      : 15/15 real
  pelacakan dan perekaman data | tidak_relevan: 15/15 real
  pelacakan dan perekaman data | relevan      : 15/15 real
  prestasi mahasiswa           | tidak_relevan: 15/15 real
  prestasi mahasiswa           | relevan      : 15/15 real
  karya dengan hak kekayaan in | tidak_relevan: 15/15 real
  karya dengan hak kekayaan in | relevan      : 15/15 real
  kesesuaian bidang kerja      | tidak_relevan: 15/15 real
  kesesuaian bidang kerja      | relevan      : 15/15 real

config.json:   0%|          | 0.00/1.53k [00:00<?, ?B/s]

pytorch_model.bin:   0%|          | 0.00/498M [00:00<?, ?B/s]

Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]

  epoch 1/5:   0%|          | 0/15 [00:00<?, ?it/s]

model.safetensors:   0%|          | 0.00/498M [00:00<?, ?B/s]

  epoch 1: train_loss=4.9322

  epoch 2/5:   0%|          | 0/15 [00:00<?, ?it/s]

  epoch 2: train_loss=3.8374

  epoch 3/5:   0%|          | 0/15 [00:00<?, ?it/s]

  epoch 3: train_loss=3.2890

  epoch 4/5:   0%|          | 0/15 [00:00<?, ?it/s]

  epoch 4: train_loss=2.9521

  epoch 5/5:   0%|          | 0/15 [00:00<?, ?it/s]

  epoch 5: train_loss=2.4887

  Evaluating on held-out TEST set…

================================================================
```

```
  ACD_AK
================================================================

Aspect: capaian pembelajaran lulusan
  tidak_relevan         P=0.000  R=0.000  F1=0.000
  relevan               P=0.951  R=1.000  F1=0.975
  Macro F1 0.4875  |  Weighted F1 0.9274

Aspect: indeks prestasi kumulatif
  tidak_relevan         P=0.429  R=0.500  F1=0.462
  relevan               P=0.912  R=0.886  F1=0.899
  Macro F1 0.6800  |  Weighted F1 0.8346

Aspect: kelulusan tepat waktu
  tidak_relevan         P=0.700  R=0.538  F1=0.609
  relevan               P=0.806  R=0.893  F1=0.847
  Macro F1 0.7281  |  Weighted F1 0.7718

Aspect: masa tunggu lulusan
  tidak_relevan         P=0.800  R=0.800  F1=0.800
  relevan               P=0.935  R=0.935  F1=0.935
  Macro F1 0.8677  |  Weighted F1 0.9024

Aspect: pelacakan dan perekaman data lulusan
  tidak_relevan         P=0.727  R=0.500  F1=0.593
  relevan               P=0.733  R=0.880  F1=0.800
  Macro F1 0.6963  |  Weighted F1 0.7191

Aspect: prestasi mahasiswa
  tidak_relevan         P=0.538  R=0.778  F1=0.636
  relevan               P=0.929  R=0.812  F1=0.867
  Macro F1 0.7515  |  Weighted F1 0.8161

Aspect: karya dengan hak kekayaan intelektual
  tidak_relevan         P=0.636  R=0.467  F1=0.538
  relevan               P=0.733  R=0.846  F1=0.786
  Macro F1 0.6621  |  Weighted F1 0.6953

Aspect: kesesuaian bidang kerja
  tidak_relevan         P=0.800  R=0.615  F1=0.696
  relevan               P=0.839  R=0.929  F1=0.881
  Macro F1 0.7885  |  Weighted F1 0.8225

  MEAN Macro F1: 0.7077  |  MEAN Weighted F1: 0.8111

Saved -> /content/drive/MyDrive/results/baseline/cm_ACD_AK.png

Saved -> /content/drive/MyDrive/results/baseline/training_ACD_AK.png
  Saved 41 test comments + predictions -> /content/drive/MyDrive/results/base
```

```
line/predictions_ACD_AK_weighted_baseline.xlsx

##################################################################
  ACD_AL  [ACD / weighted-class baseline / 80-20 / few-shot 15]
##################################################################
Loaded Hasil Ground Truth_ACD AL_clean.xlsx  shape=(200, 12)
  [min-2] moved 2 row(s) train->test to satisfy per-class minimum
  Split: total=200 | train=157 | test=43  (NO validation)
  Saved splits ->
/content/drive/MyDrive/results/baseline/splits/ACD_AL_train.xlsx  &
ACD_AL_test.xlsx
  Few-shot fit set: 120 unique comments (target 15/class)
  capaian pembelajaran lulusan | tidak_relevan:  1/15 real
  capaian pembelajaran lulusan | relevan      : 15/15 real
  indeks prestasi kumulatif    | tidak_relevan: 13/15 real
  indeks prestasi kumulatif    | relevan      : 15/15 real
  kelulusan tepat waktu        | tidak_relevan: 15/15 real
  kelulusan tepat waktu        | relevan      : 15/15 real
  masa tunggu lulusan          | tidak_relevan: 15/15 real
  masa tunggu lulusan          | relevan      : 15/15 real
  pelacakan dan perekaman data | tidak_relevan: 15/15 real
  pelacakan dan perekaman data | relevan      : 15/15 real
  prestasi mahasiswa           | tidak_relevan: 15/15 real
  prestasi mahasiswa           | relevan      : 15/15 real
  karya dengan hak kekayaan in | tidak_relevan: 15/15 real
  karya dengan hak kekayaan in | relevan      : 15/15 real
  kesesuaian bidang kerja      | tidak_relevan: 15/15 real
  kesesuaian bidang kerja      | relevan      : 15/15 real
Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]
  epoch 1/5:   0%|          | 0/15 [00:00<?, ?it/s]
  epoch 1: train_loss=4.7581
  epoch 2/5:   0%|          | 0/15 [00:00<?, ?it/s]
  epoch 2: train_loss=3.9465
  epoch 3/5:   0%|          | 0/15 [00:00<?, ?it/s]
  epoch 3: train_loss=3.2544
  epoch 4/5:   0%|          | 0/15 [00:00<?, ?it/s]
  epoch 4: train_loss=2.6656
  epoch 5/5:   0%|          | 0/15 [00:00<?, ?it/s]
  epoch 5: train_loss=2.2397

  Evaluating on held-out TEST set…
```

```
=================================================================
  ACD_AL
=================================================================

Aspect: capaian pembelajaran lulusan
  tidak_relevan         P=1.000  R=0.500  F1=0.667
  relevan               P=0.976  R=1.000  F1=0.988
  Macro F1 0.8273  |  Weighted F1 0.9730

Aspect: indeks prestasi kumulatif
  tidak_relevan         P=0.667  R=0.667  F1=0.667
  relevan               P=0.946  R=0.946  F1=0.946
  Macro F1 0.8063  |  Weighted F1 0.9070

Aspect: kelulusan tepat waktu
  tidak_relevan         P=0.750  R=0.900  F1=0.818
  relevan               P=0.968  R=0.909  F1=0.938
  Macro F1 0.8778  |  Weighted F1 0.9098

Aspect: masa tunggu lulusan
  tidak_relevan         P=0.769  R=0.769  F1=0.769
  relevan               P=0.900  R=0.900  F1=0.900
  Macro F1 0.8346  |  Weighted F1 0.8605

Aspect: pelacakan dan perekaman data lulusan
  tidak_relevan         P=0.750  R=0.600  F1=0.667
  relevan               P=0.806  R=0.893  F1=0.847
  Macro F1 0.7571  |  Weighted F1 0.7844

Aspect: prestasi mahasiswa
  tidak_relevan         P=0.733  R=0.846  F1=0.786
  relevan               P=0.929  R=0.867  F1=0.897
  Macro F1 0.8411  |  Weighted F1 0.8630

Aspect: karya dengan hak kekayaan intelektual
  tidak_relevan         P=0.800  R=0.800  F1=0.800
  relevan               P=0.893  R=0.893  F1=0.893
  Macro F1 0.8464  |  Weighted F1 0.8605

Aspect: kesesuaian bidang kerja
  tidak_relevan         P=0.769  R=0.909  F1=0.833
  relevan               P=0.967  R=0.906  F1=0.935
  Macro F1 0.8844  |  Weighted F1 0.9094

  MEAN Macro F1: 0.8344  |  MEAN Weighted F1: 0.8834

Saved -> /content/drive/MyDrive/results/baseline/cm_ACD_AL.png

Saved -> /content/drive/MyDrive/results/baseline/training_ACD_AL.png
```

```
  Saved 43 test comments + predictions -> /content/drive/MyDrive/results/base
line/predictions_ACD_AL_weighted_baseline.xlsx

##################################################################
  ASC_AK  [ASC / weighted-class baseline / 80-20 / few-shot 15]
##################################################################
Loaded Hasil Ground Truth_ASC AK_clean.xlsx  shape=(200, 12)
  Split: total=200 | train=158 | test=42  (NO validation)
  Saved splits ->
/content/drive/MyDrive/results/baseline/splits/ASC_AK_train.xlsx  &
ASC_AK_test.xlsx
  Few-shot fit set: 134 unique comments (target 15/class)
  capaian pembelajaran lulusan | positif     : 15/15 real
  capaian pembelajaran lulusan | negatif     : 15/15 real
  capaian pembelajaran lulusan | netral      : 15/15 real
  indeks prestasi kumulatif    | positif     : 15/15 real
  indeks prestasi kumulatif    | negatif     : 11/15 real
  indeks prestasi kumulatif    | netral      : 15/15 real
  kelulusan tepat waktu        | positif     : 15/15 real
  kelulusan tepat waktu        | negatif     :  9/15 real
  kelulusan tepat waktu        | netral      : 15/15 real
  masa tunggu lulusan          | positif     : 15/15 real
  masa tunggu lulusan          | negatif     : 10/15 real
  masa tunggu lulusan          | netral      : 15/15 real
  pelacakan dan perekaman data | positif     : 15/15 real
  pelacakan dan perekaman data | negatif     :  8/15 real
  pelacakan dan perekaman data | netral      : 15/15 real
  prestasi mahasiswa           | positif     : 15/15 real
  prestasi mahasiswa           | negatif     : 14/15 real
  prestasi mahasiswa           | netral      : 15/15 real
  karya dengan hak kekayaan in | positif     : 15/15 real
  karya dengan hak kekayaan in | negatif     : 12/15 real
  karya dengan hak kekayaan in | netral      : 15/15 real
  kesesuaian bidang kerja      | positif     : 15/15 real
  kesesuaian bidang kerja      | negatif     : 10/15 real
  kesesuaian bidang kerja      | netral      : 15/15 real

Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]

  epoch 1/5:   0%|          | 0/17 [00:00<?, ?it/s]

  epoch 1: train_loss=9.0780

  epoch 2/5:   0%|          | 0/17 [00:00<?, ?it/s]

  epoch 2: train_loss=8.5834

  epoch 3/5:   0%|          | 0/17 [00:00<?, ?it/s]

  epoch 3: train_loss=8.1590

  epoch 4/5:   0%|          | 0/17 [00:00<?, ?it/s]
```

```
  epoch 4: train_loss=7.6994

  epoch 5/5:   0%|          | 0/17 [00:00<?, ?it/s]

  epoch 5: train_loss=6.9389

  Evaluating on held-out TEST set…

================================================================
  ASC_AK
================================================================

Aspect: capaian pembelajaran lulusan
  positif               P=0.769  R=0.526  F1=0.625
  negatif               P=0.250  R=0.167  F1=0.200
  netral                P=0.417  R=0.625  F1=0.500
  Macro F1 0.4417  |  Weighted F1 0.5140

Aspect: indeks prestasi kumulatif
  positif               P=0.800  R=0.632  F1=0.706
  negatif               P=1.000  R=0.333  F1=0.500
  netral                P=0.500  R=0.750  F1=0.600
  Macro F1 0.6020  |  Weighted F1 0.6503

Aspect: kelulusan tepat waktu
  positif               P=0.765  R=0.684  F1=0.722
  negatif               P=0.111  R=0.500  F1=0.182
  netral                P=0.500  R=0.143  F1=0.222
  Macro F1 0.3754  |  Weighted F1 0.5586

Aspect: masa tunggu lulusan
  positif               P=0.667  R=0.333  F1=0.444
  negatif               P=0.000  R=0.000  F1=0.000
  netral                P=0.318  R=0.700  F1=0.438
  Macro F1 0.2940  |  Weighted F1 0.3992

Aspect: pelacakan dan perekaman data lulusan
  positif               P=0.786  R=0.611  F1=0.688
  negatif               P=0.000  R=0.000  F1=0.000
  netral                P=0.333  R=0.400  F1=0.364
  Macro F1 0.3504  |  Weighted F1 0.5677

Aspect: prestasi mahasiswa
  positif               P=0.917  R=0.611  F1=0.733
  negatif               P=0.000  R=0.000  F1=0.000
  netral                P=0.462  R=0.750  F1=0.571
  Macro F1 0.4349  |  Weighted F1 0.5924

Aspect: karya dengan hak kekayaan intelektual
```

```
  positif               P=0.571  R=0.286  F1=0.381
  negatif               P=0.000  R=0.000  F1=0.000
  netral                P=0.353  R=0.667  F1=0.462
  Macro F1 0.2808  |  Weighted F1 0.3649

Aspect: kesesuaian bidang kerja
  positif               P=0.615  R=0.533  F1=0.571
  negatif               P=0.000  R=0.000  F1=0.000
  netral                P=0.400  R=0.545  F1=0.462
  Macro F1 0.3443  |  Weighted F1 0.4874

  MEAN Macro F1: 0.3904  |  MEAN Weighted F1: 0.5168

Saved -> /content/drive/MyDrive/results/baseline/cm_ASC_AK.png

Saved -> /content/drive/MyDrive/results/baseline/training_ASC_AK.png
  Saved 42 test comments + predictions -> /content/drive/MyDrive/results/base
line/predictions_ASC_AK_weighted_baseline.xlsx

##################################################################
  ASC_AL  [ASC / weighted-class baseline / 80-20 / few-shot 15]
##################################################################
Loaded Hasil Ground Truth_ASC AL_clean.xlsx  shape=(200, 12)
  [min-2] moved 3 row(s) train->test to satisfy per-class minimum
  Split: total=200 | train=156 | test=44  (NO validation)
  Saved splits ->
/content/drive/MyDrive/results/baseline/splits/ASC_AL_train.xlsx  &
ASC_AL_test.xlsx
  Few-shot fit set: 132 unique comments (target 15/class)
  capaian pembelajaran lulusan | positif      : 15/15 real
  capaian pembelajaran lulusan | negatif      : 12/15 real
  capaian pembelajaran lulusan | netral       : 15/15 real
  indeks prestasi kumulatif    | positif      : 15/15 real
  indeks prestasi kumulatif    | negatif      :  5/15 real
  indeks prestasi kumulatif    | netral       : 15/15 real
  kelulusan tepat waktu        | positif      : 15/15 real
  kelulusan tepat waktu        | negatif      : 10/15 real
  kelulusan tepat waktu        | netral       : 15/15 real
  masa tunggu lulusan          | positif      : 15/15 real
  masa tunggu lulusan          | negatif      : 10/15 real
  masa tunggu lulusan          | netral       : 15/15 real
  pelacakan dan perekaman data | positif      : 15/15 real
  pelacakan dan perekaman data | negatif      :  4/15 real
  pelacakan dan perekaman data | netral       : 15/15 real
  prestasi mahasiswa           | positif      : 15/15 real
  prestasi mahasiswa           | negatif      :  7/15 real
  prestasi mahasiswa           | netral       : 15/15 real
  karya dengan hak kekayaan in | positif      : 15/15 real
  karya dengan hak kekayaan in | negatif      :  3/15 real
```

```
  karya dengan hak kekayaan in | netral       : 15/15 real
  kesesuaian bidang kerja      | positif      : 15/15 real
  kesesuaian bidang kerja      | negatif      :  2/15 real
  kesesuaian bidang kerja      | netral       : 15/15 real

Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]

  epoch 1/5:   0%|          | 0/17 [00:00<?, ?it/s]

  epoch 1: train_loss=8.6836

  epoch 2/5:   0%|          | 0/17 [00:00<?, ?it/s]

  epoch 2: train_loss=7.9509

  epoch 3/5:   0%|          | 0/17 [00:00<?, ?it/s]

  epoch 3: train_loss=7.2582

  epoch 4/5:   0%|          | 0/17 [00:00<?, ?it/s]

  epoch 4: train_loss=6.7324

  epoch 5/5:   0%|          | 0/17 [00:00<?, ?it/s]

  epoch 5: train_loss=6.3625

  Evaluating on held-out TEST set…

================================================================
  ASC_AL
================================================================

Aspect: capaian pembelajaran lulusan
  positif               P=0.882  R=0.517  F1=0.652
  negatif               P=0.500  R=0.571  F1=0.533
  netral                P=0.316  R=0.750  F1=0.444
  Macro F1 0.5433  |  Weighted F1 0.5955

Aspect: indeks prestasi kumulatif
  positif               P=0.500  R=0.286  F1=0.364
  negatif               P=0.000  R=0.000  F1=0.000
  netral                P=0.360  R=0.600  F1=0.450
  Macro F1 0.2712  |  Weighted F1 0.3786

Aspect: kelulusan tepat waktu
  positif               P=1.000  R=0.077  F1=0.143
  negatif               P=0.300  R=0.750  F1=0.429
  netral                P=0.538  R=0.700  F1=0.609
  Macro F1 0.3934  |  Weighted F1 0.4255

Aspect: masa tunggu lulusan
  positif               P=0.750  R=0.316  F1=0.444
```

```
  negatif                P=0.250  R=0.667  F1=0.364
  netral                 P=0.389  R=0.583  F1=0.467
  Macro F1 0.4249  |  Weighted F1 0.4452

Aspect: pelacakan dan perekaman data lulusan
  positif                P=0.733  R=0.524  F1=0.611
  negatif                P=0.000  R=0.000  F1=0.000
  netral                 P=0.368  R=0.583  F1=0.452
  Macro F1 0.3542  |  Weighted F1 0.5215

Aspect: prestasi mahasiswa
  positif                P=0.882  R=0.682  F1=0.769
  negatif                P=0.200  R=0.500  F1=0.286
  netral                 P=0.455  R=0.556  F1=0.500
  Macro F1 0.5183  |  Weighted F1 0.6665

Aspect: karya dengan hak kekayaan intelektual
  positif                P=0.875  R=0.700  F1=0.778
  negatif                P=0.000  R=0.000  F1=0.000
  netral                 P=0.500  R=0.875  F1=0.636
  Macro F1 0.4714  |  Weighted F1 0.6882

Aspect: kesesuaian bidang kerja
  positif                P=0.538  R=0.389  F1=0.452
  negatif                P=0.500  R=0.500  F1=0.500
  netral                 P=0.450  R=0.600  F1=0.514
  Macro F1 0.4886  |  Weighted F1 0.4812

  MEAN Macro F1: 0.4332  |  MEAN Weighted F1: 0.5253

Saved -> /content/drive/MyDrive/results/baseline/cm_ASC_AL.png

Saved -> /content/drive/MyDrive/results/baseline/training_ASC_AL.png
  Saved 44 test comments + predictions -> /content/drive/MyDrive/results/base
line/predictions_ASC_AL_weighted_baseline.xlsx

Done.  Mean macro-F1 / weighted-F1:
  ACD_AK    macro=0.7077  weighted=0.8111
  ACD_AL    macro=0.8344  weighted=0.8834
  ASC_AK    macro=0.3904  weighted=0.5168
  ASC_AL    macro=0.4332  weighted=0.5253
```

7.18 Results summary

[ ]:
```
summary = pd.DataFrame([
    {'experiment': k,
     'mean_macro_f1':   round(v['metrics']['mean_macro_f1'], 4),
     'mean_weighted_f1':round(v['metrics']['mean_weighted_f1'], 4),
```

```
     'n_fit':  v['n_fit'], 'n_test': v['n_test']}
    for k, v in ALL_RESULTS.items()
]).set_index('experiment')
print(summary.to_string())

os.makedirs(f'{BASE_PATH}/results/baseline', exist_ok=True)
summary.to_excel(f'{BASE_PATH}/results/baseline/summary_weighted_baseline.xlsx')
print('\nSaved -> {}/results/baseline/summary_weighted_baseline.xlsx'.
 ↪format(BASE_PATH))
```

```
            mean_macro_f1  mean_weighted_f1  n_fit  n_test
experiment
ACD_AK             0.7077            0.8111    119      41
ACD_AL             0.8344            0.8834    120      43
ASC_AK             0.3904            0.5168    134      42
ASC_AL             0.4332            0.5253    132      44

Saved ->
/content/drive/MyDrive/results/baseline/summary_weighted_baseline.xlsx
```

8 Tahap Few-Shot Fine-Tuning — Tesaurus (IndoBERT)

Tahap ini melatih model IndoBERT (`indobenchmark/indobert-base-p2`) untuk ACD dan ASC dengan augmentasi data menggunakan penggantian sinonim dari tesaurus domain yang dibangun pada tahap sebelumnya. Tesaurus digunakan untuk mengisi kekurangan data pada kelas aspek yang kurang terwakili melalui variasi leksikal yang tetap mempertahankan label asli. Konfigurasi ini menggunakan split data uji yang persis sama dengan tahap baseline, sehingga perbedaan performa dapat sepenuhnya diatribusikan pada sumber augmentasi. Hasil (prediksi, metrik precision/recall/F1, dan confusion matrix) disimpan untuk dibandingkan dengan konfigurasi baseline.

8.1 Install & Import

[ ]:
```
!pip install -q transformers datasets scikit-learn pandas openpyxl matplotlib␣
↪seaborn tqdm iterative-stratification
```

[ ]:
```
import os, re, random, warnings, gc
import numpy as np
import pandas as pd
from copy import deepcopy
from tqdm.auto import tqdm

import torch
import torch.nn as nn
from torch.optim import AdamW
from torch.utils.data import Dataset, DataLoader

from transformers import (
    BertTokenizer, BertModel,
    set_seed,
)
from sklearn.utils.class_weight import compute_class_weight
from sklearn.metrics import (
    precision_recall_fscore_support, f1_score,
    confusion_matrix, accuracy_score,
)
import matplotlib.pyplot as plt
import seaborn as sns

warnings.filterwarnings('ignore')
sns.set_style('whitegrid')
print('Libraries loaded.')
```

```
Libraries loaded.
```

8.2 Configuration

[ ]:
```python
MODEL_CHECKPOINT = 'indobenchmark/indobert-base-p2'

# Hyperparameters
LEARNING_RATE = 2e-5
BATCH_SIZE    = 8
DROPOUT       = 0.1
EPOCHS        = 5
WEIGHT_DECAY  = 0.3
MAX_SEQ_LEN   = 512
SEED          = 42

# Few-shot sampling & split
N_PER_CLASS = 15
TEST_FRAC   = 0.20
MIN_TEST_PER_CLASS = 2   # enforce >= this many test examples per class, per aspect
STRATIFY    = True       # multilabel-stratified split (keeps per-aspect class mix)

# Labels
ACD_LABEL_MAP   = {'relevan': 1, 'tidak relevan': 0}
ACD_LABEL_NAMES = ['tidak_relevan', 'relevan']

ASC_LABEL_MAP   = {'positif': 0, 'negatif': 1, 'netral': 2}
ASC_LABEL_NAMES = ['positif', 'negatif', 'netral']

# Aspect columns (8 fixed aspects, '(manual)' gold columns)
ASPECT_COLUMNS = [
    'capaian pembelajaran lulusan (manual)',
    'indeks prestasi kumulatif (manual)',
    'kelulusan tepat waktu (manual)',
    'masa tunggu lulusan (manual)',
    'pelacakan dan perekaman data lulusan (manual)',
    'prestasi mahasiswa (manual)',
    'karya dengan hak kekayaan intelektual (manual) ',
    'kesesuaian bidang kerja (manual)',
]
NUM_ASPECTS   = len(ASPECT_COLUMNS)
ASPECT_SHORTS = [c.replace(' (manual)', '').strip() for c in ASPECT_COLUMNS]


# Paths (Colab / Google Drive)
BASE_PATH = '/content/drive/MyDrive'

ACD_EXPERIMENTS = [
```

```python
    {'name': 'ACD_AK', 'file': 'Hasil Ground Truth_ACD AK_clean.xlsx', 'task': 'ACD', 'text_col': 'komentar_AK'},
    {'name': 'ACD_AL', 'file': 'Hasil Ground Truth_ACD AL_clean.xlsx', 'task': 'ACD', 'text_col': 'komentar_AL'},
]
ASC_EXPERIMENTS = [
    {'name': 'ASC_AK', 'file': 'Hasil Ground Truth_ASC AK_clean.xlsx', 'task': 'ASC', 'text_col': 'komentar_AK'},
    {'name': 'ASC_AL', 'file': 'Hasil Ground Truth_ASC AL_clean.xlsx', 'task': 'ASC', 'text_col': 'komentar_AL'},
]

print(f'Model        : {MODEL_CHECKPOINT}')
print(f'LR {LEARNING_RATE} | batch {BATCH_SIZE} | dropout {DROPOUT} | epochs {EPOCHS} | max_len {MAX_SEQ_LEN}')
print(f'Few-shot     : {N_PER_CLASS} per (aspect,class), THESAURUS augmentation to fill')
print(f'Split        : {int((1-TEST_FRAC)*100)}/{int(TEST_FRAC*100)}, NO validation | min {MIN_TEST_PER_CLASS} test/class/aspect')

# Folder that holds the thesaurus union files (defaults to BASE_PATH).
ACD_THESAURUS_DIR = BASE_PATH
ASC_THESAURUS_DIR = f'{BASE_PATH}/Final Thesaurus/ASC/8 aspects_filtered'
# ACD: ONE cleaned union file (covers both AK and AL). ASC: 8 per-aspect files, filtered by 'Dataset' column per stage.
ACD_THESAURUS_FILES = {
    'ACD_AK': ['ACD_union_cleaned.xlsx'],
    'ACD_AL': ['ACD_union_cleaned.xlsx'],
}
ASC_ASPECT_FILES = [
    'capaian_pembelajaran_lulusan_filtered.xlsx',
    'indeks_prestasi_kumulatif_filtered.xlsx',
    'kelulusan_tepat_waktu_filtered.xlsx',
    'masa_tunggu_lulusan_filtered.xlsx',
    'pelacakan_dan_perekaman_data_lulusan_filtered.xlsx',
    'prestasi_mahasiswa_filtered.xlsx',
    'karya_dengan_hak_kekayaan_intelektual_filtered.xlsx',
    'kesesuaian_bidang_kerja_filtered.xlsx',
]
ASC_THESAURUS_FILES = {
    'ASC_AK': ASC_ASPECT_FILES,   # same 8 files; rows filtered by Dataset == 'ASC_AK'
    'ASC_AL': ASC_ASPECT_FILES,   # same 8 files; rows filtered by Dataset == 'ASC_AL'
}
```

```
# Thesaurus term filtering (keep concise, interchangeable phrases; drop noise/
↪meta notes)
SYN_MIN_CH, SYN_MAX_CH, SYN_MAX_WORDS = 3, 70, 8
_BAD_SUBSTR = ['tidak ditemukan', 'tidak ada', 'tidak relevan', 'tidak␣
↪terdapat',
               'akan diisi', 'sinonim', 'variasi', 'mengacu', 'nan']
SPLITS_DIR = f'{BASE_PATH}/results/baseline/splits'   # written by the tahap␣
↪baseline
```

```
Model         : indobenchmark/indobert-base-p2
LR 2e-05 | batch 8 | dropout 0.1 | epochs 5 | max_len 512
Few-shot      : 15 per (aspect,class), THESAURUS augmentation to fill
Split         : 80/20, NO validation | min 2 test/class/aspect
```

8.3 Device & Reproducibility

[ ]:
```
def set_all_seeds(seed):
    random.seed(seed); np.random.seed(seed)
    torch.manual_seed(seed); torch.cuda.manual_seed_all(seed)
    set_seed(seed)

set_all_seeds(SEED)
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print('Device:', device)
if torch.cuda.is_available():
    print('  GPU:', torch.cuda.get_device_name(0))
```

```
Device: cuda
  GPU: NVIDIA L4
```

8.4 Mount Google Drive

[ ]:
```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

8.5 Tokenizer

[ ]:
```
tokenizer = BertTokenizer.from_pretrained(MODEL_CHECKPOINT)
print('Tokenizer loaded:', MODEL_CHECKPOINT, '| vocab', tokenizer.vocab_size)
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a
HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated
requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and
faster downloads.
```

```
tokenizer_config.json:   0%|          | 0.00/2.00 [00:00<?, ?B/s]
```

```
vocab.txt:   0%|          | 0.00/229k [00:00<?, ?B/s]

special_tokens_map.json:   0%|          | 0.00/112 [00:00<?, ?B/s]

Tokenizer loaded: indobenchmark/indobert-base-p2 | vocab 30521
```

8.6 Preprocessing (singkatan + typo normalization)

[ ]:
```
import re
# Singkatan + typo normalization dictionaries (provided by user).
# NOTE: typo_dict2 arrived truncated at the end ("...,\"amik\": \"amik\","} );
# the corrupted trailing fragment was dropped and the dict closed cleanly.

replace_weird_words1 = {
"bbrp":"beberapa","bhw":"bahwa","bk":"bidang_keahlian","bl":"bulan","blm":
↪"belum","bln":"bulan",
"bnsp":"badan_nasional_sertifikasi_profesi","bpc":
↪"bidang_pengembang_kompetensi","bpjs":
↪"badan_penyelenggara_jaminan_sosial","brp":"berapa",
"cmpk":"capaian_pembelajaran_mata_kuliah","cp":"capaian_pembelajaran","cs":
↪"computer_science","csd":"center_of_student_development","dcs":
↪"diploma_computer_science","dg":"dengan","dgn":"dengan","dll":
↪"dan_lain_lain","dlm":"dalam","dr":"dari","dsb":"dan_sebagainya",
"dtpr":"dosen_tetap_penghitung_rasio","dtprmhs":
↪"dosen_tetap_penghitung_rasio_mahasiswa","fcns":
↪"foresec_certified_network_security",
"fgd":"forum_group_discussion","ft":"fakultas_teknik","gbr":"gambar",
"hk":"hak_kekayaan_intelektual","hkl":"hak_kekayaan_intelektual","hlm":
↪"halaman","jl":"jumlah","jml":"jumlah","jwd":"junior_web_developer",
"kbk":"kurikulum_berbasis_kompetensi","kbm":"kegiatan_belajar_mengajar","kklp":
↪"kuliah_kerja_lapangan_pengganti","kkn":"kuliah_kerja_nyata","kp":
↪"kerja_praktek","kpt":"konsultan_pendidikan_tinggi","krn":"karena",
"ktw":"kelulusan_tepat_waktu","lck":"laporan_capaian_kinerja","lkps":
↪"laporan_kinerja_program_studi","lm":"layanan_mahasiswa","lpdp":
↪"lembaga_pengelola_dana_pendidikan","lpk":"indeks_prestasi_kumulatif",
"lppm":"lembaga_penelitian_dan_pengabdian_masyarakat","lsp":
↪"lembaga_sertifikasi_profesi","lspk":
↪"lembaga_sertifikasi_profesi_komputer","lt":"information_technology","lttq":
↪"lembaga_tahfizh_dan_tilawah_alquran",
"mbkm":"merdeka_belajar_kampus_merdeka","mhn":"mohon","mhs":"mahasiswa","mhsw":
↪"mahasiswa","mk":"mata_kuliah","mkbm":"merdeka_belajar_kampus_merdeka","ms":
↪"mahasiswa","ntb":"nusa_tenggara_barat",
"pd":"pada","pk":"prestasi_kumulatif","pkm":
↪"program_kreativitas_mahasiswa","pkmk":
↪"program_kreativitas_mahasiswi_kewirausahaan","pkmkw":
↪"program_kreativitas_mahasiswi_kewirausahaan","pkmm":
↪"program_kreativitas_mahasiswi_pengabdian_masyarakat",
```

```
"pkmpm":"program_kreativitas_mahasiswi_pengabdian_masyarakat","pl":
 ↪"pembelajaran_lanjutan","pmmb":"program_magang_mahasiswa_bersertifikat",
"pp":"pimpinan_pusat","ppm":"pusat_penjaminan_mutu","ps":"program_studi","psbb":
 ↪"pembatasan_sosial_berskala_besar",
"pspsk":"pusat_studi_pendidikan_dan_kebijakan","ptnpts":
 ↪"perguruan_tinggi_negeri_perguruan_tinggi_swasta","ptpn":
 ↪"perkebunan_tanaman_nusantara","rpl":"rekayasa_perangkat_lunak","rpm":
 ↪"rencana_pembelajaran_mingguan","rpp":"rencana_pelaksanaan_pembelajaran",
"rps":"rencana_pembelajaran_semester","sbb":"sebagai_berikut","sbg":
 ↪"sebagai","sbgmn":"sebagaimana","scr":"secara","sd":"sampai_dengan",
"sdh":"sudah","sdlc":"system_development_life_cycle","sdm":
 ↪"sumber_daya_manusia","shg":"sehingga","sk":"surat_keputusan",
"sks":"satuan_kredit_semester","sn":"standar_nasional","snpt":
 ↪"standar_nasional_pendidikan_tinggi","spp":
 ↪"surat_perjanjian_pemanfaatan","str":"sarjana_terapan","stt":
 ↪"sekolah_tinggi_teknologi","tdk":"tidak","tgl":"tanggal","th":"tahun","thn":
 ↪"tahun",
"thd":"terhadap","tkj":"teknik_komputer_dan_jaringan","tkk":
 ↪"teknik_komputer_kontrol","tkt":"tingkat","trk":
 ↪"teknologi_rekayasa_komputer",
"trkj":"teknologi_rekayasa_komputer","ts":"tracer_study","tsb":
 ↪"tersebut","tsbt":"tersebut","ttg":"tentang",
"vm":"visi_misi","vmts":"visi_misi_tujuan_dan_strategi","wt":
 ↪"waktu_tunggu","prodi":"program_studi","ik":"ilmu_komputer","al":
 ↪"asesmen_lapangan","ak":"asesmen_kecukupan","rtm":"papat_tinjauan_manajemen",
"cdc":"career_development_center","spm":"sistem_penjaminan_mutu",
"phb":"program_hibah_binaan","pt":"perguruan_tinggi",
"dtps":"dosen_tetap_program_studi","lrs":"laporan_rencana_studi","lpks":
 ↪"lembaga_pengembangan_keterampilan_dan_sertifikasi",
"smk":"sekolah_menengah_kejuruan","dst":"dan_seterusnya","smp":"sampai","lpm":
 ↪"lembaga_penjaminan_mutu",
"std":"standar","bpm":"badan_penjaminan_mutu","bhs":"bahasa",
"kkmk":"kriteria_kinerja_manajemen_kualitas","pkl":
 ↪"rpaktek_kerja_lapangan","psms":"program_studi_manajemen_sistem",
"kk":"kode_klasifikasi","rtl":"rencana_tindak_lanjut","smt":"semester",
"pbl":"problem_based_learning","rpsm":"rencana_pembelajaran_semester",
"lp":"lembaga_penjaminan","lpd":"laporan_penelitian_dosen","pppm":
 ↪"pengembangan_pendidikan_mahasiswa",
"bppm":"badan_penelitian_pengabdian_masyarakat","lpppm":
 ↪"lembaga_penelitian_pengabdian_masyarakat","rpjp":
 ↪"rencana_pembangunan_jangka_panjang",
"bpp":"biaya_penyelenggaraan_pendidikan","st":"sarjana_teknik","sp":
 ↪"surat_peringatan",
```

```
"dpp":"dosen_pendamping_program","upps":"unit_pengelola_program_studi","pts":
 ↪"perguruan_tinggi_swasta","ptn":"perguruan_tinggi_negeri","pr":
 ↪"penelitian_riset","ttd":"tanda_tangan","pllm":
 ↪"lembaga_penelitian_dan_pengabdian_masyarakat",
"rp":"rencana_pembelajaran","drpm":"direktorat_riset_pengabdian_masyarakat",
"bph":"badan_pengurus_harian","ttp":"tetap","llppm":
 ↪"lembaga_penelitian_pengabdian_masyarakat",
"cfp":"certified_financial_planner","lppmbppm":
 ↪"lembaga_penelitian_pengabdian_masyarakat_badan_penjaminan_mutu_pendidikan",
"spk":"surat_perjanjian_kerja","pm":"penjaminan_mutu","dptr":
 ↪"dosen_tetap_penghitung_rasio","kpd":"kepada","dtpt":
 ↪"dosen_tetap_perguruan_tinggi","bgm":"bagaimana",
"dtp":"dosen_tetap_program","kts":"kurikulum_tingkat_satuan",
"jg":"juga","dgx":"dengan","rpjmn":
 ↪"rencana_pembangunan_jangka_menengah_nasional","ppkm":
 ↪"program_pengembangan_keterampilan_mahasiswa",
"dprpm":"direktorat_penelitian_pengabdian_masyarakat","bg":"bagaimana","spmpt":
 ↪"sistem_penjaminan_mutu_pendidikan_tinggi","pnbp":
 ↪"penerimaan_negara_bukan_pajak",
"wkt":"waktu","csr":"corporate_social_responsibility",
"kkp":"kualitas_kinerja_program","pkmdtpr":
 ↪"program_kreativitas_mahasiswa_dosen_tetap_penghitung_rasio",
"lppp":"lembaga_penelitian_pengabdian_masyarakat",
"dtr":"dosen_tetap_penghitung_rasio","ln":"luar_negeri","tmps":
 ↪"tim_penjaminan_mutu_program_studi","tpmj":"tim_penjaminan_mutu_jurusan",
"pmk":"penjaminan_mutu_kinerja","bkn":"bukan","pkms":
 ↪"program_kreativitas_mahasiswa_sosial","skripsita":"skripsi_tugas_akhir",
"pstp":"program_studi_teknik_pertambangan","ppkmdm":
 ↪"program_pengembangan_keterampilan_mahasiswa_dan_dosen",
"plp":"program_latihan_profesi","skl":"surat_keterangan_lulus",
"cplcpmk":
 ↪"capaian_pembelajaran_lulusan_dan_capaian_pembelajaran_mata_kuliah","cpmk":
 ↪"capaian_pembelajaran_mata_kuliah","rkps":
 ↪"rencana_kegiatan_pembelajaran_semester",
"sbgmn":"sebagaimana","jmlh":"jumlah","sms":"semester","lplppm":
 ↪"lembaga_penelitian_pengabdian_masyarakat","pstg":
 ↪"program_studi_teknik_geologi","pstk":"program_studi_teknik_kimia",
"skp":"surat_kerja_praktek","pblpbm":
 ↪"problem_based_learning_project_based_learning","prn":
 ↪"program_riset_nasional","ppg":"program_profesi_guru","psbm":
 ↪"program_studi_bisnis_dan_manajemen",
"lpj":"laporan_pertanggungjawaban","kkt":"kualitas_kinerja_tim",
"pjj":"pendidikan_jarak_jauh","pmp":"pemetaan_mutu_pendidikan","bpmp":
 ↪"balai_penjaminan_mutu_pendidikan","bppmp":
 ↪"balai_penjaminan_mutu_pendidikan","bdp":"balai_guru_penggerak",
```

```
"bkd":"beban_kerja_dosen","bkk":"bursa_kerja_khusus","bmw":
 ↪"belajar_melanjut_wirausaha","bnpb":
 ↪"badan_nasional_penanggulangan_bencana","bnpp":
 ↪"badan_nasional_pengelola_perbatasan","bnpt":
 ↪"badan_nasional_penanggulangan_terorisme",
"bp":"bimbingan_penyuluhan","bpk":"badan_pengurus_keuangan","bpl":
 ↪"badang_pengelola_latihan","bssn":"badan_siber_dan_sandi_negara",
"cpns":"calon_pegawai_negeri_sipil","crf":
 ↪"center_for_research_and_facilitation","crm":
 ↪"sistem_manajemen_hubungan_pelanggan","csrpkbl":
 ↪"corporate_social_responsibility_program_kemitraan_bina_lingkungan","pkbl":
 ↪"program_kemitraan_bina_lingkungan",
"ct":"computational_thinking","ctf":"career_and_technology_foundations","cv":
 ↪"curriculum_vitae",
"ddtpr":"dosen_tetap_penghitung_rasio","dkk":"dan_kawan_kawan","dn":"dan","dng":
 ↪"dengan","dpkm":"direktorat_pengabdian_masyarakat","dpl":
 ↪"dosen_pembimbing_lapangan","dpm":"dewan_perwakilan_mahasiswa","dppm":
 ↪"direktorat_penelitian_pengabdian_masyarakat",
"dpr":"laporan_kemajuan_gelar","dpt":"daftar_pemilih_tetap",
"hki":"hak_kekayaan_intelektual","hku":"hak_kekayaan_usaha",
"iku":"indikator_kinerja_utama","ikt":"indikator_kinerja_tambahan","ikuikt":
 ↪"indikator_kinerja_utama_indikator_kinerja_tambahan","cbr":
 ↪"community_based_research","cbt":"computer_based_test","cl":
 ↪"collaborative_learning",
}

replace_weird_words2 = {
"tpg":"tunjangan_profesi_guru","dtprmhs":
 ↪"dosen_tetap_penghitung_rasio_mahasiswa","mhs":"mahasiswa",
"upps":"unit_pengelola_program_studi","spmi":
 ↪"sistem_penjaminan_mutu_internal","renstra":"rencana_strategis",
"menemu":"menemukan","isbn":"international_standard_book_number","prosentase":
 ↪"presentase","yg":"yang","haki":"hak_atas_kekayaan_intelektual",
"led":"laporan_evaluasi_diri","danatau":"dan_atau","peneltian":
 ↪"penelitian","utk":"untuk","ta":"tugas_akhir","dharma":"tridharma","amik":
 ↪"akademi_manajemen_informatika_dan_komputer",
"pgri":"persatuan_guru_republik_indonesia","umkm":
 ↪"usaha_mikro_kecil_menengah","abdimas":"pengabdian_masyarakat","uppsps":
 ↪"unit_pengelola_program_studi",
"berdasar":"berdasarkan","ti":"teknologi_informasi","dtprmahasiswa":
 ↪"dosen_tetap_penghitung_rasio_mahasiswa","ppepp":
 ↪"penetapan_pelaksanaan_evaluasi_pengendalian_dan_peningkatan",
"si":"sitem_informasi","kaprodi":"kepala_program_studi","dipa":
 ↪"daftar_isian_pelaksanaan_anggaran","itb":"institut_teknologi_bandung",
```

```
"kebijakanstandar":"kebijakan_standar","renop":
↪"rencana_operasional","lemabagaunit":"lembaga_unit","akakom":
↪"akademi_aplikasi_komputer",
"permendikbud":"peraturan_menteri_pendidikan",
"standard":"standar","simlitabmas":
↪"sistem_informasi_penelitian_dan_pengabdian_masyarakat","pob":
↪"pedoman_operasional_baku",
"penelitan":"penelitian","fti":"fakultas_teknologi_industri","puslit":
↪"pusat_penelitian","ri":"republik_indonesia","dosenmahasiswa":
↪"dosen_mahasiswa",
"kkni":"kerangka_kualifikasi_nasional_indonesia",
"kkn":"kuliah_kerja_nyata","pkmgft":
↪"program_kreativitas_mahasiswa_gagasan_futuristik_tertulis","pkmgfk":
↪"program_kreativitas_mahasiswa_gagasan_futuristik_konstruktif",
}

replace_weird_words3 = {
"prodi":"program_studi","ik":"ilmu_komputer","ps":"program_studi","al":
↪"asesmen_lapangan","ak":"asesmen_kecukupan","pkm":
↪"program_kreativitas_mahasiswa","rtm":"papat_tinjauan_manajemen",
"ppm":"pengabdian_dan_pemberdayaan_masyarakat","cp":"capaian_pembelajaran",
"lkps":"laporan_kinerja_program_studi",
"lrs":"laporan_rencana_studi","kp":"kerja_praktek","lpks":
↪"lembaga_pengembangan_keterampilan_dan_sertifikasi",
"smk":"sekolah_menengah_kejuruan","dst":"dan_seterusnya","smp":"sampai","pk":
↪"penilaian_kinerja","lpm":"lembaga_penjaminan_mutu",
"kkmk":"kriteria_kinerja_manajemen_kualitas","pkl":
↪"rpaktek_kerja_lapangan","psms":"program_studi_manajemen_sistem",
"kk":"kode_klasifikasi","tkk":"tenaga_kependidikan_khusus",
"vmts":"visi_misi_tujuan_dan_sasaran","kbk":
↪"kurikulum_berbasis_kompetensi","pbl":"problem_based_learning","rpsm":
↪"rencana_pembelajaran_semester","ft":"fakultas_teknik","ttg":"tentang",
"snpt":"standar_nasional_perguruan_tinggi","lp":"lembaga_penjaminan","lpd":
↪"laporan_penelitian_dosen","pppm":"pengembangan_pendidikan_mahasiswa",
"rpjp":"rencana_pembangunan_jangka_panjang",
"st":"sarjana_teknik","sp":"surat_peringatan",
"dpp":"dosen_pendamping_program","pr":"penelitian_riset","ttd":
↪"tanda_tangan","pllm":"lembaga_penelitian_dan_pengabdian_masyarakat",
"rp":"rencana_pembelajaran","drpm":"direktorat_riset_pengabdian_masyarakat",
"ttp":"tetap","llppm":"lembaga_penelitian_pengabdian_masyarakat",
"cfp":"certified_financial_planner","lppmbppm":
↪"lembaga_penelitian_pengabdian_masyarakat_badan_penjaminan_mutu_pendidikan",
"spk":"surat_perjanjian_kerja","pm":"penjaminan_mutu","dptr":
↪"dosen_tetap_penghitung_rasio","kpd":"kepada","dtpt":
↪"dosen_tetap_perguruan_tinggi","bgm":"bagaimana",
"dtp":"dosen_tetap_program","kts":"kurikulum_tingkat_satuan",
```

```
"jg":"juga","dgx":"dengan","rpjmn":
 ↪"rencana_pembangunan_jangka_menengah_nasional","ppkm":
 ↪"program_pengembangan_keterampilan_mahasiswa",
"dprpm":"direktorat_penelitian_pengabdian_masyarakat","bg":"bagaimana","spmpt":
 ↪"sistem_penjaminan_mutu_pendidikan_tinggi","pnbp":
 ↪"penerimaan_negara_bukan_pajak",
"wkt":"waktu","csr":"corporate_social_responsibility","fgd":
 ↪"focus_group_discussion",
"kkp":"kualitas_kinerja_program","pkmdtpr":
 ↪"program_kreativitas_mahasiswa_dosen_tetap_penghitung_rasio",
"lppp":"lembaga_penelitian_pengabdian_masyarakat","rpp":
 ↪"rencana_pembelajaran_program",
"dtr":"dosen_tetap_penghitung_rasio","ln":"luar_negeri","tmps":
 ↪"tim_penjaminan_mutu_program_studi","tpmj":"tim_penjaminan_mutu_jurusan",
"pmk":"penjaminan_mutu_kinerja","bkn":"bukan","pkms":
 ↪"program_kreativitas_mahasiswa_sosial",
"pstp":"program_studi_teknik_pertambangan","ppkmdm":
 ↪"program_pengembangan_keterampilan_mahasiswa_dan_dosen",
"plp":"program_latihan_profesi","bnsp":
 ↪"badan_nasional_sertifikasi_profesi","skl":"surat_keterangan_lulus",
"cplcpmk":
 ↪"capaian_pembelajaran_lulusan_dan_capaian_pembelajaran_mata_kuliah","cpmk":
 ↪"capaian_pembelajaran_mata_kuliah","rkps":
 ↪"rencana_kegiatan_pembelajaran_semester","mhsw":"mahasiswa",
"jl":"jalan","sbgmn":"sebagaimana","jmlh":"jumlah","sms":"semester","lplppm":
 ↪"lembaga_penelitian_pengabdian_masyarakat","pstg":
 ↪"program_studi_teknik_geologi","pstk":"program_studi_teknik_kimia",
"kpt":"kualitas_pendidikan_tinggi","skp":"surat_kerja_praktek","rata2":
 ↪"rata_rata","mk":"mata_kuliah","pblpbm":
 ↪"problem_based_learning_project_based_learning","prn":
 ↪"program_riset_nasional","ppg":"program_profesi_guru","psbm":
 ↪"program_studi_bisnis_dan_manajemen",
"lpj":"laporan_pertanggungjawaban","dpts":"dosen_tetap_program_studi","kkt":
 ↪"kualitas_kinerja_tim",
"pjj":"pendidikan_jarak_jauh","pmp":"pemetaan_mutu_pendidikan","ts1":
 ↪"tracer_study_1","ts2":"tracer_study_2","ts3":"tracer_study_3","ts4":
 ↪"tracer_study_4","ts5":"tracer_study_5","ts6":"tracer_study_6",
}

typo_dict1 = {
"undri":"undur_diri","detil":"detail","luarancapaian":
 ↪"luaran_capaian","sebasar":"sebesar","komptetensi":"kompetensi",
"luluipk":"lulus_indeks_prestasi_kumulatif","perkulihan":"perkuliahan","dirjen":
 ↪"direktorat_jenderal","belmawa":"pembelajaran_dan_kemahasiswaan",
"systemaplikasi":"system_aplikasi","deskripsidatanarasi":
 ↪"deskripsi_data_narasi","informasidata":"informasi_data",
```

```
"egovernment":"government","ipkprestasi":
 ↪"indeks_prestasi_kumulatif_prestasi","perkaman":"perekaman","ydapat":"dapat",
"mahasiswat":"mahasiswa","lululusan":"lulusan","senasif":
 ↪"seminar_nasional_informasi",
"prestai":"prestasi","standart":"standar","kessusuaian":"kesesuaian","kwerja":
 ↪"kerja","mendaoat":"mendapat",
"maksimun":"maksimum","berbading":"berbanding","kebuman":
 ↪"kebumen","diperolehlah":"diperoleh","tigatahunan":"tiga_tahunan",
"namum":"namun","detailriil":"detail_riil","dtprdosen":
 ↪"dosen_tetap_penghitung_rasio_dosen","klarfikiasi":"klarifikasi",
"kepuasanan":"kepuasan","fluaktif":"fluktuatif","ketidakkonsitenan":
 ↪"ketidakkonsistenan","simkatmawa":"sistem_pemeringkatan_kemahasiswaan",
"ekplisit":"eksplisit","terlenuhi":"terpenuhi","diihat":
 ↪"dilihat","pujiansebesar":"pujian_sebesar","pddikti":
 ↪"pangkalan_data_pendidikan_tinggi",
"atunggu":"tunggu","masy":"masyarakat","ditumkan":"ditemukan","reseponden":
 ↪"responden","kesesuaiaan":"kesesuaian",
"denagn":"dengan","berturutan":"berurutan","mnempuh":"menempuh","emenuhan":
 ↪"pemenuhan","mahaswasiswa":"mahasiswa",
"melampui":"melampaui","dilaporakan":"dilaporkan","laporaan":"laporan","elama":
 ↪"lama","mskor":"skor","erlacak":"terlacak",
"lulusanlulusan":"lulusan_lulusan","daapat":"dapat","reepositori":
 ↪"repositori","bidanginfokom":"bidang_infokom",
"jumlh":"jumlah","menyarakan":"menyarankan","polinas":
 ↪"politeknik_informatika_nasional","inggirs":"inggris",
"datajumlah":"data_jumlah","aktf":"aktif","kementrian":
 ↪"kementerian","eratcukup":"erat_cukup","pprodi":"program_studi",
"sertiikat":"sertifikat","terakahir":"terakhir","berkerjaberwirausaha":
 ↪"bekerja_berwirausaha",
"lokalwilayahberwirausaha":"lokal_wilayah_berusaha","depkominfo":
 ↪"departemen_komunikasi_dan_informatika",
"linktracer":"link_tracer","dtidak":"tidak","berhuubng":"berhubung","gkerja":
 ↪"kerja","subcpmk":"sub_capaian_pembelajaran_mata_kuliah",
"saaat":"saat","danstr":"dan_sarjana_terapan","hkiteknologi":
 ↪"hak_kekayaan_intelektual_teknologi",
"gunaprodukbuku":"guna_produk_buku","kurikulumlink":
 ↪"kurikulum_link","ketersedian":"ketersediaan",
"sikappengetahuanketerampilan":"sikap_pengetahuan_keterampilan","penilainnya":
 ↪"penilaiannya","lulusantabel":"lulusan_tabel",
"kerampilan":"keterampilan","tugaskuis":"tugas_kuis","pda":"pada","ket":
 ↪"keterangan","healthkathon":"health_kathon",
"defenisi":"definisi","perekeman":"perekaman","oprestasi":"prestasi","jkarya":
 ↪"karya","academyc":"academic",
"dibandikan":"dibandingkan","telahbelum":"telah_belum","sem":"semester","kkni":
 ↪"kerangka_kualifikasi_nasional_indonesia",
```

```
"diharapka":"diharapkan","penuruanan":"penurunan","renatan":"rentan","anatar":"antar","dilaksanakantarcer":"dilaksanakan_tracer",
"dasboard":"dashboard","dalammkurun":"dalam_kurun","kelulusah":"kelulusan","dtingkat":"tingkat","sekprodi":"sekretaris_program_studi",
"sertau":"serta","unipa":"universitas_papua","ppemenuhan":"pemenuhan","ditelusur":"ditelusuri",
"sebelumnyasetiap":"sebelumnya_setiap","diberikansedangkan":"diberikan_sedangkan","bbrapa":"beberapa",
"mahasiswabelum":"mahasiswa_belum","rekruitmen":"rekrutmen","hakidibawah":"hak_kekayaan_intelektual_di_bawah",
"menghasillkan":"menghasilkan","disosialisasiakn":"disosialisasikan","nasionalpada":"nasional_pada","infromatika":"informatika",
"ratrata":"rata_rata","dtpsmahasiswa":"dosen_tetap_program_studi_mahasiswa","tahunberdasarkan":"tahun_berdasarkan",
"eventevent":"event_event","erdasarkan":"berdasarkan","nyapelaksanaan":"nya_pelaksanaan","temanteman":"teman_teman",
"studyunikama":"study_unikama","skore":"skor","menjacapi":"mencapai","acehsumatera":"aceh_sumatera",
"internasiona":"internasional","tercpai":"tercapai","renstra":"rencana_strategis",
"renstrarenoprkat":"rencana_strategis_rencana_operasional_rencana_kerja_dan_anggaran_tahunan",
"kebermanfaatanaspek":"kebermanfaatan_aspek","tunggunnya":"tunggunya","silahkan":"silakan","oresec":"foresec",
"mutinasionalinternasional":"multinasional_internasional","tercapaikesesuain":"tercapai_kesesuaian","pelacakanperekaman":"pelacakan_perekaman","matakuliah":"mata_kuliah",
"stimik":"sekolah_tinggi_ilmu_manajemen_informatika_dan_komputer","sepnop":"sepuluh_nopember","ikuikt":"indikator_kinerja_utama_indikator_kinerja_tambahan",
"pelbelajaran":"pembelajaran","dijelaskantetapi":"dijelaskan_tetapi","tingat":"tingkat","katagari":"kategori",
"keberadaanya":"keberadaannya","adalahdiperoleh":"adalah_diperoleh","pertaa":"pertama","dikatagorikan":"dikatagorikan",
"utuk":"untuk","dsetkan":"disertakan","standeriku":"standar_indikator_kinerja_utama","prosentase":"persentase",
"nasionalsedangkan":"nasional_sedangkan","tekait":"terkait","crosscek":"cross_check","kiurang":"kurang",
"kompetisilomba":"kompetisi_lomba","acc":"accept","diuukur":"diukur","keterlaksannan":"keterlaksanaan","kemabali":"kembali",
"kesesuaiakn":"kesesuaian","psnya":"program_studi","utaraperingkat":"utara_peringkat","gbkperingkat":"gbk_peringkat","bahw":"bahwa","rerarta":"rerata",
```

```
"mendeskripsi":"mendeskripsikan","padalink":"pada_link","tercapaian":
 ↪"ketercapaian","pa":"pembimbing_akademik",
"regionallokal":"regional_lokal","intektual":"intelektual","pdki":
 ↪"pangkalan_data_kekayaan_intelektual","butirbutir":"butir_butir",
"penejelasan":"penjelasan","ilkom":"ilmu_komputer","padangsidimpuan":
 ↪"padang_sidimpuan","didata":"di_data","kehlian":"keahlian","lulusn":
 ↪"lulusan",
"finalisdan":"finalis_dan","usni":
 ↪"universitas_satya_negara_indonesia","ekternal":"eksternal","khususkarena":
 ↪"khusus_karena","runnerup":"runner_up",
"edupreuner":"edupreneur","sebagainyatidak":"sebagainya_tidak","led":
 ↪"laporan_evaluasi_diri","ledlkps":
 ↪"laporan_evaluasi_diri_laporan_kinerja_program_studi",
"meghasilkan":"menghasilkan","pemenuhana":"pemenuhan","sehingg":
 ↪"sehingga","kekesahihan":"kesahihan","skalan":"skala","ditetapkansemua":
 ↪"ditetapkan_semua",
"ckelulusan":"kelulusan","sebanyal":"sebanyak","drivecek":
 ↪"drive_cek","lolakwilayah":"lolak_wilayah","adakelulusan":
 ↪"ada_kelulusan","proditeknologi":"program_studi_teknologi",
"terpehuni":"terpenuhi","terpenhi":"terpenuhi","mahassiwa":"mahasiswa","tungg":
 ↪"tunggu","lokalnasionalinternasional":
 ↪"lokal_nasional_internasional","piagamsertifikat":"piagam_sertifikat",
"kelulusa":"kelulusan","dokumem":"dokumen","ratarataipk":
 ↪"rata_rata_indeks_prestasi_kumulatif","berhki":
 ↪"yang_memiliki_hak_kekayaan_intelektual","upi":
 ↪"universitas_pendidikan_indonesia",
"univ":"universitas","akhiir":"akhir","capaiantabel":"capaian_tabel","peilaian":
 ↪"penilaian","pekerjaanketerlaksanaan":
 ↪"pekerjaan_keterlaksanaan","juarapesertapanitia":"juara_peserta_panitia",
"aklademiknon":"akademik_non","kaademik":"akademik","keseuain":
 ↪"kesesuaian","kalrfikasi":"klarifikasi","kkniobeskkni":
 ↪"kerangka_kualifikasi_nasional_indonesia_outcome_based_education_standar_kompetensi_kerja_na
"samapai":"sampai","kesesuaikan":"kesesuaian","tautantautan":
 ↪"tautan_tautan","engisi":"mengisi","akreditas":"akreditasi","pembelajran":
 ↪"pembelajaran","dikur":"diukur","presntasi":"presentasi","unuk":
 ↪"untuk","sepeti":"seperti",
"pelaksaan":"pelaksanaan","pendidika":"pendidikan","thun":"tahun","lylusan":
 ↪"lulusan","keterlaksanaanya":"keterlaksanaannya",
"pihakpihak":"pihak_pihak","unitama":
 ↪"universitas_teknologi_akba_makassar","memilik":"memiliki","unsika":
 ↪"universitas_singaperbangsa_karawang",
"darisama":"dari_sama","terkahir":"terakhir","studyratarata":
 ↪"study_rata_rata","almuni":"alumni","angakatan":
 ↪"angkatan","terlacakterdapat":"terlacak_terdapat",
"aspekapa":"aspek_apa","pertanyaaan":"pertanyaan","infokomps":
 ↪"infokom_program_studi","mengujur":"mengukur","miyaitu":"mi_yaitu",
```

```
"dilaksananakan":"dilaksanakan","tuingkat":"tingkat","nasonal":
 ↪"nasional","presteasi":"prestasi",
"tetang":"tentang","terjad":"terjadi","pemurunan":"penurunan","menadi":
 ↪"menjadi","peremkaman":"perekaman","rendahkurang":"rendah_kurang","perekama":
 ↪"perekaman","llusan":"lulusan","lulusanpada":"lulusan_pada","siginifikan":
 ↪"signifikan","karenkan":"karenakan","unesa":
 ↪"universitas_negeri_surabaya","perkerjaan":"pekerjaan","pemutahiran":
 ↪"pemutakhiran",
"dilakukankonfirmasi":"dilakukan_konfirmasi","mahsa":"masa","rataata":
 ↪"rata_rata",
"pelaksnaan":"pelaksanaan","tidaada":"tidak_ada","askpek":"aspek","tentan":
 ↪"tentang","kemanpuan":"kemampuan","bulanadapun":"bulan_adapun","ulusan":
 ↪"lulusan","infokomkepuasan":"infokom_kepuasan","wilayahnasional":
 ↪"wilayah_nasional",
"nformatika":"informatika","daengan":"dengan","dilaksakan":
 ↪"dilaksanakan","kuisoner":"kuesioner","adala":"adalah","masayang":
 ↪"masa_yang","pemerintahdalam":"pemerintah_dalam","amsa":"masa","tudi":
 ↪"studi","mendpatkan":"mendapatkan","lkti":
 ↪"lomba_karya_tulis_ilmiah","psychoferia":"psychofrenia","sigra":
 ↪"sistem_terintegrasi","kalrifikasin":"klarifikasi","terupsat":
 ↪"terpusat","diaksesbroken":"diakses_broken","keilmuankeahlian":
 ↪"keilmuan_keahlian",
"dilingkungan":"di_lingkungan","dimatakuliah":"di_mata_kuliah","tracertubacid":
 ↪"tracer_ub_ac_id","rataarat":"rata_rata",
"dilakukanperlu":"dilakukan_perlu","emndapat":"mendapat","mendapar":
 ↪"mendapat","tpi":"tapi","diidentifkasi":"diidentifikasi","pad":"pada","oba":
 ↪"outcome_based_assessment","acmis":
 ↪"academic_management_information_system","almni":"alumni","wirausaa":
 ↪"wirausaha","mekanime":"mekanisme",
"pemenuhancapaianpembelajaranlulusancplratarataipkprestasi":
 ↪"pemenuhan_capaian_pembelajaran_lulusan_rata_rata_indeks_prestasi_kumulatif_prestasi","reke
 ↪"rekayasa","komputerbekerja":"komputer_bekerja","akademiknonakademik":
 ↪"akademik_non_akademik","upp":"unit_pengelola_program_studi",
"keterlaksaan":"keterlaksanaan","idikator":"indikator","penetatpan":
 ↪"penetapan","disajika":"disajikan","memperolah":"memperoleh","nadhlatul":
 ↪"nahdlatul","tachnopreneur":"technopreneur","shasil":"hasil","waktuvisual":
 ↪"waktu","bersma":"bersama","isuisu":"isu_isu","semeter":
 ↪"semester","sedangkah":"sedangkan","peenuhan":"pemenuhan",
"careersitsacidtracer":"careersits_ac_id_tracer","standariku":
 ↪"standar_indikator_kinerja_utama","pelakanaan":"pelaksanaan","taraget":
 ↪"target","melaksankan":"melaksanakan","salama":"selama","kesesuaianya":
 ↪"kesesuaiannya","kesesusian":"kesesuaian",
"icts":
 ↪"international_conference_on_information_and_communication_technology_and_systems","sharepe
 ↪"sharepedia_id",
```

```
"kompetisikompetisi":"kompetisi_kompetisi","wa":"whatsapp","ebanyak":
 ↪"sebanyak","tsnya":"tracer_study_nya",
"lipihki":
 ↪"lembaga_inovasi_penulisan_ilmiah_dan_hak_kekayaan_intelektual","standarikuikt":
 ↪"standar_indikator_kinerja_utama_indikator_kinerja_tambahan","negaranegara":
 ↪"negara_negara","bidangbidang":"bidang_bidang","teknologiteknologi":
 ↪"teknologi_teknologi","kegiatankegiatan":"kegiatan_kegiatan",
}

typo_dict2 = {
"di konfirmasi":"dikonfirmasi","di validasi":"divalidasi","ter indentifikasi":
 ↪"terindentifikasi","ter sematkan":"tersematkan","pdik":
 ↪"program_studi_pendidikan","ilmu iahnya":"ilmiahnya","psmtif":
 ↪"program_studi_magister_teknik_informatika","perdosentahun":
 ↪"per_dosen_tahun",
"lacak":"lacakan","di tandai":"ditandai","ter pusat":"terpusat","per baikan":
 ↪"perbaikan","ber kualitas":"berkualitas","mkom":"magister_komputer","fmipa":
 ↪"fakultas_matematika_dan_ilmu_pengetahuan_alam","intelligence":
 ↪"kecerdasan","puslit":"pusat_penelitian","unmuh":
 ↪"universitas_muhammadiyah","pbi":
 ↪"pendidikan_bahasa_inggris","hilirisasipenerapan":
 ↪"hilirisasi_penerapan","matakuliah":"mata_kuliah",
"unsiq":"universitas_sains_al_quran","uppsprodi":
 ↪"unit_pengelola_program_studi","keseuaian":"kesesuaian","kebijakankebijakan":
 ↪"kebijakan_kebijakan","indicator":"indikator","procedure":
 ↪"prosedur","programprogram":"program_program","ketrampilan":
 ↪"keterampilan","pengmas":"pengabdian_masyarakat",
"ugn":"universitas_graha_nusantara","mtelah":"telah","kepahahaman":
 ↪"kepahaman","meningkatkanmengoptimalisasi":
 ↪"meningkatkan_mengoptimalisasi","noninfokom":"non_infokom","bindonesia":
 ↪"bahasa_indonesia",
"mi":"manajemen_informatika","renop":"rencana_operasi","sndikti":
 ↪"standar_nasional_pendidikan_tinggi",
}

DICTS = [replace_weird_words1, replace_weird_words2, replace_weird_words3,␣
 ↪typo_dict1, typo_dict2]


# Merge dicts, drop empty / self-mapping entries
DICTS = [replace_weird_words1, replace_weird_words2, replace_weird_words3,␣
 ↪typo_dict1, typo_dict2]
NORM_MAP = {}
for _d in DICTS:
    for _k, _v in _d.items():
        _k = _k.strip().lower(); _v = _v.strip()
```

```python
        if _k and _v and _k != _v:
            NORM_MAP[_k] = _v

# Longest keys first so multi-word phrases match before their parts.
_NORM_KEYS = sorted(NORM_MAP.keys(), key=len, reverse=True)
_NORM_RE = re.compile(r'\b(' + '|'.join(re.escape(k) for k in _NORM_KEYS) +␣
↪r')\b', re.IGNORECASE)
print(f'Normalization map: {len(NORM_MAP)} entries ({sum(1 for k in NORM_MAP if␣
↪" " in k)} multi-word).')

def normalize_singkatan_typo(text):
    return _NORM_RE.sub(lambda m: NORM_MAP[m.group(0).lower()], text)

def clean_text(text):
    if pd.isna(text) or text == '':
        return ''
    text = str(text)
    text = re.sub(r'http\S+|www\S+|https\S+', '', text, flags=re.MULTILINE)
    text = re.sub(r'@\w+', '', text)
    text = _emoji.sub('', text)
    text = ' '.join(text.split())
    return text.strip()

def preprocess_pipeline(text):
    """clean -> lowercase -> singkatan/typo normalization (the provided␣
↪dictionaries)."""
    if pd.isna(text) or text == '':
        return ''
    return normalize_singkatan_typo(clean_text(str(text)).lower()).strip()

# quick check
_ex = 'IPK mhs prodi TI sdh sesuai dgn standart, namum detil ts belum lengkap.'
print('Example:')
print('  in :', _ex)
print('  out:', preprocess_pipeline(_ex))
```

```
Normalization map: 701 entries (9 multi-word).
Example:
  in : IPK mhs prodi TI sdh sesuai dgn standart, namum detil ts belum lengkap.
  out: ipk mahasiswa program_studi teknologi_informasi sudah sesuai dengan
standar, namun detail tracer_study belum lengkap.
```

8.7 Load the baseline's saved splits

[ ]:
```
# Load the EXACT train/test split the baseline saved to disk.
def load_split(name, task, which):
    path = f'{SPLITS_DIR}/{name}_{which}.xlsx'        # e.g. .../ACD_AK_train.
↪xlsx
    df = pd.read_excel(path)                          # read that saved split␣
↪back in
    for col in ASPECT_COLUMNS:
        df[col] = pd.to_numeric(df[col], errors='coerce')  # ints for ACD,␣
↪NaN-able floats for ASC
        if task == 'ACD':
            df[col] = df[col].astype(int)             # ACD labels are strict 0/
↪1
    print(f'  loaded {which}: {path}  ({len(df)} rows)')
    return df.reset_index(drop=True)
```

8.8 Thesaurus + augmentation + few-shot fill

[ ]:
```
import re as _re
from collections import defaultdict

def _tnorm(v): return ' '.join(str(v).strip().split()).lower()   # normalise␣
↪text: trim + collapse spaces + lowercase

def _good_term(v):
    n = _tnorm(v)
    if not n or len(n) < SYN_MIN_CH or len(n) > SYN_MAX_CH: return False   #␣
↪too short / too long
    if len(n.split()) > SYN_MAX_WORDS: return False                        #␣
↪too many words
    if any(ch in str(v) for ch in '[]*#'): return False                   #␣
↪contains markup junk
    if any(b in n for b in _BAD_SUBSTR): return False                     #␣
↪contains a banned substring
    return True

_CMAP = {'positif': 0, 'negatif': 1, 'netral': 2}   # sentiment word -> class␣
↪number

# Build the ACD (synonym) thesaurus: for each aspect, the list of synonym␣
↪phrases from the Excel files.
def build_acd_thesaurus(files):
    """{aspect_short: [synonym phrases]} from Sinonim/Padanan grouped by␣
↪Istilah Aspek."""
```

```python
    thes = {s: [] for s in ASPECT_SHORTS}; seen = {s: set() for s in ASPECT_SHORTS}
    for fn in files:                                            # each validated thesaurus file
        df = pd.read_excel(f'{ACD_THESAURUS_DIR}/{fn}')
        for _, r in df.iterrows():                              # each synonym row
            asp = _tnorm(r.get('Istilah Aspek'))                # which aspect this row belongs to
            if asp not in thes: continue                        # skip aspects we don't track
            for v in [r.get('Sinonim/Padanan'), r.get('Istilah Aspek')]:  # the synonym + the term itself
                if _good_term(v) and _tnorm(v) not in seen[asp]:
                    seen[asp].add(_tnorm(v)); thes[asp].append(str(v).strip()) # add if new + valid
    thes = {k: sorted(v, key=len, reverse=True) for k, v in thes.items()}   # longest first (match greedily later)
    print('  ACD thesaurus:', {k[:18]: len(v) for k, v in thes.items()})
    return thes

# Build the ASC (sentiment) thesaurus: per aspect + class, pairs of (specific indicator -> general expression).
def build_asc_thesaurus(files, dataset_name):
    """{aspect_short: {class: [(indikator, generalisasi)]}} - specific -> general expressions.
    files: the 8 per-aspect Excel files (same set for AK & AL).
    dataset_name: 'ASC_AK' or 'ASC_AL' - each file's rows are filtered to this stage via its 'Dataset' column.
    """
    thes = defaultdict(lambda: {0: [], 1: [], 2: []}); seen = set()
    for fn in files:
        df = pd.read_excel(f'{ASC_THESAURUS_DIR}/{fn}')
        if 'Dataset' in df.columns:                                         # keep only this stage's rows
            df = df[df['Dataset'].astype(str).str.strip().str.upper() == dataset_name.upper()]
        for _, r in df.iterrows():
            asp = _tnorm(r.get('Istilah Aspek')); cls = _tnorm(r.get('Kelas Sentimen'))
            if asp not in ASPECT_SHORTS or cls not in _CMAP: continue     # skip unknown aspect/class
            ind = str(r.get('Ekspresi Indikator Sentimen')).strip()       # the specific phrase in comments
            gen = str(r.get('Ekspresi Generalisasi')).strip()             # its generalised form
```

```python
            if not _good_term(ind) or not _good_term(gen) or _tnorm(ind) == _tnorm(gen): continue
            key = (asp, _CMAP[cls], _tnorm(ind))
            if key in seen: continue                                          # skip duplicates
            seen.add(key); thes[asp][_CMAP[cls]].append((ind, gen))
    for a in ASPECT_SHORTS: thes[a]                                           # make sure every aspect key exists
    print('  ASC thesaurus:', {a[:18]: (len(thes[a][0]), len(thes[a][1]), len(thes[a][2])) for a in ASPECT_SHORTS})
    return thes

# Swap the longest matching term in a comment for a different synonym that isn't already there.
def _syn_replace(comment, terms, rng):
    """Replace the LONGEST present term with another synonym not already in the comment."""
    low = comment.lower()
    present = sorted([t for t in terms if t.lower() in low], key=len, reverse=True)  # terms found, longest first
    for t in present:
        alts = [a for a in terms if a.lower() != t.lower() and a.lower() not in low] # candidate replacements
        rng.shuffle(alts)                                                     # random choice
        if alts:
            aug = _re.sub(_re.escape(t), alts[0], comment, count=1, flags=_re.IGNORECASE)  # replace one occurrence
            if aug.lower() != comment.lower():                                # only if it changed
                return aug
    return None                                                               # nothing to swap -> None

# ACD augmentation: swap one keyword belonging to a relevan aspect of this comment.
def augment_acd(comment, src_row, target_aidx, acd_thes, rng):
    """Take the comment's relevan aspects (target first if relevan) and swap one keyword."""
    relevan = [j for j, col in enumerate(ASPECT_COLUMNS) if int(src_row[col]) == 1]   # aspects marked relevan
    order = ([target_aidx] if target_aidx in relevan else []) + [j for j in relevan if j != target_aidx]   # try target first
    for j in order:
        out = _syn_replace(comment, acd_thes.get(ASPECT_SHORTS[j], []), rng)  # attempt a swap
```

```python
        if out is not None:
            return out        # success
    return None        # no swap possible

# ASC augmentation: replace a specific sentiment phrase with its generalised version.
def augment_asc(comment, target_aidx, lv, asc_thes, rng):
    """Replace an Ekspresi Indikator Sentimen (this aspect + class) with its Ekspresi Generalisasi."""
    pairs = asc_thes.get(ASPECT_SHORTS[target_aidx], {}).get(lv, [])       # (indicator, general) pairs for this cell
    low = comment.lower()
    present = sorted([(i, g) for (i, g) in pairs if i.lower() in low], key=lambda x: -len(x[0]))  # ones found, longest first
    for ind, gen in present:
        if gen.lower() in low:                                             # skip if the general form is already there
            continue
        aug = _re.sub(_re.escape(ind), gen, comment, count=1, flags=_re.IGNORECASE)  # do the replacement
        if aug.lower() != comment.lower():
            return aug
    return None

# BUILD THE THESAURUS FIT SET.
# Pass 1 is IDENTICAL to the baseline (same seed -> same sampled comments).
# Pass 2 is the ONLY difference: it tops up any cell still under n with augmented sentences.
def build_fit_rows(train_df, task, text_col, n, seed, thes):
    rng = random.Random(seed)                                 # SAME seed as baseline -> pass 1 picks the same rows
    pool = train_df.reset_index(drop=True)
    label_vals = [0, 1] if task == 'ACD' else [0, 1, 2]
    names = ACD_LABEL_NAMES if task == 'ACD' else ASC_LABEL_NAMES

    # pass 1: real per-(aspect,class) union
    real_keep = set()
    for a, col in enumerate(ASPECT_COLUMNS):
        for lv in label_vals:
            idxs = pool.index[pool[col] == (lv if task == 'ACD' else float(lv))].tolist()  # rows with this class
            if not idxs: continue
            real_keep.update(rng.sample(idxs, n) if len(idxs) >= n else idxs)          # up to n, union
```

```python
    real_df = pool.loc[sorted(real_keep)].reset_index(drop=True)            # the shared real fit set

    # pass 2: fill deficits with augmentation  (ONLY the starved cells get extra rows)
    aug_rows, aug_log = [], []
    print(f'\n  --- augmenting to fill few-shot cells ({task}) ---')
    for a, (col, short) in enumerate(zip(ASPECT_COLUMNS, ASPECT_SHORTS)):
        for lv in label_vals:
            have = int((real_df[col] == (lv if task == 'ACD' else float(lv))).sum())       # real examples already present
            deficit = n - have            # how many short of n we are
            if deficit <= 0: continue            # already at/above n -> skip
            src_idxs = pool.index[pool[col] == (lv if task == 'ACD' else float(lv))].tolist()  # sources to augment from
            if not src_idxs: continue            # nothing to base augmentation on
            for k in range(deficit):            # make one augmented row per missing slot
                src = pool.loc[src_idxs[k % len(src_idxs)]]             # cycle through the few available sources
                ori = str(src['comment_original'])            # the original comment
                aug = augment_acd(ori, src, a, thes, rng) if task == 'ACD' else augment_asc(ori, a, lv, thes, rng)  # try to reword
                tag = '' if aug is not None else '  (duplicate: no thesaurus match)'        # note if no swap was possible
                if aug is None:
                    aug = ori            # fallback: keep the original (a duplicate)
                row = {'comment_original': aug, text_col: preprocess_pipeline(aug)}        # build the new training row
                for c2 in ASPECT_COLUMNS:
                    row[c2] = src[c2]             # copy ALL 8 labels from the source
                aug_rows.append(row)
                aug_log.append({'aspect': short, 'class': names[lv], 'original': ori, 'augmented': aug})  # log for the thesis
                print(f'  [{short} / {names[lv]}]{tag}')
                print(f'      ori: {ori}')
                print(f'      aug: {aug}')
    fit = pd.concat([real_df, pd.DataFrame(aug_rows)], ignore_index=True) if aug_rows else real_df  # real + augmented
```

```
    print(f'\n  Fit set: {len(real_df)} real + {len(aug_rows)} augmented =␣
↪{len(fit)} rows')
    return fit, aug_log
```

8.9 Dataset

[ ]:
```
class ABSAMultiHeadDataset(Dataset):
    '''One comment -> input_ids/mask + label vector of length NUM_ASPECTS.
       ASC absent-aspect (NaN) -> -100 so it is ignored by that head's loss.'''
    def __init__(self, df, text_col, task, tokenizer, max_len):
        self.texts = df[text_col].astype(str).tolist()    # the cleaned comment␣
↪texts
        self.task = task
        labels = []
        for _, r in df.iterrows():                         # build one label row␣
↪per comment
            row = []
            for col in ASPECT_COLUMNS:                     # one entry per␣
↪aspect (8 total)
                v = r[col]
                if isinstance(v, float) and np.isnan(v):
                    row.append(-100)                       # -100 = "no label␣
↪here" -> loss will skip this head
                else:
                    row.append(int(v))                     # otherwise the class␣
↪index
            labels.append(row)
        self.labels = np.array(labels, dtype=np.int64)
        self.tokenizer = tokenizer; self.max_len = max_len
    def __len__(self):
        return len(self.texts)                             # how many comments␣
↪in this dataset
    def __getitem__(self, i):
        enc = self.tokenizer(self.texts[i], add_special_tokens=True,␣
↪max_length=self.max_len,
                             padding='max_length', truncation=True,␣
↪return_tensors='pt')  # tokenise one comment
        return {'input_ids': enc['input_ids'].flatten(),          # token IDs␣
↪the model reads
                'attention_mask': enc['attention_mask'].flatten(),# 1 for real␣
↪tokens, 0 for padding
                'labels': torch.tensor(self.labels[i], dtype=torch.long)}  #␣
↪the 8-aspect label vector
```

8.10 Model — shared IndoBERT + per-aspect heads

[ ]:
```
class IndoBERT_MultiHead_ABSA(nn.Module):
    def __init__(self, num_aspects, num_labels_per_head, dropout_rate=0.3):
        super().__init__()
        self.bert = BertModel.from_pretrained(MODEL_CHECKPOINT)   # the shared␣
↪IndoBERT encoder
        h = self.bert.config.hidden_size                           # size of␣
↪BERT's output vector (768)
        self.dropout = nn.Dropout(dropout_rate)                    #␣
↪regularisation before the heads
        # ONE small linear classifier per aspect (8 heads). Each head decides␣
↪for its own aspect only.
        self.heads = nn.ModuleList([nn.Linear(h, num_labels_per_head) for _ in␣
↪range(num_aspects)])
    def forward(self, input_ids, attention_mask):
        pooled = self.bert(input_ids=input_ids, attention_mask=attention_mask).
↪pooler_output  # encode the comment ONCE
        pooled = self.dropout(pooled)                              # apply␣
↪dropout to that shared vector
        return [head(pooled) for head in self.heads]               # each head␣
↪reads the SAME vector -> 8 sets of logits
```

8.11 Per-aspect class weights

[ ]:
```
def compute_head_weights(fit_df, task):
    '''Balanced class weights per aspect head, computed on available (non -100)␣
↪labels.'''
    n_lab = 2 if task == 'ACD' else 3                     # 2 classes for ACD, 3␣
↪for ASC
    weights = []
    for col in ASPECT_COLUMNS:                            # compute a weight␣
↪vector for each aspect head
        y = fit_df[col].dropna().astype(int).values      # the labels actually␣
↪present for this aspect
        present = np.unique(y)                            # which classes appear␣
↪at all
        w = np.ones(n_lab, dtype=np.float32)              # default weight 1.0␣
↪for every class
        if len(present) >= 2:                             # only balance if at␣
↪least 2 classes are present
            cw = compute_class_weight('balanced', classes=present, y=y)  #␣
↪rarer class -> bigger weight
            for c, val in zip(present, cw):
                w[c] = val                                # store the computed␣
↪weight in its class slot
```

```
        weights.append(torch.tensor(w, dtype=torch.float32).to(device))  # move␣
↪to GPU/CPU as a tensor
    return weights                                      # one weight vector per␣
↪aspect, used in the loss
```

8.12 Training (weighted CE, no validation)

```
[ ]: def train_model(fit_df, task, text_col, seed):
    set_all_seeds(seed)                                 # reset all RNGs so␣
↪training is reproducible
    n_lab = 2 if task == 'ACD' else 3                   # classes per head: 2␣
↪(ACD) or 3 (ASC)
    ds = ABSAMultiHeadDataset(fit_df, text_col, task, tokenizer, MAX_SEQ_LEN) ␣
↪# wrap the fit set
    loader = DataLoader(ds, batch_size=BATCH_SIZE, shuffle=True)             ␣
↪# feed it in shuffled batches

    model = IndoBERT_MultiHead_ABSA(NUM_ASPECTS, n_lab, DROPOUT).to(device)  ␣
↪# build the 8-head model
    head_weights = compute_head_weights(fit_df, task)                        ␣
↪# per-aspect class weights
    # One weighted cross-entropy loss per aspect. ignore_index=-100 -> absent␣
↪labels don't contribute.
    loss_fns = [nn.CrossEntropyLoss(weight=head_weights[a], ignore_index=-100)
                for a in range(NUM_ASPECTS)]

    optimizer = AdamW(model.parameters(), lr=LEARNING_RATE,␣
↪weight_decay=WEIGHT_DECAY)  # flat LR, no scheduler

    history = []
    for epoch in range(EPOCHS):                         # train for a fixed number␣
↪of epochs (5)
        model.train(); run = 0.0                        # training mode; running␣
↪loss accumulator
        pbar = tqdm(loader, desc=f'  epoch {epoch+1}/{EPOCHS}', leave=False)
        for batch in pbar:                              # one mini-batch of␣
↪comments at a time
            ids  = batch['input_ids'].to(device)      # token IDs -> GPU
            mask = batch['attention_mask'].to(device)# attention mask -> GPU
            lab  = batch['labels'].to(device)         # true labels, shape␣
↪[batch, 8]
            logits_list = model(ids, mask)            # forward pass -> 8 heads'␣
↪logits
            # total loss = sum of each head's loss (each head compares its␣
↪column of labels)
```

```
            loss = sum(loss_fns[a](logits_list[a], lab[:, a]) for a in range(NUM_ASPECTS))
            optimizer.zero_grad(); loss.backward()   # clear old gradients, back-propagate
            optimizer.step()                         # update the weights
            run += loss.item(); pbar.set_postfix(loss=loss.item())
        ep_loss = run / len(loader)                  # average loss this epoch
        history.append(ep_loss)
        print(f'  epoch {epoch+1}: train_loss={ep_loss:.4f}')
    return model, history                            # trained model + loss curve
```

8.13 Prediction

[ ]:

```
@torch.no_grad()                                     # no gradients needed when only predicting
def predict(model, df, text_col, task):
    model.eval()                                     # evaluation mode (turns off dropout)
    ds = ABSAMultiHeadDataset(df, text_col, task, tokenizer, MAX_SEQ_LEN)
    loader = DataLoader(ds, batch_size=BATCH_SIZE, shuffle=False)  # keep order so preds line up with rows
    preds, labels = [], []
    for batch in loader:
        ids  = batch['input_ids'].to(device)
        mask = batch['attention_mask'].to(device)
        logits_list = model(ids, mask)               # 8 heads' logits for this batch
        # for each head take the highest-scoring class -> one prediction per aspect -> [batch, 8]
        p = torch.stack([torch.argmax(l, dim=1) for l in logits_list], dim=1)
        preds.append(p.cpu().numpy())                # collect predictions
        labels.append(batch['labels'].numpy())       # collect the true labels
    return np.vstack(preds), np.vstack(labels)       # stack all batches into full arrays
```

8.14 Metrics — macro-F1 + weighted-F1

[ ]:

```
def _valid(true, pred, task):
    if task == 'ASC':
        m = true != -100
        return true[m], pred[m]
    return true, pred

def print_metrics(preds, labels, task, name):
    label_names = ACD_LABEL_NAMES if task == 'ACD' else ASC_LABEL_NAMES
```

```python
    idxs = list(range(len(label_names)))
    print(f'\n{"="*64}\n  {name}\n{"="*64}')
    macro_list, weighted_list = [], []
    for a, asp in enumerate(ASPECT_SHORTS):
        true, pred = _valid(labels[:, a], preds[:, a], task)
        if len(true) == 0:
            print(f'\nAspect: {asp}  (no labels)'); continue
        p, r, f, _ = precision_recall_fscore_support(true, pred, labels=idxs, average=None, zero_division=0)
        mf1 = f1_score(true, pred, average='macro',    zero_division=0)
        wf1 = f1_score(true, pred, average='weighted', zero_division=0)
        macro_list.append(mf1); weighted_list.append(wf1)
        print(f'\nAspect: {asp}')
        for i, ln in enumerate(label_names):
            print(f'  {ln:<22} P={p[i]:.3f}  R={r[i]:.3f}  F1={f[i]:.3f}')
        print(f'  Macro F1 {mf1:.4f}  |  Weighted F1 {wf1:.4f}')
    mean_macro = float(np.mean(macro_list)) if macro_list else float("nan")
    mean_weight = float(np.mean(weighted_list)) if weighted_list else float("nan")
    print(f'\n  MEAN Macro F1: {mean_macro:.4f}  |  MEAN Weighted F1: {mean_weight:.4f}')
    return {'mean_macro_f1': mean_macro, 'mean_weighted_f1': mean_weight,
            'per_aspect_macro': dict(zip(ASPECT_SHORTS, macro_list)),
            'per_aspect_weighted': dict(zip(ASPECT_SHORTS, weighted_list))}

def plot_cms(preds, labels, task, name):
    label_names = ACD_LABEL_NAMES if task == 'ACD' else ASC_LABEL_NAMES
    idxs = list(range(len(label_names)))
    fig, axes = plt.subplots(2, 4, figsize=(24, 12))
    for a, (asp, ax) in enumerate(zip(ASPECT_SHORTS, axes.flatten())):
        true, pred = _valid(labels[:, a], preds[:, a], task)
        if len(true) == 0:
            ax.set_title(f'{asp}\n(no data)', fontsize=9); ax.axis('off'); continue
        cm = confusion_matrix(true, pred, labels=idxs)
        sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
                    xticklabels=label_names, yticklabels=label_names, ax=ax,
                    cbar=True, linewidths=0.5, linecolor='white')
        ax.set_title(asp, fontsize=9); ax.set_xlabel('Pred', fontsize=8); ax.set_ylabel('True', fontsize=8)
        ax.tick_params(labelsize=7)
    plt.suptitle(f'Confusion Matrices - {name}', fontsize=13, y=1.01)
    plt.tight_layout()
    fname = f'{BASE_PATH}/results/thesaurus/cm_{name}.png'; os.makedirs(os.path.dirname(fname), exist_ok=True); plt.savefig(fname, dpi=150, bbox_inches='tight'); plt.show()
```

```python
    print('Saved ->', fname)

def plot_training(history, name):
    plt.figure(figsize=(7, 4))
    plt.plot(range(1, len(history) + 1), history, marker='o', linewidth=2, color='#2b6cb0')
    plt.title(f'Training Loss - {name}', fontsize=12)
    plt.xlabel('Epoch'); plt.ylabel('Summed weighted CE loss')
    plt.xticks(range(1, len(history) + 1)); plt.grid(True, alpha=0.3)
    fname = f'{BASE_PATH}/results/thesaurus/training_{name}.png'
    os.makedirs(os.path.dirname(fname), exist_ok=True)
    plt.tight_layout(); plt.savefig(fname, dpi=150, bbox_inches='tight'); plt.show()
    print('Saved ->', fname)
```

8.15 Save predictions to Excel

[ ]:

```python
def save_predictions_excel(test_df, preds, labels, text_col, task, name):
    names = ACD_LABEL_NAMES if task == 'ACD' else ASC_LABEL_NAMES
    nm = lambda v: '' if int(v) == -100 else names[int(v)]
    rows = []
    for i in range(len(test_df)):
        r = {'comment': str(test_df.iloc[i][text_col])}
        for a, asp in enumerate(ASPECT_SHORTS):
            r[f'{asp} | true'] = nm(labels[i, a])
            r[f'{asp} | pred'] = nm(preds[i, a])
        rows.append(r)
    out = pd.DataFrame(rows)
    os.makedirs(f'{BASE_PATH}/results/thesaurus', exist_ok=True)
    path = f'{BASE_PATH}/results/thesaurus/predictions_{name}_weighted_baseline.xlsx'
    out.to_excel(path, index=False)
    print(f'  Saved {len(out)} test comments + predictions -> {path}')
    return out
```

8.16 Runner

[ ]:

```python
def run_experiment(exp):
    name, task, text_col = exp['name'], exp['task'], exp['text_col']
    print(f'\n{"#"*66}\n  {name}  [{task} / THESAURUS / few-shot {N_PER_CLASS} / same test set]\n{"#"*66}')

    # Load the SAME train/test split the baseline saved -> guarantees an apples-to-apples comparison.
    train_df = load_split(name, task, 'train')
    test_df  = load_split(name, task, 'test')
```

```
    # Build this stage's thesaurus, then fill the few-shot cells with␣
↪augmentation.
    files = (ACD_THESAURUS_FILES if task == 'ACD' else␣
↪ASC_THESAURUS_FILES)[name]  # which thesaurus files for this run
    thes  = build_acd_thesaurus(files) if task == 'ACD' else␣
↪build_asc_thesaurus(files, name)
    fit_df, aug_log = build_fit_rows(train_df, task, text_col, N_PER_CLASS,␣
↪SEED, thes)  # real + augmented rows

    model, history = train_model(fit_df, task, text_col, SEED)          #␣
↪fine-tune on the augmented fit set

    print('\n  Evaluating on the shared TEST set...')
    preds, labels = predict(model, test_df, text_col, task)             # SAME␣
↪test set as baseline
    metrics = print_metrics(preds, labels, task, name)
    plot_cms(preds, labels, task, name)
    plot_training(history, name)
    save_predictions_excel(test_df, preds, labels, text_col, task, name)

    # Save every augmented sentence so they can be inspected / reported in the␣
↪thesis.
    if aug_log:
        os.makedirs(f'{BASE_PATH}/results/thesaurus', exist_ok=True)
        pd.DataFrame(aug_log).to_excel(f'{BASE_PATH}/results/thesaurus/
↪augmented_{name}_thesaurus.xlsx', index=False)
        print(f'  Saved {len(aug_log)} augmented sentences -> results/
↪augmented_{name}_thesaurus.xlsx')

    del model; gc.collect()                                             # free␣
↪memory for the next run
    if torch.cuda.is_available(): torch.cuda.empty_cache()
    return {'metrics': metrics, 'preds': preds, 'labels': labels, 'n_fit':␣
↪len(fit_df), 'n_test': len(test_df)}
```

8.17 Run all four experiments (ACD AK/AL, ASC AK/AL)

[ ]:
```
ALL_RESULTS = {}
for exp in ACD_EXPERIMENTS + ASC_EXPERIMENTS:
    ALL_RESULTS[exp['name']] = run_experiment(exp)

print('\nDone.  Mean macro-F1 / weighted-F1:')
for k, v in ALL_RESULTS.items():
    m = v['metrics']
    print(f'  {k:<8}  macro={m["mean_macro_f1"]:.4f} ␣
↪weighted={m["mean_weighted_f1"]:.4f}  (+{v["n_aug"]} aug)')
```

```
####################################################################
  ACD_AK  [ACD / THESAURUS / few-shot 15 / same test set]
####################################################################
  loaded train:
/content/drive/MyDrive/results/baseline/splits/ACD_AK_train.xlsx  (159 rows)
  loaded test:
/content/drive/MyDrive/results/baseline/splits/ACD_AK_test.xlsx  (41 rows)
  ACD thesaurus: {'capaian pembelajar': 110, 'indeks prestasi ku': 53,
'kelulusan tepat wa': 125, 'masa tunggu lulusa': 96, 'pelacakan dan pere': 147,
'prestasi mahasiswa': 192, 'karya dengan hak k': 49, 'kesesuaian bidang ': 195}

  --- augmenting to fill few-shot cells (ACD) ---
  Fit set: 119 real + 13 augmented = 132 rows

config.json:   0%|          | 0.00/1.53k [00:00<?, ?B/s]

pytorch_model.bin:   0%|          | 0.00/498M [00:00<?, ?B/s]

Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]

  epoch 1/5:   0%|          | 0/17 [00:00<?, ?it/s]

model.safetensors:   0%|          | 0.00/498M [00:00<?, ?B/s]

  epoch 1: train_loss=4.6045

  epoch 2/5:   0%|          | 0/17 [00:00<?, ?it/s]

  epoch 2: train_loss=3.5114

  epoch 3/5:   0%|          | 0/17 [00:00<?, ?it/s]

  epoch 3: train_loss=2.7536

  epoch 4/5:   0%|          | 0/17 [00:00<?, ?it/s]

  epoch 4: train_loss=2.6339

  epoch 5/5:   0%|          | 0/17 [00:00<?, ?it/s]

  epoch 5: train_loss=2.2488

  Evaluating on the shared TEST set…

==================================================================
  ACD_AK
==================================================================

Aspect: capaian pembelajaran lulusan
  tidak_relevan         P=0.000  R=0.000  F1=0.000
  relevan               P=0.951  R=1.000  F1=0.975
  Macro F1 0.4875  |  Weighted F1 0.9274

Aspect: indeks prestasi kumulatif
```

```
  tidak_relevan          P=0.571  R=0.667  F1=0.615
  relevan                P=0.941  R=0.914  F1=0.928
  Macro F1 0.7715  |  Weighted F1 0.8819

Aspect: kelulusan tepat waktu
  tidak_relevan          P=0.778  R=0.538  F1=0.636
  relevan                P=0.812  R=0.929  F1=0.867
  Macro F1 0.7515  |  Weighted F1 0.7936

Aspect: masa tunggu lulusan
  tidak_relevan          P=0.875  R=0.700  F1=0.778
  relevan                P=0.909  R=0.968  F1=0.938
  Macro F1 0.8576  |  Weighted F1 0.8985

Aspect: pelacakan dan perekaman data lulusan
  tidak_relevan          P=0.667  R=0.750  F1=0.706
  relevan                P=0.826  R=0.760  F1=0.792
  Macro F1 0.7488  |  Weighted F1 0.7582

Aspect: prestasi mahasiswa
  tidak_relevan          P=0.429  R=1.000  F1=0.600
  relevan                P=1.000  R=0.625  F1=0.769
  Macro F1 0.6846  |  Weighted F1 0.7321

Aspect: karya dengan hak kekayaan intelektual
  tidak_relevan          P=0.667  R=0.533  F1=0.593
  relevan                P=0.759  R=0.846  F1=0.800
  Macro F1 0.6963  |  Weighted F1 0.7241

Aspect: kesesuaian bidang kerja
  tidak_relevan          P=0.643  R=0.692  F1=0.667
  relevan                P=0.852  R=0.821  F1=0.836
  Macro F1 0.7515  |  Weighted F1 0.7826

  MEAN Macro F1: 0.7187  |  MEAN Weighted F1: 0.8123

Saved -> /content/drive/MyDrive/results/thesaurus/cm_ACD_AK.png

Saved -> /content/drive/MyDrive/results/thesaurus/training_ACD_AK.png
  Saved 41 test comments + predictions -> /content/drive/MyDrive/results/thes
aurus/predictions_ACD_AK_weighted_baseline.xlsx
  Saved 13 augmented sentences -> results/augmented_ACD_AK_thesaurus.xlsx

##################################################################
  ACD_AL  [ACD / THESAURUS / few-shot 15 / same test set]
##################################################################
  loaded train:
/content/drive/MyDrive/results/baseline/splits/ACD_AL_train.xlsx  (157 rows)
  loaded test:
```

```
/content/drive/MyDrive/results/baseline/splits/ACD_AL_test.xlsx  (43 rows)
  ACD thesaurus: {'capaian pembelajar': 110, 'indeks prestasi ku': 53,
'kelulusan tepat wa': 125, 'masa tunggu lulusa': 96, 'pelacakan dan pere': 147,
'prestasi mahasiswa': 192, 'karya dengan hak k': 49, 'kesesuaian bidang ': 195}

  --- augmenting to fill few-shot cells (ACD) ---
  Fit set: 120 real + 16 augmented = 136 rows
Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]
  epoch 1/5:   0%|          | 0/17 [00:00<?, ?it/s]
  epoch 1: train_loss=4.2914
  epoch 2/5:   0%|          | 0/17 [00:00<?, ?it/s]
  epoch 2: train_loss=3.0152
  epoch 3/5:   0%|          | 0/17 [00:00<?, ?it/s]
  epoch 3: train_loss=2.5054
  epoch 4/5:   0%|          | 0/17 [00:00<?, ?it/s]
  epoch 4: train_loss=1.9329
  epoch 5/5:   0%|          | 0/17 [00:00<?, ?it/s]
  epoch 5: train_loss=1.6833

  Evaluating on the shared TEST set…

================================================================
  ACD_AL
================================================================

Aspect: capaian pembelajaran lulusan
  tidak_relevan          P=1.000  R=0.500  F1=0.667
  relevan                P=0.976  R=1.000  F1=0.988
  Macro F1 0.8273  |  Weighted F1 0.9730

Aspect: indeks prestasi kumulatif
  tidak_relevan          P=0.667  R=0.667  F1=0.667
  relevan                P=0.946  R=0.946  F1=0.946
  Macro F1 0.8063  |  Weighted F1 0.9070

Aspect: kelulusan tepat waktu
  tidak_relevan          P=0.692  R=0.900  F1=0.783
  relevan                P=0.967  R=0.879  F1=0.921
  Macro F1 0.8516  |  Weighted F1 0.8885

Aspect: masa tunggu lulusan
  tidak_relevan          P=0.733  R=0.846  F1=0.786
```

```
  relevan              P=0.929  R=0.867  F1=0.897
  Macro F1 0.8411  |  Weighted F1 0.8630

Aspect: pelacakan dan perekaman data lulusan
  tidak_relevan        P=0.750  R=0.600  F1=0.667
  relevan              P=0.806  R=0.893  F1=0.847
  Macro F1 0.7571  |  Weighted F1 0.7844

Aspect: prestasi mahasiswa
  tidak_relevan        P=0.733  R=0.846  F1=0.786
  relevan              P=0.929  R=0.867  F1=0.897
  Macro F1 0.8411  |  Weighted F1 0.8630

Aspect: karya dengan hak kekayaan intelektual
  tidak_relevan        P=0.857  R=0.800  F1=0.828
  relevan              P=0.897  R=0.929  F1=0.912
  Macro F1 0.8699  |  Weighted F1 0.8827

Aspect: kesesuaian bidang kerja
  tidak_relevan        P=0.750  R=0.818  F1=0.783
  relevan              P=0.935  R=0.906  F1=0.921
  Macro F1 0.8516  |  Weighted F1 0.8853

  MEAN Macro F1: 0.8308  |  MEAN Weighted F1: 0.8809
Saved -> /content/drive/MyDrive/results/thesaurus/cm_ACD_AL.png
Saved -> /content/drive/MyDrive/results/thesaurus/training_ACD_AL.png
  Saved 43 test comments + predictions -> /content/drive/MyDrive/results/thes
aurus/predictions_ACD_AL_weighted_baseline.xlsx
  Saved 16 augmented sentences -> results/augmented_ACD_AL_thesaurus.xlsx

##################################################################
  ASC_AK  [ASC / THESAURUS / few-shot 15 / same test set]
##################################################################
  loaded train:
/content/drive/MyDrive/results/baseline/splits/ASC_AK_train.xlsx  (158 rows)
  loaded test:
/content/drive/MyDrive/results/baseline/splits/ASC_AK_test.xlsx  (42 rows)
  ASC thesaurus: {'capaian pembelajar': (266, 134, 87), 'indeks prestasi ku':
(298, 66, 198), 'kelulusan tepat wa': (165, 139, 209), 'masa tunggu lulusa':
(195, 85, 188), 'pelacakan dan pere': (195, 147, 142), 'prestasi mahasiswa':
(340, 83, 58), 'karya dengan hak k': (211, 118, 75), 'kesesuaian bidang ': (273,
94, 129)}

  --- augmenting to fill few-shot cells (ASC) ---
  Fit set: 134 real + 31 augmented = 165 rows
Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]
```

```
epoch 1/5:   0%|          | 0/21 [00:00<?, ?it/s]

epoch 1: train_loss=8.8574

epoch 2/5:   0%|          | 0/21 [00:00<?, ?it/s]

epoch 2: train_loss=7.9761

epoch 3/5:   0%|          | 0/21 [00:00<?, ?it/s]

epoch 3: train_loss=6.9743

epoch 4/5:   0%|          | 0/21 [00:00<?, ?it/s]

epoch 4: train_loss=5.8708

epoch 5/5:   0%|          | 0/21 [00:00<?, ?it/s]

epoch 5: train_loss=4.8105

Evaluating on the shared TEST set…

================================================================
  ASC_AK
================================================================

Aspect: capaian pembelajaran lulusan
  positif                P=0.607  R=0.895  F1=0.723
  negatif                P=0.750  R=0.500  F1=0.600
  netral                 P=0.556  R=0.312  F1=0.400
  Macro F1 0.5745  |  Weighted F1 0.5791

Aspect: indeks prestasi kumulatif
  positif                P=0.667  R=0.737  F1=0.700
  negatif                P=0.000  R=0.000  F1=0.000
  netral                 P=0.462  R=0.500  F1=0.480
  Macro F1 0.3933  |  Weighted F1 0.5606

Aspect: kelulusan tepat waktu
  positif                P=0.696  R=0.842  F1=0.762
  negatif                P=1.000  R=0.500  F1=0.667
  netral                 P=0.250  R=0.143  F1=0.182
  Macro F1 0.5368  |  Weighted F1 0.6101

Aspect: masa tunggu lulusan
  positif                P=0.737  R=0.778  F1=0.757
  negatif                P=0.000  R=0.000  F1=0.000
  netral                 P=0.300  R=0.300  F1=0.300
  Macro F1 0.3523  |  Weighted F1 0.5362

Aspect: pelacakan dan perekaman data lulusan
  positif                P=0.727  R=0.889  F1=0.800
```

```
  negatif               P=0.000  R=0.000  F1=0.000
  netral                P=0.000  R=0.000  F1=0.000
  Macro F1 0.2667  |  Weighted F1 0.5760

Aspect: prestasi mahasiswa
  positif               P=0.750  R=0.833  F1=0.789
  negatif               P=0.333  R=0.250  F1=0.286
  netral                P=0.429  R=0.375  F1=0.400
  Macro F1 0.4917  |  Weighted F1 0.6184

Aspect: karya dengan hak kekayaan intelektual
  positif               P=0.667  R=0.571  F1=0.615
  negatif               P=0.000  R=0.000  F1=0.000
  netral                P=0.462  R=0.667  F1=0.545
  Macro F1 0.3869  |  Weighted F1 0.5202

Aspect: kesesuaian bidang kerja
  positif               P=0.588  R=0.667  F1=0.625
  negatif               P=0.000  R=0.000  F1=0.000
  netral                P=0.300  R=0.273  F1=0.286
  Macro F1 0.3036  |  Weighted F1 0.4471

  MEAN Macro F1: 0.4132  |  MEAN Weighted F1: 0.5560

Saved -> /content/drive/MyDrive/results/thesaurus/cm_ASC_AK.png

Saved -> /content/drive/MyDrive/results/thesaurus/training_ASC_AK.png
  Saved 42 test comments + predictions -> /content/drive/MyDrive/results/thes
aurus/predictions_ASC_AK_weighted_baseline.xlsx
  Saved 31 augmented sentences -> results/augmented_ASC_AK_thesaurus.xlsx

##################################################################
  ASC_AL  [ASC / THESAURUS / few-shot 15 / same test set]
##################################################################
  loaded train:
/content/drive/MyDrive/results/baseline/splits/ASC_AL_train.xlsx  (156 rows)
  loaded test:
/content/drive/MyDrive/results/baseline/splits/ASC_AL_test.xlsx  (44 rows)
  ASC thesaurus: {'capaian pembelajar': (166, 48, 33), 'indeks prestasi ku':
(182, 24, 115), 'kelulusan tepat wa': (103, 71, 129), 'masa tunggu lulusa':
(117, 31, 110), 'pelacakan dan pere': (133, 44, 86), 'prestasi mahasiswa': (243,
23, 17), 'karya dengan hak k': (152, 26, 52), 'kesesuaian bidang ': (185, 33,
71)}

  --- augmenting to fill few-shot cells (ASC) ---
  Fit set: 132 real + 67 augmented = 199 rows

Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]

  epoch 1/5:   0%|          | 0/25 [00:00<?, ?it/s]
```

```
  epoch 1: train_loss=7.9725

  epoch 2/5:   0%|          | 0/25 [00:00<?, ?it/s]

  epoch 2: train_loss=5.8686

  epoch 3/5:   0%|          | 0/25 [00:00<?, ?it/s]

  epoch 3: train_loss=4.7693

  epoch 4/5:   0%|          | 0/25 [00:00<?, ?it/s]

  epoch 4: train_loss=3.9816

  epoch 5/5:   0%|          | 0/25 [00:00<?, ?it/s]

  epoch 5: train_loss=3.3354

  Evaluating on the shared TEST set…

================================================================
  ASC_AL
================================================================

Aspect: capaian pembelajaran lulusan
  positif                P=0.905  R=0.655  F1=0.760
  negatif                P=0.400  R=0.286  F1=0.333
  netral                 P=0.389  R=0.875  F1=0.538
  Macro F1 0.5439  |  Weighted F1 0.6518

Aspect: indeks prestasi kumulatif
  positif                P=0.556  R=0.476  F1=0.513
  negatif                P=0.000  R=0.000  F1=0.000
  netral                 P=0.368  R=0.467  F1=0.412
  Macro F1 0.3082  |  Weighted F1 0.4459

Aspect: kelulusan tepat waktu
  positif                P=0.545  R=0.462  F1=0.500
  negatif                P=0.250  R=0.250  F1=0.250
  netral                 P=0.591  R=0.650  F1=0.619
  Macro F1 0.4563  |  Weighted F1 0.5373

Aspect: masa tunggu lulusan
  positif                P=0.769  R=0.526  F1=0.625
  negatif                P=0.000  R=0.000  F1=0.000
  netral                 P=0.429  R=0.750  F1=0.545
  Macro F1 0.3902  |  Weighted F1 0.5418

Aspect: pelacakan dan perekaman data lulusan
  positif                P=0.765  R=0.619  F1=0.684
  negatif                P=0.000  R=0.000  F1=0.000
  netral                 P=0.375  R=0.500  F1=0.429
```

```
  Macro F1 0.3709  |  Weighted F1 0.5575

Aspect: prestasi mahasiswa
  positif                P=0.900  R=0.409  F1=0.562
  negatif                P=0.333  R=0.500  F1=0.400
  netral                 P=0.400  R=0.889  F1=0.552
  Macro F1 0.5047  |  Weighted F1 0.5497

Aspect: karya dengan hak kekayaan intelektual
  positif                P=0.889  R=0.800  F1=0.842
  negatif                P=0.000  R=0.000  F1=0.000
  netral                 P=0.545  R=0.750  F1=0.632
  Macro F1 0.4912  |  Weighted F1 0.7298

Aspect: kesesuaian bidang kerja
  positif                P=0.667  R=0.778  F1=0.718
  negatif                P=0.000  R=0.000  F1=0.000
  netral                 P=0.643  R=0.600  F1=0.621
  Macro F1 0.4462  |  Weighted F1 0.6352

  MEAN Macro F1: 0.4390  |  MEAN Weighted F1: 0.5811
Saved -> /content/drive/MyDrive/results/thesaurus/cm_ASC_AL.png
Saved -> /content/drive/MyDrive/results/thesaurus/training_ASC_AL.png
  Saved 44 test comments + predictions -> /content/drive/MyDrive/results/thes
aurus/predictions_ASC_AL_weighted_baseline.xlsx
  Saved 67 augmented sentences -> results/augmented_ASC_AL_thesaurus.xlsx

Done.  Mean macro-F1 / weighted-F1:
  ACD_AK    macro=0.7187  weighted=0.8123  (+13 aug)
  ACD_AL    macro=0.8308  weighted=0.8809  (+16 aug)
  ASC_AK    macro=0.4132  weighted=0.5560  (+31 aug)
  ASC_AL    macro=0.4390  weighted=0.5811  (+67 aug)
```

8.18 Results summary

[ ]:
```python
summary = pd.DataFrame([
    {'experiment': k,
     'mean_macro_f1':    round(v['metrics']['mean_macro_f1'], 4),
     'mean_weighted_f1': round(v['metrics']['mean_weighted_f1'], 4),
     'n_fit': v['n_fit'], 'n_aug': v['n_aug'], 'n_test': v['n_test']}
    for k, v in ALL_RESULTS.items()
]).set_index('experiment')
print(summary.to_string())

os.makedirs(f'{BASE_PATH}/results/thesaurus', exist_ok=True)
summary.to_excel(f'{BASE_PATH}/results/thesaurus/summary_thesaurus.xlsx')
```

```python
print('\nSaved -> {}/results/thesaurus/summary_thesaurus.xlsx'.
 ↪format(BASE_PATH))
```

```
            mean_macro_f1  mean_weighted_f1  n_fit  n_aug  n_test
experiment
ACD_AK             0.7187            0.8123    132     13      41
ACD_AL             0.8308            0.8809    136     16      43
ASC_AK             0.4132            0.5560    165     31      42
ASC_AL             0.4390            0.5811    199     67      44

Saved -> /content/drive/MyDrive/results/thesaurus/summary_thesaurus.xlsx
```

9 Tahap Analisis Metrik Terpadu - Baseline vs Tesaurus

Tahap ini menggabungkan (pooling) hasil prediksi dari kedua tahap fine-tuning sebelumnya (baseline dan tesaurus) di seluruh delapan aspek dan kedua tahap asesmen (AK, AL), lalu menghitung ulang precision, recall, dan F1-score per kelas langsung dari confusion matrix gabungan — bukan dengan merata-ratakan skor per-run — sehingga hasil akhir mencerminkan performa aktual pada seluruh data uji sekaligus. Confusion matrix pooled yang lengkap (bukan hanya diagonal) turut disimpan untuk memeriksa pola kesalahan klasifikasi, misalnya berapa banyak komentar berlabel ‘positif‘ yang justru diprediksi sebagai ‘netral‘.

9.1 Mount Drive

[ ]:
```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

9.2 Config (must match the training stages)

[ ]:
```
BASE_PATH = '/content/drive/MyDrive'    # <-- same base path the training stages␣
 ↪used

# label order is fixed by the training config (cell 4 there)
ACD_LABEL_NAMES = ['tidak_relevan', 'relevan']
ASC_LABEL_NAMES = ['positif', 'negatif', 'netral']

# the 8 aspects, exactly as they appear in the prediction-file column names
ASPECT_SHORTS = [
    'capaian pembelajaran lulusan',
    'indeks prestasi kumulatif',
    'kelulusan tepat waktu',
    'masa tunggu lulusan',
    'pelacakan dan perekaman data lulusan',
    'prestasi mahasiswa',
    'karya dengan hak kekayaan intelektual',
    'kesesuaian bidang kerja',
]

EXPERIMENTS = ['ACD_AK', 'ACD_AL', 'ASC_AK', 'ASC_AL']

# where each model's predictions live on Drive
DIRS = {'baseline':  f'{BASE_PATH}/results/baseline',
        'thesaurus': f'{BASE_PATH}/results/thesaurus'}

# prediction filenames carry the '_weighted_baseline' suffix in BOTH folders
FILE = 'predictions_{name}_weighted_baseline.xlsx'
```

9.3 Pooling functions

[ ]:
```python
import os
import numpy as np, pandas as pd
from sklearn.metrics import precision_recall_fscore_support, confusion_matrix

def _task(name):  return name.split('_')[0]                              # 'ASC_AK' -> 'ASC'
def _names(task): return ACD_LABEL_NAMES if task == 'ACD' else ASC_LABEL_NAMES

def pooled_from_excel(path, task):
    """Flatten the 8 aspect columns of a predictions file into one pooled
    (true, pred) vector. Empty cells ('') are masked-out ASC 'absent' rows and
    are dropped -- exactly as the training stage's _valid() drops -100."""
    names = _names(task); idx = {n: i for i, n in enumerate(names)}
    df = pd.read_excel(path); t_all, p_all, skipped = [], [], 0
    for asp in ASPECT_SHORTS:
        for tv, pv in zip(df[f'{asp} | true'].fillna(''), df[f'{asp} | pred'].fillna('')):
            tv, pv = str(tv).strip(), str(pv).strip()
            if tv == '':                          # absent aspect -> excluded from scoring
                continue
            if tv not in idx or pv not in idx:  # guard; should never fire
                skipped += 1; continue
            t_all.append(idx[tv]); p_all.append(idx[pv])
    if skipped: print(f'  [warn] {skipped} unrecognised label cell(s) skipped in {os.path.basename(path)}')
    return np.array(t_all), np.array(p_all)

def report_pooled(true, pred, task, title, show=True):
    """Pooled per-class precision/recall/F1 + the raw TP/support counts."""
    names, ids = _names(task), list(range(len(_names(task))))
    cm = confusion_matrix(true, pred, labels=ids)                  # rows=true, cols=pred
    P, R, F, S = precision_recall_fscore_support(true, pred, labels=ids,
                                                  average=None, zero_division=0)
    if show:
        print(f'\n=== {title}   (pooled over {len(ASPECT_SHORTS)} aspects, n={len(true)}) ===')
        print(f'{"class":<15}{"TP/supp":>10}{"prec":>8}{"recall":>8}{"F1":>8}')
        for i, n in enumerate(names):
            print(f'{n:<15}{f"{cm[i,i]}/{int(S[i])}":>10}{P[i]:>8.3f}{R[i]:>8.3f}{F[i]:>8.3f}')
        print(f'{"pooled macro-F1":<15}{F.mean():>34.3f}')
        print(f'{"pooled wtd-F1":<15}{np.average(F, weights=S):>34.3f}')
```

```
        print('  (headline macro-F1 on the slides = MEAN of the 8 per-aspect␣
↪macro-F1,')
        print('   a different aggregation from this pooled macro-F1)')
    return {names[i]: dict(tp=int(cm[i, i]), support=int(S[i]),
                           precision=float(P[i]), recall=float(R[i]),␣
↪f1=float(F[i]))
            for i in range(len(names))}
```

9.4 Run it — baseline vs thesaurus, side by side

[ ]:
```
rows = []   # collect a tidy table to save at the end

for name in EXPERIMENTS:
    task = _task(name)
    pb = f'{DIRS["baseline"]}/{FILE.format(name=name)}'
    pt = f'{DIRS["thesaurus"]}/{FILE.format(name=name)}'
    try:
        b = report_pooled(*pooled_from_excel(pb, task), task, f'{name} ␣
↪BASELINE')
        t = report_pooled(*pooled_from_excel(pt, task), task, f'{name} ␣
↪THESAURUS')
    except FileNotFoundError as e:
        print(f'\n[skip] {name}: missing file -> {e.filename}'); continue
    print(f'\n  {name}  baseline -> thesaurus:')
    for cls in _names(task):
        print(f'    {cls:<15} recall {b[cls]["recall"]:.3f} ->␣
↪{t[cls]["recall"]:.3f}'
              f'   |  F1 {b[cls]["f1"]:.3f} -> {t[cls]["f1"]:.3f}'
              f'   |  TP {b[cls]["tp"]}/{b[cls]["support"]} -> {t[cls]["tp"]}/
↪{t[cls]["support"]}')
        for model, d in (('baseline', b), ('thesaurus', t)):
            rows.append(dict(experiment=name, model=model, cls=cls,
                             tp=d[cls]['tp'], support=d[cls]['support'],
                             precision=round(d[cls]['precision'], 3),
                             recall=round(d[cls]['recall'], 3),
                             f1=round(d[cls]['f1'], 3)))
    print('-' * 78)
```

```
=== ACD_AK  BASELINE   (pooled over 8 aspects, n=328) ===
class              TP/supp    prec   recall      F1
tidak_relevan        48/84   0.667    0.571   0.615
relevan            220/244   0.859    0.902   0.880
pooled macro-F1                               0.748
pooled wtd-F1                                 0.812
  (headline macro-F1 on the slides = MEAN of the 8 per-aspect macro-F1,
   a different aggregation from this pooled macro-F1)
```

```
=== ACD_AK  THESAURUS   (pooled over 8 aspects, n=328) ===
class             TP/supp    prec  recall      F1
tidak_relevan       56/84   0.629   0.667   0.647
relevan           211/244   0.883   0.865   0.874
pooled macro-F1                             0.761
pooled wtd-F1                               0.816
  (headline macro-F1 on the slides = MEAN of the 8 per-aspect macro-F1,
   a different aggregation from this pooled macro-F1)

  ACD_AK  baseline -> thesaurus:
    tidak_relevan   recall 0.571 -> 0.667   |  F1 0.615 -> 0.647   |  TP 48/84
-> 56/84
    relevan         recall 0.902 -> 0.865   |  F1 0.880 -> 0.874   |  TP 220/244
-> 211/244
------------------------------------------------------------------------------

=== ACD_AL  BASELINE   (pooled over 8 aspects, n=344) ===
class             TP/supp    prec  recall      F1
tidak_relevan       66/85   0.759   0.776   0.767
relevan           238/259   0.926   0.919   0.922
pooled macro-F1                             0.845
pooled wtd-F1                               0.884
  (headline macro-F1 on the slides = MEAN of the 8 per-aspect macro-F1,
   a different aggregation from this pooled macro-F1)

=== ACD_AL  THESAURUS   (pooled over 8 aspects, n=344) ===
class             TP/supp    prec  recall      F1
tidak_relevan       66/85   0.750   0.776   0.763
relevan           237/259   0.926   0.915   0.920
pooled macro-F1                             0.842
pooled wtd-F1                               0.882
  (headline macro-F1 on the slides = MEAN of the 8 per-aspect macro-F1,
   a different aggregation from this pooled macro-F1)

  ACD_AL  baseline -> thesaurus:
    tidak_relevan   recall 0.776 -> 0.776   |  F1 0.767 -> 0.763   |  TP 66/85
-> 66/85
    relevan         recall 0.919 -> 0.915   |  F1 0.922 -> 0.920   |  TP 238/259
-> 237/259
------------------------------------------------------------------------------

=== ASC_AK  BASELINE   (pooled over 8 aspects, n=243) ===
class             TP/supp    prec  recall      F1
positif            75/140   0.750   0.536   0.625
negatif              3/25   0.115   0.120   0.118
netral              47/78   0.402   0.603   0.482
pooled macro-F1                             0.408
```

```
pooled wtd-F1                                0.527
  (headline macro-F1 on the slides = MEAN of the 8 per-aspect macro-F1,
   a different aggregation from this pooled macro-F1)

=== ASC_AK  THESAURUS   (pooled over 8 aspects, n=243) ===
class              TP/supp    prec  recall      F1
positif            110/140   0.679   0.786   0.728
negatif               5/25   0.417   0.200   0.270
netral               27/78   0.391   0.346   0.367
pooled macro-F1                              0.455
pooled wtd-F1                                0.565
  (headline macro-F1 on the slides = MEAN of the 8 per-aspect macro-F1,
   a different aggregation from this pooled macro-F1)

  ASC_AK  baseline -> thesaurus:
    positif         recall 0.536 -> 0.786   |  F1 0.625 -> 0.728   |  TP 75/140
-> 110/140
    negatif         recall 0.120 -> 0.200   |  F1 0.118 -> 0.270   |  TP 3/25 ->
5/25
    netral          recall 0.603 -> 0.346   |  F1 0.482 -> 0.367   |  TP 47/78
-> 27/78
------------------------------------------------------------------------------

=== ASC_AL  BASELINE   (pooled over 8 aspects, n=286) ===
class              TP/supp    prec  recall      F1
positif             75/163   0.758   0.460   0.573
negatif              11/24   0.314   0.458   0.373
netral               64/99   0.421   0.646   0.510
pooled macro-F1                              0.485
pooled wtd-F1                                0.534
  (headline macro-F1 on the slides = MEAN of the 8 per-aspect macro-F1,
   a different aggregation from this pooled macro-F1)

=== ASC_AL  THESAURUS   (pooled over 8 aspects, n=286) ===
class              TP/supp    prec  recall      F1
positif             97/163   0.752   0.595   0.664
negatif               4/24   0.250   0.167   0.200
netral               65/99   0.461   0.657   0.542
pooled macro-F1                              0.469
pooled wtd-F1                                0.583
  (headline macro-F1 on the slides = MEAN of the 8 per-aspect macro-F1,
   a different aggregation from this pooled macro-F1)

  ASC_AL  baseline -> thesaurus:
    positif         recall 0.460 -> 0.595   |  F1 0.573 -> 0.664   |  TP 75/163
-> 97/163
    negatif         recall 0.458 -> 0.167   |  F1 0.373 -> 0.200   |  TP 11/24
-> 4/24
```

```
   netral          recall 0.646 -> 0.657   |  F1 0.510 -> 0.542   |  TP 64/99
-> 65/99
-------------------------------------------------------------------------------
```

9.5 (optional) Save the pooled table to Drive

```
[ ]: pooled = pd.DataFrame(rows)
     out = f'{BASE_PATH}/results/pooled_per_class_metrics.xlsx'
     pooled.to_excel(out, index=False)
     print('Saved ->', out)
     pooled
```

```
Saved -> /content/drive/MyDrive/TA/results/pooled_per_class_metrics.xlsx
```

```
[ ]:    experiment      model          cls   tp  support  precision  recall  \
     0      ACD_AK   baseline  tidak_relevan   48       84      0.667   0.571
     1      ACD_AK  thesaurus  tidak_relevan   56       84      0.629   0.667
     2      ACD_AK   baseline        relevan  220      244      0.859   0.902
     3      ACD_AK  thesaurus        relevan  211      244      0.883   0.865
     4      ACD_AL   baseline  tidak_relevan   66       85      0.759   0.776
     5      ACD_AL  thesaurus  tidak_relevan   66       85      0.750   0.776
     6      ACD_AL   baseline        relevan  238      259      0.926   0.919
     7      ACD_AL  thesaurus        relevan  237      259      0.926   0.915
     8      ASC_AK   baseline        positif   75      140      0.750   0.536
     9      ASC_AK  thesaurus        positif  110      140      0.679   0.786
     10     ASC_AK   baseline        negatif    3       25      0.115   0.120
     11     ASC_AK  thesaurus        negatif    5       25      0.417   0.200
     12     ASC_AK   baseline         netral   47       78      0.402   0.603
     13     ASC_AK  thesaurus         netral   27       78      0.391   0.346
     14     ASC_AL   baseline        positif   75      163      0.758   0.460
     15     ASC_AL  thesaurus        positif   97      163      0.752   0.595
     16     ASC_AL   baseline        negatif   11       24      0.314   0.458
     17     ASC_AL  thesaurus        negatif    4       24      0.250   0.167
     18     ASC_AL   baseline         netral   64       99      0.421   0.646
     19     ASC_AL  thesaurus         netral   65       99      0.461   0.657

            f1
     0   0.615
     1   0.647
     2   0.880
     3   0.874
     4   0.767
     5   0.763
     6   0.922
     7   0.920
     8   0.625
     9   0.728
     10  0.118
```

```
11   0.270
12   0.482
13   0.367
14   0.573
15   0.664
16   0.373
17   0.200
18   0.510
19   0.542
```

9.6 Full pooled confusion matrices (the REAL off-diagonal counts)

The per-class table above only shows the diagonal (TP) and the recall/precision margins. From those alone the *misrouting* cells (e.g. how many `positif` went to `netral`) are **not** uniquely determined. This section prints the complete pooled matrix straight from the prediction files, so every off-diagonal count is a real, readable number — nothing inferred.

```python
def pooled_confusion(true, pred, task, title, show=True):
    """Full pooled confusion matrix (rows = TRUE, cols = PREDICTED) with margins.
    Off-diagonal cells are the REAL misrouting counts, read directly from the
    prediction files -- not inferred from precision/recall. recall = row diagonal
    / row total; precision = col diagonal / col total."""
    names = _names(task); ids = list(range(len(names)))
    cm = confusion_matrix(true, pred, labels=ids)          # rows=true, cols=pred
    row_tot = cm.sum(axis=1)                                 # support per true class
    col_tot = cm.sum(axis=0)                                 # times each class predicted
    diag    = np.diag(cm)
    recall    = np.divide(diag, row_tot, out=np.zeros(len(ids), float), where=row_tot > 0)
    precision = np.divide(diag, col_tot, out=np.zeros(len(ids), float), where=col_tot > 0)
    if show:
        print(f'\n=== {title}  confusion matrix (pooled, n={int(cm.sum())}) ===')
        print('  rows = TRUE, cols = PREDICTED\n')
        head = f'{"true \\ pred":<16}' + ''.join(f'{n:>14}' for n in names) + f'{"| support":>12}{"recall":>9}'
        print(head)
        for i, n in enumerate(names):
            body = ''.join(f'{cm[i, j]:>14}' for j in range(len(names)))
            print(f'{n:<16}{body}   |{row_tot[i]:>8}{recall[i]:>9.3f}')
        print('-' * len(head))
```

```
        print(f'{"pred total":<16}' + ''.join(f'{col_tot[j]:>14}' for j in␣
↪range(len(names))))
        print(f'{"precision":<16}' + ''.join(f'{precision[j]:>14.3f}' for j in␣
↪range(len(names))))
    dfm = pd.DataFrame(cm, index=[f'true_{n}' for n in names],
                           columns=[f'pred_{n}' for n in names])
    dfm['support'] = row_tot
    dfm['recall']  = recall.round(3)
    return dfm
```

[ ]:
```
CM_TABLES = {}   # (experiment, model) -> labelled confusion-matrix DataFrame

for name in EXPERIMENTS:
    task = _task(name)
    paths = {'baseline':  f'{DIRS["baseline"]}/{FILE.format(name=name)}',
             'thesaurus': f'{DIRS["thesaurus"]}/{FILE.format(name=name)}'}
    for model, path in paths.items():
        try:
            true, pred = pooled_from_excel(path, task)
        except FileNotFoundError as e:
            print(f'\n[skip] {name} {model}: missing file -> {e.filename}');␣
↪continue
        CM_TABLES[(name, model)] = pooled_confusion(true, pred, task, f'{name} ␣
↪{model.upper()}')
    print('=' * 78)
```

```
=== ACD_AK  BASELINE  confusion matrix (pooled, n=328) ===
  rows = TRUE, cols = PREDICTED

true \ pred      tidak_relevan        relevan   | support   recall
tidak_relevan               48             36   |      84    0.571
relevan                     24            220   |     244    0.902
-----------------------------------------------------------------
pred total                  72            256
precision                0.667          0.859

=== ACD_AK  THESAURUS  confusion matrix (pooled, n=328) ===
  rows = TRUE, cols = PREDICTED

true \ pred      tidak_relevan        relevan   | support   recall
tidak_relevan               56             28   |      84    0.667
relevan                     33            211   |     244    0.865
-----------------------------------------------------------------
pred total                  89            239
precision                0.629          0.883
==============================================================================
```

```
=== ACD_AL  BASELINE  confusion matrix (pooled, n=344) ===
  rows = TRUE, cols = PREDICTED

true \ pred      tidak_relevan       relevan   | support   recall
tidak_relevan               66            19   |      85    0.776
relevan                     21           238   |     259    0.919
-----------------------------------------------------------------
pred total                  87           257
precision                0.759         0.926

=== ACD_AL  THESAURUS  confusion matrix (pooled, n=344) ===
  rows = TRUE, cols = PREDICTED

true \ pred      tidak_relevan       relevan   | support   recall
tidak_relevan               66            19   |      85    0.776
relevan                     22           237   |     259    0.915
-----------------------------------------------------------------
pred total                  88           256
precision                0.750         0.926
==============================================================================

=== ASC_AK  BASELINE  confusion matrix (pooled, n=243) ===
  rows = TRUE, cols = PREDICTED

true \ pred            positif       negatif        netral   | support   recall
positif                     75            13            52   |     140    0.536
negatif                      4             3            18   |      25    0.120
netral                      21            10            47   |      78    0.603
-------------------------------------------------------------------------------
pred total                 100            26           117
precision                0.750         0.115         0.402

=== ASC_AK  THESAURUS  confusion matrix (pooled, n=243) ===
  rows = TRUE, cols = PREDICTED

true \ pred            positif       negatif        netral   | support   recall
positif                    110             2            28   |     140    0.786
negatif                      6             5            14   |      25    0.200
netral                      46             5            27   |      78    0.346
-------------------------------------------------------------------------------
pred total                 162            12            69
precision                0.679         0.417         0.391
==============================================================================

=== ASC_AL  BASELINE  confusion matrix (pooled, n=286) ===
  rows = TRUE, cols = PREDICTED
```

```
true \ pred              positif       negatif        netral   | support   recall
positif                       75             7            81   |     163    0.460
negatif                        6            11             7   |      24    0.458
netral                        18            17            64   |      99    0.646
-------------------------------------------------------------------------------
pred total                    99            35           152
precision                  0.758         0.314         0.421

=== ASC_AL  THESAURUS  confusion matrix (pooled, n=286) ===
  rows = TRUE, cols = PREDICTED

true \ pred              positif       negatif        netral   | support   recall
positif                       97             3            63   |     163    0.595
negatif                        7             4            13   |      24    0.167
netral                        25             9            65   |      99    0.657
-------------------------------------------------------------------------------
pred total                   129            16           141
precision                  0.752         0.250         0.461
==============================================================================
```

[ ]:
```python
# Save every pooled confusion matrix to one multi-sheet workbook.
cm_out = f'{BASE_PATH}/results/pooled_confusion_matrices.xlsx'
with pd.ExcelWriter(cm_out) as xl:
    for (name, model), dfm in CM_TABLES.items():
        dfm.to_excel(xl, sheet_name=f'{name}_{model}'[:31])   # 31 = Excel␣
 ↪sheet-name limit
print('Saved ->', cm_out)
```

```
Saved -> /content/drive/MyDrive/TA/results/pooled_confusion_matrices.xlsx
```

9.7 (optional) Heatmaps — mirrors the appendix `asc_cm` / `acd_cm` figures

[ ]:
```python
import matplotlib.pyplot as plt

def plot_pooled_cm(dfm, task, title, save_path=None):
    names = _names(task)
    cm = dfm[[f'pred_{n}' for n in names]].values
    fig, ax = plt.subplots(figsize=(4.2, 3.6))
    im = ax.imshow(cm, cmap='Blues')
    ax.set_xticks(range(len(names))); ax.set_xticklabels(names, rotation=45,␣
 ↪ha='right')
    ax.set_yticks(range(len(names))); ax.set_yticklabels(names)
    ax.set_xlabel('predicted'); ax.set_ylabel('true'); ax.set_title(title)
    thr = cm.max() / 2 if cm.max() else 0.5
    for i in range(len(names)):
        for j in range(len(names)):
            ax.text(j, i, int(cm[i, j]), ha='center', va='center',
                    color='white' if cm[i, j] > thr else 'black')
```

```
    fig.tight_layout()
    if save_path:
        fig.savefig(save_path, dpi=150, bbox_inches='tight')
    plt.show()

os.makedirs(f'{BASE_PATH}/results/cm_figs', exist_ok=True)
for (name, model), dfm in CM_TABLES.items():
    plot_pooled_cm(dfm, _task(name), f'{name}  {model}',
                   save_path=f'{BASE_PATH}/results/cm_figs/cm_{name}_{model}.png')
```